NATURE'S BUSINESS MODEL

For Building Better Organizations



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Nature's Business Model

52 Principles from Nature for Building Better Organizations

Ron Taylor

An Executive Class Series

INVEST IN YOUR SUCCESS

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How to Use This Book

This book takes fifty-two mechanisms from the natural world — how bees choose a home, how starlings turn as one, how a forest recovers from fire — and asks what each one can teach an organization. Every chapter starts with the biology, as the research describes it, and then tests how far the business lesson really holds.

How each chapter works

Evidence Level. Each chapter rates two things separately: the *biological claim* and the *business application*. **Level 1**: strong scientific evidence for the biology, applied to business as an analogy. **Level 2**: documented biomimicry, where the biology was actually transferred into a technology or design, or biology that is real but still debated. **Level 3**: descriptive biology used as a metaphor. The business application is always an analogy; the chapters say where it breaks down.

Executive Class Lens turns the mechanism into a question about your own work. **Key Takeaways** and the **Summary** give the main points in plain English. **Science Check** separates what is *established*, what is *inferred* and what remains *uncertain* — several popular nature stories are better or worse than their reputation.

Vocabulary & Phrasal Verbs and **Practice** give you the language to discuss the idea, and the **Vocabulary Index** at the back lists every term. **Sources** are graded: **A** for peer-reviewed research and primary sources, **B** for books and long-form science writing, **C** for general reference and commentary.

This Connects To links chapters across the book. **Interview & Workplace Practice** asks you first to explain the principle, then to make it your own

story using the STAR method: the **Situation**, your **Task**, the **Action** you took and the **Result**.

The Six Themes

At a glance, in the order they appear.

WEEKS 1–9 **PART I**

Decision-Making & Collective Intelligence

How does nature make good decisions?

WEEKS 10–18 **PART II**

Leadership & Teams

How does nature organize groups?

WEEKS 19–27 **PART III**

Resilience & Adaptation

How does nature survive disruption?

WEEKS 28–36 **PART IV**

Innovation & Design

How does nature solve problems?

WEEKS 37–43 **PART V**

Organizational Design & Systems

How does nature build sustainable systems?

WEEKS 44–52 **PART VI**

Life, Strategy & Personal Success

What can nature teach us about living well?

PART I

Decision-Making & Collective Intelligence

How does nature make good decisions?

1

Ants: Order Without a Manager

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

An ant is not clever. A single ant has a tiny brain, poor eyesight and almost no idea what is happening more than a few centimetres away. And yet an ant colony can find the shortest route to a food source, share out work between thousands of individuals, and change its plans when the environment changes. Nobody is in charge. There is no manager.

The mechanism is simple. Ants leave a chemical trail behind them as they walk. A shorter route gets travelled more often, so its trail builds up faster, so more ants choose it — and the colony gradually settles on a good answer without anyone ever comparing the options. Bad routes fade away on their own, because the chemical evaporates when nobody reinforces it.

This idea turned out to be so powerful that computer scientists borrowed it. "Ant colony optimization" is now used to work out delivery schedules, telecommunications routing and logistics problems that are too complex to solve by calculating every possibility.

This was demonstrated with a piece of apparatus so simple it is still used in teaching. In the late 1980s a Belgian team gave Argentine ants two bridges between the nest and a food source, one twice as long as the other. Ants set out along both at random. Those on the short bridge returned sooner, so its trail was reinforced twice as fast, and within minutes almost all traffic had transferred to it. Nobody compared the bridges. The colony's answer was a side-effect of the time each journey took.

The same apparatus revealed the failure mode. When the short bridge was only made available after the long trail was already established, the colony frequently carried on using the long route — the reinforcement that found the answer now defended it.

There is a catch, and it is instructive. If a trail becomes strong early, the colony can lock onto a route that is merely good rather than best, because the reinforcement that finds the answer also defends it. This is why the chemical must evaporate, and why the algorithms modelled on ants build evaporation in deliberately: without a force that steadily weakens established paths, the system stops exploring and simply repeats itself.

Notice what a system like this actually requires, because organizations usually install one part of it and expect the result. It needs enough parallel attempts that a better option can be discovered at all; a signal that grows stronger when something works; and a rule that weakens options nobody is choosing. A company running a single pilot has no parallel attempts. A company where success is announced but never reallocates budget has no reinforcing signal. A company that never retires an initiative has no decay — and, like the ants on the long bridge, will defend whatever it happened to try first.

In the algorithms this is made explicit. A simulated ant chooses its next step probabilistically, so there is always some chance of taking the less-marked path — and if that term is removed, the algorithm locks onto the first decent solution and stops improving. Few companies write the equivalent term into their own systems. Budget follows the strongest signal, attention follows budget, and nothing at all is reserved for the option with no evidence behind it yet.

The lesson for organizations is uncomfortable. We often assume that good decisions come from clever people at the top. Ants suggest something different: good decisions can come from a good system — one that lets many

small experiments run at once, strengthens what works, and quietly abandons what doesn't.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Who in your organization is allowed to try a second option without asking permission?*

KEY TAKEAWAYS

1. Complex, high-quality decisions can emerge from simple individuals following simple rules.
2. Success should feed back into the system so that good options automatically become stronger.
3. Weak options must be allowed to fade away; nobody needs to formally cancel them.

SUMMARY

Ant colonies solve difficult problems without any central control, by reinforcing successful paths and letting unsuccessful ones disappear. This principle has been turned into real optimization algorithms. For organizations, it suggests that a well-designed system can produce better decisions than a single brilliant decision-maker.

SCIENCE CHECK

ESTABLISHED

Ants select shorter routes through pheromone reinforcement; this has been shown experimentally and formalized as an algorithm.

INFERRED

That organizations should be designed as reinforcement systems rather than around a central decision-maker.

UNCERTAIN

How far the analogy extends: human 'trails' carry status and budget, which do not evaporate on their own.

VOCABULARY & PHRASAL VERBS

colony (n.) a large group of animals of the same kind living together. The colony survives the winter by living off stored food.

trail (n.) a line or mark left behind that others can follow. The consultants left a trail of unread reports behind them.

to be in charge (phrase) to have control or responsibility. Nobody was quite sure who was in charge of the launch.

build up (phrasal verb) to increase gradually over time. Pressure built up for months before anyone raised it.

fade away (phrasal verb) to become weaker and slowly disappear. Interest in the initiative faded away once the funding stopped.

settle on (phrasal verb) to finally choose one option after considering others. After three meetings we finally settled on a supplier.

work out (phrasal verb) to calculate or find the answer to something. It took the team a week to work out the real cause.

borrow (v.) here: to take an idea from another field and use it in your own. Software engineers borrowed the idea from biology.

PRACTICE

Where the Analogy Breaks

An ant that follows a weak trail loses a few minutes. A person who follows a failing strategy may lose a job, so humans defend paths for reasons ants never have.

Use the Language

Make sentences using these three words: settle on, fade away and build up.

Discuss

Does your organization run several small experiments at once, or one large plan? Which does it claim to do?

Write

Describe in 120 words a decision your team made well without anyone senior deciding it.

SOURCES

A Goss, S., Aron, S., Deneubourg, J.-L. & Pasteels, J. M., "Self-organized shortcuts in the Argentine ant", Naturwissenschaften 76, 579–581 (1989).

B Dorigo, M. & Stützle, T., Ant Colony Optimization, MIT Press (2004).

THE CONTRAST

Story 1: Ants: Order Without a Manager vs. Story 12: Elephants: What the Oldest Female Knows

Story 1: Ants: Order Without a Manager: Good decisions emerge from many simple individuals following local rules, with no one holding the full picture.

Story 12: Elephants: What the Oldest Female Knows: Good decisions come from one individual holding decades of accumulated, specific knowledge that the group defers to.

Is a good decision better produced by distributed simple rules, or by concentrated deep experience?

THIS CONNECTS TO

Story 4: Fish: The Informed Minority SAME MECHANISM
Both show a group reaching a good outcome without every member needing to know the answer — a small informed part, or a simple reinforced signal, carries the rest.

Story 8: Slime Mould: Build Wide, Then Prune SAME MECHANISM
Both use reinforcement of what works and quiet abandonment of what doesn't, rather than advance planning, to build an efficient network.

Story 17: Ants: Thresholds, Not Assignments LIMITATION
This chapter shows decisions emerging from reinforcement over time. Thresholds shows a faster, second mechanism in the same species — individuals responding instantly to visible unmet need — proving colonies use more than one coordination method depending on the problem.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

An ant colony finds the shortest route to a food source without any individual ant knowing the map.

What is the mechanism?

Ants laying a pheromone trail reinforce whichever path returns fastest, since more ants complete a short trip in the same time and lay more pheromone on it; slower paths fade as the scent evaporates.

What organizational problem does it explain?

How can an organization make good routing or resource-allocation decisions without a central planner who has to model the whole system in advance?

Level 2 — Make It Your Story

> *Tell me about a time when a good outcome emerged from a decentralized process rather than a single decision-maker's plan.*

Use STAR to answer. Think about: a team process, a market mechanism, a decision that improved once you stopped trying to control it centrally.

TAKE THE IDEA FURTHER

> *What decision in your organization currently requires central approval that might actually improve if it were made locally and simply reinforced when it worked?*

2

Honeybees: Quorum, Not Consensus

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

When a honeybee colony grows too large, roughly two-thirds to three-quarters of the workers leave with the old queen and hang in a cluster on a branch while they decide where to live next. The decision matters enormously: choose a poor cavity and the whole swarm may die in winter. They usually get it right.

Several hundred scout bees fly off independently and inspect possible sites. Each scout assesses a site on her own — its volume, its entrance, its dryness — and then returns and performs a waggle dance whose vigour reflects how good she thinks the site is. Crucially, a scout does not simply copy her neighbours. She goes and looks for herself first.

Other scouts then visit the advertised sites and form their own opinion. Support builds for the better options, and the swarm commits only once enough scouts agree on a single site — a quorum, not a majority vote and not the first idea proposed. The biologist Thomas Seeley, who spent decades studying this, argues that the bees achieve something many human committees fail at: they gather genuinely independent judgments before they converge.

We know the detail because of a decade of work on Appledore Island, off the coast of Maine — a treeless place with almost no natural nest cavities, where a researcher can put out wooden boxes of known quality and be confident the bees have nothing else to choose from. Thomas Seeley's team marked

individual scouts with paint, filmed the dances on the surface of the swarm, and reconstructed which bee visited which box and how hard she argued for it.

What the recordings showed was that a swarm offered several boxes of different quality usually ends up in the best one available, and that the scouts advertising the poorer sites are not defeated in argument. They simply stop dancing. A scout returns from a mediocre site with a shorter dance each time until she falls silent — while a scout who keeps finding the site as good as she remembered keeps recruiting.

The assessment itself is more rigorous than the word "inspect" suggests. A scout entering a candidate cavity spends the better part of an hour inside it, walking the interior walls and making short flights across the space, and from that survey she arrives at an estimate of its volume — a judgment she makes alone, in the dark, about a space she has never seen whole.

It is worth naming what this costs. The swarm may inspect a dozen or more sites and hang exposed on its branch for hours or days while it does so, unfed and vulnerable to weather. Independent assessment is not the fast option. The bees accept a real delay in exchange for a decision they only get to make once.

The commitment, when it comes, is not announced. Scouts begin producing a high-pitched piping signal that prompts the resting bees to warm their flight muscles, then run through the cluster jostling their neighbours until the whole swarm is ready to fly. Only then does it lift off — with a small number of informed scouts streaking repeatedly through the airborne cloud to indicate the direction. The deliberation is slow and the execution is fast, which is the opposite of most organizational timetables.

Most bad meetings fail at exactly this point. Someone senior speaks first, everyone adjusts to that view, and the room mistakes agreement for accuracy.

The bees separate the two steps: first collect independent information, then converge.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *In your last important meeting, did anyone form an opinion before hearing the senior person's?*

KEY TAKEAWAYS

1. Independent assessment must come before group discussion, not after it.
2. A quorum threshold prevents a group from committing too early to the first proposal.
3. Agreement that arrives too quickly is not evidence that the decision is good.

SUMMARY

Honeybee swarms choose a new home by sending out scouts who inspect sites independently and then advertise them, with the colony committing only when enough scouts converge on one option. The process protects independent judgment before consensus forms. For teams, it warns against confusing consensus with conformity.

SCIENCE CHECK

ESTABLISHED

Swarms gather independent assessments and commit at a quorum, usually choosing the best available site.

INFERRED

That human groups should separate independent judgment from group discussion.

UNCERTAIN

Whether the quality of bee decisions depends on conditions rarely present in meetings — no hierarchy, no careers, no cost to dissent.

VOCABULARY & PHRASAL VERBS

swarm (n.) a large group of bees moving together. A swarm of about ten thousand bees left the hive that morning.

scout (n.) someone sent ahead to look for information. She acts as a scout for the company, testing new markets first.

cavity (n.) a hollow space inside something. The bees chose a dry cavity in an old oak.

quorum (n.) the minimum number of supporters needed before a decision is valid. The committee could not vote because there was no quorum.

to converge (v.) to come together towards one point or one opinion. The two teams slowly converged on a single design.

conformity (n.) behaving the same as everyone else because of social pressure. Conformity in meetings is often mistaken for agreement.

fly off / go off (phrasal verb) to leave, usually quickly. He went off to check the figures himself.

look into (phrasal verb) to investigate something. Could you look into why the delivery was late?

come up with (phrasal verb) to produce an idea or suggestion. She came up with a solution nobody had considered.

PRACTICE

Where the Analogy Breaks

Bees have no careers, no reputations and nothing to lose by disagreeing. Human independence is expensive in a way bee independence is not.

Use the Language

Make sentences using these three words: converge, conformity and quorum.

Discuss

Is it safe in your workplace to disagree with the most senior person present? What makes it safe or unsafe?

Write

Write a short protocol (five steps) for running a meeting so that opinions are collected before they are compared.

SOURCES

A "Collective decisions and cognition in bees", Nature. https://www.nature.com/articles/17047

B Seeley, T. D., Honeybee Democracy, Princeton University Press (2010).

THE CONTRAST

Story 2: Honeybees: Quorum, Not Consensus vs. Story 18: Bees: Dissent That Improves the Choice

Story 2: Honeybees: Quorum, Not Consensus: The group needs independent scouts to assess options separately, before comparing notes.

Story 18: Bees: Dissent That Improves the Choice: The group needs competing options to stay alive and contested for a while, rather than converging too fast.

Is the bigger risk in group decisions premature agreement, or not having independently gathered information to agree on in the first place?

THIS CONNECTS TO

Story 3: Locusts: Why Alignment Resists Change LIMITATION
Honeybee quorum sensing shows coordination working well. Locusts show the same underlying mechanism — individuals responding to neighbours — producing the opposite outcome: coordination so strong it resists correction.

Story 18: Bees: Dissent That Improves the Choice SAME MECHANISM
Both chapters describe the same species using a decision process built to prevent premature agreement, whether through independent scouting or deliberately sustained disagreement.

Story 6: Ants: Weighing Two Sources of Truth SAME MECHANISM
Both show a colony combining multiple independent signals rather than simply following the loudest or first one, to reach a more reliable collective judgment.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A honeybee swarm choosing a new nest site reaches a fast, reliable group decision without ever taking a vote in the way we'd recognize one.

What is the mechanism?

Scout bees independently assess candidate sites, then advertise the best ones through a waggle dance whose intensity reflects site quality; scouts recruit others to visit that site, and a decision is reached once enough independent scouts converge on it — a quorum, not universal agreement.

What organizational problem does it explain?

How can a group reach a fast, high-quality decision without either forcing early consensus or letting the most persuasive voice dominate before others have assessed the options themselves?

Level 2 — Make It Your Story

> *Tell me about a time your team reached a good decision because people assessed options independently before comparing notes, rather than discussing first.*

Use STAR to answer. Think about: a hiring decision, a design choice, a vote that worked better because opinions formed independently first.

TAKE THE IDEA FURTHER

> *Does your team's decision process ask people to form an independent view before the group discussion starts, or does the discussion happen first and shape everyone's opinion before it's really theirs?*

3

Locusts: Why Alignment Resists Change

> **Evidence Level 3**
>
> **Biological claim:** Descriptive or established
>
> **Business application:** Metaphor / inference

A locust swarm has a strange property: once it is moving in one direction, it is very hard to turn. Each individual is watching its neighbours, and each neighbour is watching it back. Direction is not chosen; it is reinforced. The same mechanism that gives a swarm its impressive coordination also makes it stubborn.

Researchers studying marching bands of locusts and other collectives have noted that alignment behaviour can lock a group into a heading long after conditions have changed. When every member is copying every other member, the group has no way to notice that the original reason for the direction has disappeared.

The alignment itself has been measured. In one widely cited experiment, juvenile locusts were placed in a ring-shaped arena at different population densities and filmed for hours. At low density, individuals moved independently and the group went nowhere in particular. Above a threshold density, marching emerged suddenly: the group locked into a single direction and stayed there. At intermediate densities the researchers saw something stranger still — the entire band would spontaneously reverse and march the other way, with no external cause at all.

That is the shape of the problem. A group like this has no smooth way to steer. It has one direction, and occasional total reversals.

The insects in a marching band are not quite the same animals that were there before it formed. Desert locusts exist in two forms — a solitary phase that avoids others, and a gregarious phase that seeks them out — and crowding converts one into the other, with repeated contact to the hind legs triggering the change through the nervous system within hours. The individual changes because the density changed. It is worth holding that beside any confident account of why people behave differently in a large organization than they did in a small one.

What breaks the lock? Sometimes it is simply individuals who stop — who become temporarily neutral rather than actively arguing for the opposite direction. Modelling work on collective behaviour suggests that uncommitted or uninformed individuals can weaken an entrenched majority and make it easier for a group to shift.

There is a practical consequence, and it concerns timing rather than courage. A pause is almost impossible to call once the march is under way, because by then it reads as disloyalty. It is easy to schedule in advance, when nobody is invested — which is why serious organizations write review points into a project at the start: a date, a named decision, and an explicit list of what would justify stopping. The point of the pre-commitment is not the review itself. It is that the pause was authorized by the group before momentum made it unsayable.

This is worth thinking about in a business context, and it is an analogy rather than a scientific finding about humans. When a company is committed to a failing strategy, the loudest opponent often makes things worse: opposition creates defensiveness, and the majority digs in. Sometimes the more effective intervention is to stop the machine — pause the project, postpone the launch, call a review — so that momentum stops doing the group's thinking for it. A pause is not indecision. It can be an active strategic tool.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What is your team still doing only because it started doing it?*

KEY TAKEAWAYS

1. The same mechanism that produces coordination can also produce collective stubbornness.
2. Momentum can keep a group moving long after the original reason has gone.
3. Deliberately pausing can be a stronger intervention than arguing against the direction.

SUMMARY

Locust swarms stay aligned because every individual copies its neighbours, which makes the group extremely hard to turn. Interrupting that copying — rather than opposing it — is what allows a change of direction. As an analogy for organizations, it suggests that a well-timed pause can be more powerful than open disagreement.

SCIENCE CHECK

ESTABLISHED

Locust bands align above a density threshold and can reverse direction spontaneously.

INFERRED

That organizations become locked into direction by mutual copying, and that pausing breaks the lock.

UNCERTAIN

The human mechanism: people align through loyalty, contracts and sunk cost, not through copying motion. The transfer is a metaphor.

VOCABULARY & PHRASAL VERBS

stubborn (adj.) refusing to change your mind or direction. He is stubborn about the deadline even though the data has changed.

alignment (n.) being arranged in the same direction or agreement. Alignment is useful until it stops anyone asking questions.

entrenched (adj.) firmly fixed and very difficult to change. The habit is entrenched; changing it will take more than an email.

momentum (n.) the force that keeps something moving once it has started. The project had so much momentum that stopping felt impossible.

indecision (n.) the state of being unable to decide. Weeks of indecision cost more than a wrong choice would have.

dig in (phrasal verb) to refuse to change your position in an argument. Both sides dug in and the discussion went nowhere.

hold off (phrasal verb) to delay doing something. Let's hold off on hiring until the results are in.

call off (phrasal verb) to cancel something that was planned. They called off the launch two days before the date.

back down (phrasal verb) to admit you were wrong and stop opposing. He backed down once he saw the customer data.

PRACTICE

Where the Analogy Breaks

Locusts align by copying motion. Humans also align through loyalty, contracts and sunk cost — mechanisms with no insect equivalent.

Use the Language

Make sentences using these three words: momentum, dig in and hold off.

Discuss

Think of a project that continued after its reason had gone. Why did nobody stop it?

Write

Write 120 words proposing a review point for a current project: the date, the decision, and what would justify stopping.

SOURCES

A Couzin, I. D. et al., "Uninformed individuals promote democratic consensus in animal groups", Science (2011).

A Buhl, J. et al., "From disorder to order in marching locusts", Science (2006).

THE CONTRAST

Story 3: Locusts: Why Alignment Resists Change vs. Story 18: Bees: Dissent That Improves the Choice

Story 3: Locusts: Why Alignment Resists Change: Strong alignment, once established, becomes very difficult to reverse — the same mechanism that creates coordination locks it in place.

Story 18: Bees: Dissent That Improves the Choice: Healthy groups deliberately preserve disagreement long enough for real alternatives to be tested before converging.

Is alignment itself the goal, or is premature alignment actually one of the biggest risks a group decision process faces?

THIS CONNECTS TO

Story 2: Honeybees: Quorum, Not Consensus LIMITATION

Honeybee quorum sensing shows the same neighbour-following mechanism producing a good outcome. Locusts show it producing the opposite: coordination so strong it resists correction.

Story 5: Starlings: Seven Neighbours Is Enough SAME MECHANISM

Both flocking behaviours emerge from individuals responding only to a small number of nearby neighbours, with no leader and no global view.

Story 4: Fish: The Informed Minority LIMITATION

A locust swarm cannot be redirected by a small informed subset once alignment locks in. Fish schools show the opposite case: a small informed minority can still steer the whole group, which suggests the difference lies in how tightly the alignment mechanism resists outside signals.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A locust swarm, once moving in one direction, becomes extremely difficult to turn or stop, even when that direction leads somewhere destructive.

What is the mechanism?

Each locust orients toward its immediate neighbours' movement and, at high densities, cannibalism risk pushes individuals to keep moving to avoid being eaten by those behind them — the same local rule that produces coordinated movement also makes that movement self-reinforcing and hard to redirect.

What organizational problem does it explain?

How does an organization recognize when its own strong alignment — usually seen as a strength — has become a liability that resists correction even after the evidence changes?

Level 2 — Make It Your Story

> *Tell me about a time your team's strong alignment around a direction made it harder, not easier, to change course when the evidence changed.*

Use STAR to answer. Think about: a strategy everyone bought into, a project with too much momentum to stop, groupthink you experienced directly.

TAKE THE IDEA FURTHER

> *What decision in your organization is moving forward mainly because everyone else is moving forward, rather than because anyone has recently checked whether the direction is still right?*

4

Fish: The Informed Minority

> **Evidence Level 1**
>
> **Biological claim:** Strong scientific evidence
>
> **Business application:** Analogy

Imagine a school of a thousand fish. A few dozen of them know where the food is. The rest know nothing at all. How accurately does the school arrive?

Surprisingly accurately. Research on collective movement in animal groups found that only a small minority of informed individuals is needed to guide a very large group towards a target — and, remarkably, the larger the group gets, the smaller the proportion of informed individuals required. The uninformed majority is not a problem to be solved. It is what makes the group cohesive enough to be guided at all.

The finding began as a model. Simulating individuals that balance staying with their neighbours against a private preferred direction, the researchers showed that accuracy rises steeply with even a small informed minority, and that the fraction needed falls as the group grows. Experiments then went looking for it in live animals — training a handful of fish to expect food in one part of a tank and releasing them into untrained shoals, and later using robotic replica fish whose behaviour could be controlled precisely. The broad prediction held: a few individuals with a consistent bias move the group.

There is a further consequence that runs against managerial instinct. Because individual errors point in different directions, they partly cancel when many individuals combine — so a larger group can be more accurate than a smaller one, and adding uninformed members does not necessarily dilute the signal.

The reflex of shrinking a decision to fewer, more senior people assumes the opposite: that quality lies in the individuals rather than in the aggregation.

Even more striking is what the informed fish do not do. They do not issue instructions. They do not identify themselves as leaders. They do not have any special status within the school. They simply swim slightly more directly towards where they know the food is, while still staying with their neighbours. That tiny bias, repeated across many individuals, steers the whole school.

The same research showed the limit of the method. When informed individuals disagreed with each other, the group compromised between their directions up to a point — and beyond that point it split. Influence therefore depends on staying inside the group's tolerance. A leader who pushes harder than the school can absorb does not steer it faster; they break it into two schools, each heading somewhere less useful.

There is an uncomfortable corollary for anyone who has ever stayed quiet in a meeting. In this model an informed individual who suppresses their bias entirely — who swims exactly like everyone else — contributes nothing to the group's accuracy. They become part of the uninformed majority. The information only counts once it is expressed as a slightly different direction of travel, which in an organization means saying the thing, making the different recommendation, or booking the meeting nobody asked for.

For managers, this reframes what leadership actually is. We tend to picture leadership as authority: the right to tell people what to do. The fish suggest an alternative model in which leadership is information plus commitment to the group. If you know something useful and you stay connected to the people around you, you will influence direction — with no title required, and no one needing to obey.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Who on your team knows something useful and has no authority to say so?*

KEY TAKEAWAYS

1. A small informed minority can guide a very large group towards an accurate outcome.
2. Influence can come from information and direction of travel rather than from authority.
3. Uninformed members are not dead weight; they provide the cohesion that makes guidance possible.

SUMMARY

Studies of animal groups on the move show that a small number of informed individuals can steer a much larger group without any signalling or hierarchy. Influence spreads simply because informed members move slightly more decisively. Leadership, on this model, is less about controlling a group than about knowing something and staying connected to it.

SCIENCE CHECK

ESTABLISHED

A small informed minority can guide a large group accurately, and the required fraction falls as the group grows.

INFERRED

That influence in organizations comes from information plus connection rather than authority.

UNCERTAIN

Whether the model's assumptions — one shared goal, no conflicting incentives — ever hold in a workplace.

VOCABULARY & PHRASAL VERBS

a school (of fish) (n.) a large group of fish swimming together. A school of small fish turned all at once beneath the boat.

cohesive (adj.) sticking together as a unit. A cohesive team disagrees loudly and still moves together.

bias (n.) a slight tendency in one direction. Her bias towards testing early has saved the company twice.

to steer (v.) to guide something in a particular direction. A good chair steers the discussion without dominating it.

dead weight (n.) something useless that others must carry. Treating quiet colleagues as dead weight is usually a mistake.

stand out (phrasal verb) to be noticeably different from others.

One recommendation stood out from all the others.

stick with (phrasal verb) to stay close to someone; to continue with something. Stick with the group even when you disagree with the route.

head for (phrasal verb) to move in the direction of a place. The department is heading for a difficult quarter.

catch on (phrasal verb) to become popular or widely adopted. The new process caught on faster than anyone expected.

PRACTICE

Where the Analogy Breaks

Fish never disagree about the destination. Human groups often contain people heading somewhere else on purpose.

Use the Language

Make sentences using these three words: steer, stick with and cohesive.

Discuss

Who in your organization has useful information but no authority? What stops it travelling?

Write

Describe a time you stayed quiet in a meeting. Write what you would say now, in three sentences.

SOURCES

A Couzin, I. D., Krause, J., Franks, N. R. & Levin, S. A., "Effective leadership and decision-making in animal groups on the move", Nature 433, 513–516 (2005). https://www.nature.com/articles/nature03236

THE CONTRAST

Story 4: Fish: The Informed Minority vs. Story 3: Locusts: Why Alignment Resists Change

Story 4: Fish: The Informed Minority: A small informed minority can still steer the whole group toward an accurate outcome.

Story 3: Locusts: Why Alignment Resists Change: Once strongly aligned, a group can become resistant to correction even by informed dissent.

Under what conditions can a small informed group still redirect the majority, and when does alignment become too strong for that to work?

THIS CONNECTS TO

Story 3: Locusts: Why Alignment Resists Change LIMITATION
A locust swarm cannot be redirected by a small informed subset once alignment locks in. Fish schools show the opposite case: a small informed minority can still steer the whole group, suggesting the difference lies in how tightly the alignment mechanism resists outside signals.

Story 1: Ants: Order Without a Manager SAME MECHANISM
Both show a group reaching a good outcome without every member needing to know the answer — a small informed part, or a simple reinforced signal, carries the rest.

Story 5: Starlings: Seven Neighbours Is Enough SAME MECHANISM
Both are cases of large-scale coordinated movement (schooling, flocking) produced entirely by individuals responding to a handful of near neighbours, with no leader.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A fish school can be steered toward food or away from danger by a small number of informed individuals, even though most fish in the school have no information at all.

What is the mechanism?

Each fish adjusts its position based on its immediate neighbours; informed individuals move with slightly more confidence toward their goal, and that small confidence differential is enough to bias the whole group's direction without any signalling or communication beyond normal schooling behaviour.

What organizational problem does it explain?

How much organizational influence does a small, genuinely informed group actually need to shift a much larger group's direction, without needing formal authority to do it?

Level 2 — Make It Your Story

> *Tell me about a time a small group with better information was able to shift the direction of a much larger team or organization.*

Use STAR to answer. Think about: a pilot program, an early adopter group, a small team whose confidence changed how others acted.

TAKE THE IDEA FURTHER

> *Where in your organization does a small, well-informed group already have more influence on direction than its size or formal authority would suggest?*

5

Starlings: Seven Neighbours Is Enough

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

On a winter evening, tens of thousands of starlings rise together over a field and move as one body — expanding, folding, splitting and reforming in seconds. It is called a murmuration, and for a long time people assumed there had to be a leader, or even some form of telepathy.

There is neither. Ballerini's team, using synchronized stereo photography, found that each starling pays attention to a fixed number of nearest neighbours — roughly six or seven — regardless of how far away they are or how dense the flock is. Each bird matches speed and direction with those few neighbours and avoids collisions. That is the entire rule set. Every dramatic manoeuvre you see is that local rule, repeated at scale, propagating across the flock faster than any central instruction ever could.

Establishing this took an unusual piece of fieldwork. A European project set up synchronized cameras on the roof of a museum in Rome, photographed flocks of up to several thousand starlings from slightly different angles, and reconstructed the three-dimensional position of every individual bird — then tracked those positions frame by frame to work out who was responding to whom. It is the kind of measurement that sounds trivial and is not: matching thousands of nearly identical black dots between two images, up to ten times a second, was most of the work.

The result overturned the standing assumption. Interaction was not governed by distance, as every earlier model had assumed. It was governed by number.

Notice why the "six or seven neighbours" detail matters. If each bird tracked everything within a fixed distance, the flock would fall apart whenever it spread out. By tracking a fixed number of relationships, cohesion survives changes in density. The structure is robust because it depends on relationships rather than on positions.

The translation is more concrete than it first appears. It suggests that what an individual needs is not visibility of the whole organization — which is what dashboards and all-hands meetings attempt — but a small, stable set of colleagues whose current state they genuinely track. Six or seven. And it suggests that the most disruptive thing a reorganization does is not changing the reporting lines. It is scattering the handful of people each person had learned to move with.

Researchers have also tracked what happens when a peregrine attacks. Dark bands of agitation travel across the flock, and analysis of the recordings shows these waves moving faster than any bird's own reaction to the falcon could account for — because each starling is responding to the movement of its neighbours rather than to the predator. An organization's response time is set the same way: not by how quickly the person who sees the problem can act, but by how quickly one person's movement changes the behaviour of the next.

Organizations reach for exactly the opposite solution. When coordination fails, we usually add reporting, approval steps and central oversight — which slows the response down at the moment when speed matters most. The starlings suggest an alternative: give people clear local rules, real-time information about what their immediate colleagues are doing, and the authority to respond without asking.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *How many approvals stand between noticing a problem and acting on it?*

KEY TAKEAWAYS

1. Large-scale coordination can be produced entirely by local rules, with no central commander.
2. Coordination based on relationships survives change better than coordination based on fixed positions.
3. Central oversight tends to slow a system down precisely when speed is most valuable.

SUMMARY

Starlings coordinate enormous flocks by each tracking only six or seven nearest neighbours and matching their movement. No leader, no signal and no global plan is involved, yet the flock responds almost instantly. Organizations become similarly agile when individuals can act on local information rather than routing every decision upward.

SCIENCE CHECK

ESTABLISHED

Starlings track a fixed number of nearest neighbours rather than all birds within a fixed distance.

INFERRED

That teams coordinate better through a small stable set of relationships than through central oversight.

UNCERTAIN

How many 'neighbours' a person can track at work, and whether the number is stable across tasks.

VOCABULARY & PHRASAL VERBS

murmuration (n.) the collective name for a flock of starlings in flight. A murmuration of starlings moved over the field like smoke.

flock (n.) a group of birds. The whole flock changed direction in less than a second.

manoeuvre (n.) a skilful or careful movement (US spelling: manoeuvre). The manoeuvre was completed without a single collision.

to propagate (v.) to spread from one place to the next. Panic propagated through the market within minutes.

robust (adj.) strong; not easily broken or disrupted. The process is robust: it works even when two people are away.

oversight (n.) supervision and monitoring by someone in authority. Adding oversight slowed the response without improving it.

fall apart (phrasal verb) to break into pieces; to stop functioning. The plan fell apart as soon as the supplier withdrew.

spread out (phrasal verb) to move apart over a wider area. The team spread out across three floors and stopped talking.

slow down (phrasal verb) to reduce speed. Approvals slow us down at exactly the wrong moment.

keep up with (phrasal verb) to move or progress at the same rate as others. It is hard to keep up with changes announced by email.

PRACTICE

Where the Analogy Breaks

Starlings share one goal and one kind of information. Organizations coordinate people with different incentives, which no local rule alone can solve.

Use the Language

Make sentences using these three words: oversight, fall apart and keep up with.

Discuss

Name the six or seven colleagues whose current work you genuinely track. Is the list the same as your reporting line?

Write

Write 120 words on one approval step you would remove, and what would go wrong if you did.

SOURCES

A Ballerini, M. et al., "Interaction ruling animal collective behaviour depends on topological rather than metric distance", PNAS (2008).

C World Economic Forum, "Nature can teach us about leadership and building resilience" (2026). https://www.weforum.org/stories/2026/04/nature-leadership-resilience-fractured-world/

THIS CONNECTS TO

Story 3: Locusts: Why Alignment Resists Change SAME MECHANISM
Both flocking behaviours emerge from individuals responding only to a small number of nearby neighbours, with no leader and no global view.

Story 4: Fish: The Informed Minority SAME MECHANISM
Both are cases of large-scale coordinated movement produced entirely by individuals responding to a handful of near neighbours, with no leader.

Story 1: Ants: Order Without a Manager LIMITATION
Starling flocks coordinate through continuous real-time position-matching with neighbours. Ant trail formation coordinates through a slower, indirect signal left in the environment — two different mechanisms for reaching the same kind of leaderless coordination.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A starling murmuration of thousands of birds moves as a single fluid shape, changing direction almost instantly, with no bird leading and no signal visibly broadcast to the whole flock.

What is the mechanism?

Each starling tracks and matches the movement of roughly seven of its nearest neighbours, regardless of the flock's total size; that fixed, small number of local relationships is enough for a change in movement to propagate through the entire flock almost instantly.

What organizational problem does it explain?

How can very large organizations stay responsive and coordinated without requiring everyone to track everyone else, or funnel every change through a central point?

Level 2 — Make It Your Story

> *Tell me about a time a large team or organization stayed coordinated because people were closely attuned to a few immediate colleagues, not because of a central broadcast.*

Use STAR to answer. Think about: a fast-moving team, an operational floor, a moment coordination happened without anyone formally directing it.

TAKE THE IDEA FURTHER

> *Is your organization trying to coordinate everyone through a small number of central broadcasts, when a starling-style model — everyone closely tracking a few immediate neighbours — might actually propagate change faster?*

6

Ants: Weighing Two Sources of Truth

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

An ant walking towards a food source has two sources of information available. The first is private: what she has learned herself, on previous trips, using her own memory and her own senses. The second is social: the pheromone trail left by everyone else. What should she do when the two disagree?

Research on ants has found that the answer depends on certainty. When an ant's private information is reliable — she has visited the site recently and repeatedly — she tends to trust herself and largely ignore the trail. When her private information is weak or out of date, she relies much more heavily on what the colony is telling her. The colony as a whole therefore weights information roughly according to how good it is.

The experiments that established this are ingenious in a slightly cruel way. Working with the black garden ant, researchers let individuals learn a route to a feeder, so that each ant carried a private memory, and then arranged for the pheromone trail to point somewhere else. Ants with a well-learned, recently confirmed memory largely ignored the conflicting trail and went their own way. When the same memory was made unreliable — the reward moved, the route changed between visits — the same ants deferred to the trail instead. The weighting is not a fixed personality trait. It tracks how good each source has recently proved to be.

That is a genuinely sophisticated system, and it is worth noticing that it requires two things at once: information must be able to move freely through the colony, and individuals must retain the right to override it.

The moment of conflict is visible to the observer. An ant whose memory and whose trail disagree will often stop, turn back on herself, and set off again — a small, legible hesitation that marks the point where two sources of information are being compared rather than blindly combined.

It is also worth noting what makes the system self-correcting. The pheromone evaporates and the memory fades, so both sources decay unless they are refreshed by a journey that confirms them. Neither can win permanently by being loud once.

It is worth asking which of your own signals decay. A dashboard that is updated automatically will keep asserting itself whether or not anyone has confirmed it lately; a colleague's confident opinion, repeated often enough, hardens into a fact nobody rechecks. The ant's arrangement is the reverse: everything weakens unless a fresh journey supports it. Information that cannot decay is not more reliable. It is merely harder to correct.

Many companies get exactly one of these two things. Some have excellent information flow and no independent judgment — everyone repeats what they heard in the last meeting. Others have strong independent experts who never share anything, so the same problem gets solved five times in five departments. The design goal is not "more communication" or "more autonomy". It is a system where good information travels quickly and the person closest to the situation is still allowed to say: I've seen this myself, and it isn't so.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *When your own experience contradicts the company view, are you allowed to act on it?*

KEY TAKEAWAYS

1. Ants weight private and social information according to how certain each one is.
2. Information systems should let knowledge circulate and let individuals override it.
3. Free-flowing information without independent judgment produces confident, shared error.

SUMMARY

Ants combine what they know personally with what the colony's trails tell them, giving more weight to whichever source is more reliable. This makes the colony's information system both fast and correctable. Organizations need the same balance: information that moves easily, and people who are still permitted to trust what they have seen themselves.

SCIENCE CHECK

ESTABLISHED

Ants weight private memory against the collective trail according to how reliable each has recently been.

INFERRED

That organizations need information to circulate and individuals permitted to override it.

UNCERTAIN

Whether humans update these weights as accurately; we are known to over-trust both confident colleagues and confident dashboards.

VOCABULARY & PHRASAL VERBS

pheromone (n.) a chemical signal released by an animal to affect others of its species. The chemical works like a pheromone, telling others where to go.

certainty (n.) the state of being sure; the opposite of doubt. We acted with more certainty than the evidence justified.

to weight (v.) to give more or less importance to something. The model weights recent data more heavily than old data.

to override (v.) to overrule or ignore a decision or signal. The manager can override the system, but rarely does.

autonomy (n.) the freedom to act independently. Give people autonomy and the information to use it well.

out of date (phrase) no longer current or accurate.

The pricing sheet was six months out of date.

rely on (phrasal verb) to depend on. Do not rely on a report nobody has checked.

pass on (phrasal verb) to give information to someone else. Please pass on the customer's comments to the design team.

check out (phrasal verb) to examine something to see if it is true or acceptable. I checked out the claim myself before repeating it.

PRACTICE

Where the Analogy Breaks

Ants cannot lie, and pheromone trails carry no reputation. Human information channels carry motives as well as facts.

Use the Language

Make sentences using these three words: rely on, override and certainty.

Discuss

When your own experience contradicts an official report, which do you act on? Which does your employer expect?

Write

Write two short paragraphs: one describing a time the data was right and you were wrong, and one the reverse.

SOURCES

A "Information certainty determines social and private information use in ants", Scientific Reports (2017). https://www.nature.com/articles/srep43607

THE CONTRAST

Story 6: Ants: Weighing Two Sources of Truth vs. Story 2: Honeybees: Quorum, Not Consensus

Story 6: Ants: Weighing Two Sources of Truth: Individuals combine private, direct information with social, secondhand information, weighted by how reliable each currently is.

Story 2: Honeybees: Quorum, Not Consensus: Individuals form an assessment independently first, and only compare notes afterward through a structured quorum process.

Is the better safeguard against bad information weighting sources by reliability as you go, or keeping assessment strictly independent until a formal comparison point?

THIS CONNECTS TO

Story 2: Honeybees: Quorum, Not Consensus SAME MECHANISM
Both show a colony combining multiple independent signals rather than simply following the loudest or first one, to reach a more reliable collective judgment.

Story 18: Bees: Dissent That Improves the Choice LIMITATION
This chapter is about weighting two sources of information correctly. Dissent That Improves the Choice is about a different safeguard entirely — preserving competing options long enough to be properly tested — showing good group judgment needs more than one kind of protection.

Story 12: Elephants: What the Oldest Female Knows SAME MECHANISM
Both cases show a group's judgment improving when a source of information — a private observation, an elder's memory — is given weight in proportion to how reliable it has proven to be.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Foraging ants use two different kinds of information to find food: their own direct exploration, and pheromone trails left by other ants — and they don't treat these two sources equally.

What is the mechanism?

An ant weights social information (a pheromone trail) more heavily when its own private information is uncertain, and relies more on direct exploration when a trail seems unreliable or contradicts what it has found itself — a dynamic balancing of source credibility rather than a fixed rule.

What organizational problem does it explain?

When should an organization trust an individual's direct, firsthand observation over the secondhand, aggregated view already circulating — and how should that trust shift as conditions change?

Level 2 — Make It Your Story

> *Tell me about a time you had to weigh your own direct observation against what everyone else around you was reporting.*

Use STAR to answer. Think about: a customer conversation that contradicted a dashboard, a firsthand read that turned out more reliable than the consensus.

TAKE THE IDEA FURTHER

> *In your organization, when private, firsthand information conflicts with the circulating social consensus, which one usually wins by default — and is that actually the right default?*

7

Octopuses: Thinking at the Edges

Evidence Level 1

Biological claim: Strong scientific evidence

Business application: Analogy

An octopus has around 500 million neurons — comparable to a dog — but only about a third of them are in its brain. The rest are distributed through its eight arms. Each arm contains a substantial nervous system of its own and can carry out complex movements, explore a crevice and identify food without waiting for a central instruction.

The result is an animal whose intelligence is genuinely distributed. Their bodies have no skeleton and therefore no fixed shape, which means an octopus cannot rely on a stored motor program the way we can. It has to work each situation out afresh.

Two lines of evidence sit behind this. The first is anatomical and behavioural: octopuses in laboratories open containers, escape through gaps barely wider than their eye, and — in one well-documented case — carry coconut shell halves across open seabed to assemble as shelter later, which is difficult to explain as fixed instinct.

The second is more pointed. When researchers stimulated the nerve of an isolated arm, it produced the animal's characteristic reaching movement — the same stereotyped wave of stiffening that travels from base to tip during a normal reach. The instruction to reach may come from the brain, but the program that performs it is held in the arm itself. The centre says what; the periphery knows how.

Sensing appears to be distributed as well as movement. Octopus skin contains light-sensitive proteins of the kind normally found in eyes, and isolated pieces of skin have been shown to respond to light by expanding their pigment cells — without any signal from the brain, and in an animal that is, by the usual tests, colour-blind. Whatever the function turns out to be, the architecture is the same: capability pushed outward to the surface, where the information arrives.

The organizational parallel is direct. Many companies are built like an animal with a large brain and numb limbs: every decision, however small and however local, has to travel up to a central point and back again. That design has a cost that is easy to miss. It is not only slow — it is also less intelligent, because the person with the most relevant information about the problem is rarely the person authorized to decide.

There is a way to test this in your own organization, and it takes about twenty minutes. List the last ten decisions that reached you. For each one, write down where the information that actually determined the answer was sitting — in your head, or in somebody else's. Most managers who do this honestly find that they were a routing step rather than a decision-maker in at least half the cases, and that the delay they added was the only thing they contributed.

An octopus can only reach into eight crevices at the same time because each arm is allowed to solve its own problem. So can your organization.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which decisions travel upward in your organization that could be made where the problem is?*

KEY TAKEAWAYS

1. Intelligence can be distributed rather than concentrated in one central place.
2. Local units that can decide for themselves respond faster and often more accurately.
3. Routing every problem upward wastes the knowledge held closest to the problem.

SUMMARY

Most of an octopus's neurons are in its arms, which can explore and solve problems semi-independently of the central brain. This distributed design allows flexible, simultaneous problem-solving. Organizations that force every decision to travel upward lose both speed and the local knowledge that makes good decisions possible.

SCIENCE CHECK

ESTABLISHED

Most of an octopus's neurons lie in its arms, and an isolated arm can produce the reaching program.

INFERRED

That decision authority should sit where the relevant information is.

UNCERTAIN

Octopus arms share one set of interests. Delegated human units frequently do not, which is the whole difficulty of decentralization.

VOCABULARY & PHRASAL VERBS

neuron (n.) a nerve cell. Two thirds of the animal's neurons are in its arms.

distributed (adj.) spread across many places rather than concentrated in one. The company runs on a distributed model with no head office.

crevice (n.) a narrow crack or opening. The octopus pushed an arm into a crevice in the rock.

instinct (n.) behaviour an animal is born with, not learned. Her instinct was right, but she still tested the assumption.

numb (adj.) unable to feel anything. After the reorganization, the regional offices went numb.

carry out (phrasal verb) to perform or complete a task.

Each arm can carry out its own search.

work out (phrasal verb) to solve or find a solution. Let the person nearest the problem work out the answer.

reach into (phrasal verb) to put your hand inside something. She reached into the drawer without looking.

figure out (phrasal verb) to understand something after thinking about it. Nobody could figure out why the process took nine days.

PRACTICE

Where the Analogy Breaks

Octopus arms share one body and one interest. Delegated human units can pursue goals that conflict with the whole.

Use the Language

Make sentences using these three words: distributed, carry out and figure out.

Discuss

Which decision that reached you last week should have been made by the person who raised it?

Write

List five decisions you made this month. For each, write where the best information actually was.

SOURCES

B Godfrey-Smith, P., Other Minds: The Octopus and the Evolution of Intelligent Life, Farrar, Straus and Giroux (2016).

A Finn, J. K., Tregenza, T. & Norman, M. D., "Defensive tool use in a coconut-carrying octopus", Current Biology (2009).

THE CONTRAST

Story 7: Octopuses: Thinking at the Edges vs. Story 12: Elephants: What the Oldest Female Knows

Story 7: Octopuses: Thinking at the Edges: Intelligence is distributed outward, to the periphery, where the relevant information actually is.

Story 12: Elephants: What the Oldest Female Knows: Intelligence is concentrated in one individual who has accumulated decades of relevant information centrally.

Is organizational intelligence better served by pushing decision authority outward to where information is, or by concentrating deep experience in a few trusted individuals?

THIS CONNECTS TO

Story 1: Ants: Order Without a Manager SAME MECHANISM
Both show complex, adaptive behaviour arising without a central controller holding the full picture — capability is pushed outward to where the relevant information is.

Story 8: Slime Mould: Build Wide, Then Prune LIMITATION
Octopus arms act with local autonomy but still share one underlying interest, since they belong to the same animal. Slime mould's network-building shows a related but distinct mechanism — exploration and pruning over time — useful for a different kind of problem: designing efficient networks, not real-time distributed action.

Story 15: Termites: Building With No Blueprint SAME MECHANISM
Both show sophisticated, coordinated outcomes arising from many local units acting on local information, with no central blueprint or controller directing the result.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Most of an octopus's neurons are located not in its brain but distributed through its eight arms, each of which can explore, grasp and solve simple problems with a degree of independence from central control.

What is the mechanism?

Each arm has its own substantial nervous system capable of directing movement without waiting for instructions to travel to the brain and back;

the brain sets broad intent, but execution and local problem-solving happen at the periphery, where the relevant sensory information already is.

What organizational problem does it explain?

How much organizational decision-making currently has to travel to a central point and back, when the people closest to the problem already have what they need to solve it themselves?

Level 2 — Make It Your Story

> *Tell me about a time decision-making authority was pushed closer to where the information actually was, and it improved the outcome.*

Use STAR to answer. Think about: a decision you were allowed to make locally, a delegation that sped things up, a bottleneck removed by trusting the periphery.

TAKE THE IDEA FURTHER

> *List your last ten decisions that required approval from above you. For how many of them was the deciding information already sitting with you, rather than with the person who approved it?*

8

Slime Mould: Build Wide, Then Prune

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

Physarum polycephalum is a single-celled organism with no brain, no nervous system and no eyes. It is, by any reasonable definition, one of the least intelligent things alive. In 2010 Tero and colleagues used it to reproduce the Tokyo rail network.

The experiment was elegant. They placed oat flakes on a wet surface in the positions of the cities around Tokyo, added a slime mould in the middle, and waited. The organism first spread out in all directions, then progressively thinned out the connections that were not carrying much traffic and thickened the ones that were. Within about 26 hours it had produced a network of tubes whose efficiency, cost and fault-tolerance were comparable to the real rail system that Japanese engineers had spent decades developing. Light was used to simulate mountains, which the mould avoided — so it was solving the problem under real geographic constraints.

The Tokyo experiment was not the first hint. A decade earlier the same laboratory had grown Physarum through a maze with food at two ends. The organism first filled the maze completely, exploring every dead end, and then withdrew from the branches that led nowhere — leaving a single thick tube along the shortest connecting path. The withdrawal, not the exploration, was the intelligent part.

Notice how it worked. The slime mould did not calculate the optimal network in advance. It overbuilt, measured what was actually used, and then removed what wasn't. It found the good design by pruning, not by planning.

What the researchers took away was not the organism but its rule. They distilled the behaviour into a mathematical model that can be run on any network problem, which is the real test of a biological insight: it works when the biology is removed. The mould also repairs itself — sever a tube and flow reroutes, then the useful detours thicken again.

It is worth being honest about why pruning is hard for us and easy for a slime mould. The mould's abandoned tubes have no owner, no job title and no case to make. In an organization, every unused connection — the report nobody reads, the standing meeting, the approval step — belongs to somebody whose work it represents. The measurement is the easy half. Removing what the measurement found is a political act, which is why it helps enormously to agree on the decay rule before you know which connections it will catch.

The model that came out of the work has since been applied well beyond biology, to problems in network design where the objective is a good balance of cost, efficiency and tolerance of failure rather than the single shortest path. That is the mark of a real insight: it survives the loss of the organism that suggested it.

The practical form of this is a standing question rather than a project. Not "what should we cut this year?" — which invites defence — but "what did we build last year that nothing has flowed through since?" The first question asks people to justify their existence. The second asks the system what it actually used, and the answer arrives without anyone having to lose an argument.

Most organizations do the reverse. We design the process first, roll it out completely, and then defend it. The slime mould suggests a different

sequence: build more connections than you need, watch which ones carry real traffic, and be ruthless about letting the rest go.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What in your process is still there because nobody has measured whether it is used?*

KEY TAKEAWAYS

1. Efficient networks can emerge from reinforcement and pruning rather than from advance planning.
2. Constraints (mountains, budgets, regulations) shape good design instead of obstructing it.
3. Overbuild, measure real usage, then remove what is not used.

SUMMARY

A brainless slime mould reproduced the efficiency of the Tokyo rail network by growing connections everywhere and then withdrawing from the ones that carried little traffic. Its method was pruning rather than planning. Organizations can design better networks the same way: create options, measure what is genuinely used, and cut the rest.

SCIENCE CHECK

ESTABLISHED

Physarum reproduces efficient, fault-tolerant networks by overbuilding and then withdrawing from low-traffic routes.

INFERRED

That organizations should create options, measure real usage, and prune the rest.

UNCERTAIN

The organism's discarded tubes have no owner. Every unused process in a company belongs to somebody, which is why pruning stalls.

VOCABULARY & PHRASAL VERBS

slime mould (n.) a simple organism that grows as a spreading mass (US: mould). A slime mould has no brain and still solves mazes.

fault-tolerance (n.) the ability to keep working when part of a system fails. Fault-tolerance means the network survives losing a node.

to prune (v.) to cut away parts that are unnecessary. We pruned the reporting list from forty items to nine.

ruthless (adj.) determined and unsentimental, without pity. Be ruthless about meetings and generous about time.

constraint (n.) a limitation or restriction. The budget constraint improved the design.

spread out (phrasal verb) to expand over a wider area. The organism spread out in every direction first.

thin out (phrasal verb) to reduce in number or density. Traffic thinned out on the routes nobody used.

roll out (phrasal verb) to introduce a new product or process officially. They rolled out the system before testing it properly.

let go of (phrasal verb) to stop holding on to something. The hardest part is letting go of a project you started.

PRACTICE

Where the Analogy Breaks

A slime mould has no staff attached to the connections it prunes. Removing an unused human process removes someone's role.

Use the Language

Make sentences using these three words: prune, roll out and let go of.

Discuss

What in your work is measured but never used? Who would object if it were removed?

Write

Write 120 words describing something worth building in excess before you know which part will matter.

SOURCES

A Tero, A. et al., "Rules for biologically inspired adaptive network design", Science 327, 439–442 (2010).

C Biomimicry Toolbox, Case Studies. https://toolbox.biomimicry.org/references/case-studies/

THE CONTRAST

Story 8: Slime Mould: Build Wide, Then Prune vs. Story 1: Ants: Order Without a Manager

Story 8: Slime Mould: Build Wide, Then Prune: Efficiency comes from overbuilding first — exploring many connections — then pruning what doesn't get used.

Story 1: Ants: Order Without a Manager: Efficiency comes from reinforcing what works as you go, letting unused options fade rather than deliberately overbuilding first.

Is it more efficient to build wide and prune afterward, or to reinforce success incrementally from the start?

THIS CONNECTS TO

Story 1: Ants: Order Without a Manager SAME MECHANISM

Both use reinforcement of what works and quiet abandonment of what doesn't, rather than advance planning, to build an efficient network.

Story 7: Octopuses: Thinking at the Edges LIMITATION

Slime mould's network-building shows exploration and pruning over time, useful for designing efficient networks. Octopus arms show a related but distinct mechanism — local autonomy in real time — useful for a different kind of problem.

Story 25: Forests: Spare Parts That Differ LIMITATION

Slime mould prunes away genuine redundancy once it's confirmed unnecessary. Forests: Spare Parts That Differ makes the opposite case — some redundancy should be deliberately preserved, not pruned, because it protects against a future that hasn't happened yet.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A slime mould exploring a food source builds a dense, overlapping network of tubes connecting every option, then progressively narrows that network down to a small number of efficient, near-optimal routes.

What is the mechanism?

The organism initially explores broadly, running many simultaneous experiments at once; routes that carry more flow are reinforced and widened, while routes that carry little flow shrink and are eventually abandoned entirely — an efficient network emerges from starting wide and pruning, not from planning the optimal layout in advance.

What organizational problem does it explain?

Should an organization commit early to what looks like the most promising option, or deliberately run several options at once and let the weaker ones fade based on real performance?

Level 2 — Make It Your Story

> *Tell me about a time you deliberately kept several options or approaches running at once before narrowing down, rather than committing early.*

Use STAR to answer. Think about: parallel prototypes, multiple vendors trialled simultaneously, a decision you deliberately delayed narrowing.

TAKE THE IDEA FURTHER

Is your organization committing to a single approach earlier than the evidence justifies, when running a few options in parallel and pruning based on real performance might produce a better outcome?

9

Dolphins: Roles No One Plays Alone

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Off the coast of Florida, bottlenose dolphins use a hunting technique called mud-ring feeding. One dolphin swims rapidly in a tight circle around a school of fish, beating its tail to throw up a wall of mud from the seabed. The fish panic and leap out of the enclosed ring — straight into the open mouths of the other dolphins, who have positioned themselves outside it, waiting.

The technique only works with defined, different roles. The dolphin creating the ring cannot also catch the fish. The catchers cannot create the ring. The behaviour is learned rather than innate, is transmitted socially within particular groups, and does not appear in dolphin populations that have never been taught it. In Laguna, Brazil, dolphins have gone further still and cooperate with human fishermen, signalling when to cast the nets — a partnership passed down on both sides for generations.

The technique is not universal, which is the most informative thing about it. Mud-ring feeding is documented in particular populations in shallow Florida waters and not in dolphins elsewhere, and within those populations individuals repeat the same role across many hunts. It is a local tradition, learned and transmitted, rather than something dolphins do.

The Brazilian partnership shows the same pattern in an even stranger form. In Laguna, a subset of the local dolphins drives mullet towards fishermen standing chest-deep in murky water and produces a distinctive movement that tells them when to throw. Recent work using drones and underwater

microphones found that the two species time their actions to each other, and that the better the synchronization, the more both sides catch. Neither the fishers nor the dolphins can see the fish. Each is relying on the other's signal.

What makes this interesting for business is not the fact of cooperation, which is common in nature. It is the specificity. These dolphins are not simply helping each other in a general, friendly way. They have built a technique that no individual dolphin can perform, that produces food none of them could obtain alone, and that requires someone to accept the role that does not personally catch anything during the manoeuvre.

Techniques like these travel along particular social lines rather than through a population at large. In Shark Bay, Western Australia, some bottlenose dolphins carry a marine sponge on the beak as a tool while foraging over rough seabed — and genetic and behavioural work indicates the practice is passed from mother to daughter, concentrated within a single matrilineal line. What looks like a species-wide capability is often a family's.

The roles are also durable. Observations of these hunts report that the same individual tends to take the ring-making position across many occasions, while others consistently wait outside it — a division that persists rather than rotating. The likely reason is unglamorous and worth transferring: the driver has become good at driving. A team that reshuffles roles every quarter in the name of fairness may be quietly trading away exactly that accumulated competence.

That is the real test of a team. Not whether people get along, but whether the group can do something that none of its members could do separately — and whether people will take on the unglamorous role that makes it work.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which role on your team makes the result possible without receiving the credit?*

KEY TAKEAWAYS

1. Genuine teamwork produces capabilities that no individual member possesses.
2. Complementary roles matter more than general goodwill.
3. Someone has to accept the role that enables the result without receiving the credit.

SUMMARY

Bottlenose dolphins use mud-ring feeding, a learned technique in which one individual traps fish while others catch them — an outcome impossible for any dolphin alone. It depends on distinct, complementary roles rather than general cooperation. The competitive advantage of a real team is what its members can achieve together that none could achieve separately.

SCIENCE CHECK

ESTABLISHED

Mud-ring feeding and the Laguna fishing partnership are learned, locally transmitted techniques requiring distinct roles.

INFERRED

That real teamwork produces capability no member has alone, and requires someone to accept the enabling role.

UNCERTAIN

Dolphin roles are not negotiated or rewarded. Human role allocation involves status and pay, which changes who volunteers.

VOCABULARY & PHRASAL VERBS

innate (adj.) inborn; present from birth rather than learned. The behaviour is learned, not innate.

to transmit (v.) to pass something from one individual to another. The technique is transmitted from older animals to younger ones.

complementary (adj.) combining well because the parts are different. The two roles are complementary rather than interchangeable.

unglamorous (adj.) not exciting or impressive, though necessary. Somebody has to do the unglamorous work that makes it possible.

goodwill (n.) friendly, cooperative feeling towards others. Goodwill is not the same as a working process.

throw up (phrasal verb) here: to raise into the air suddenly. The tail throws up a wall of mud from the seabed.

get along (phrasal verb) to have a friendly relationship. They get along, but they have never worked as a team.

take on (phrasal verb) to accept a job or responsibility. She took on the role nobody wanted.

pass down (phrasal verb) to transmit knowledge to the next generation. The skill is passed down through generations.

PRACTICE

Where the Analogy Breaks

Dolphin roles are learned but not negotiated. Humans argue about who takes the unglamorous position, and about what it is worth.

Use the Language

Make sentences using these three words: complementary, take on and unglamorous.

Discuss

What can your team achieve together that none of you could achieve alone? Be specific.

Write

Describe a role in your team that enables the result without receiving credit, and how you would recognize it.

SOURCES

A Torres, L. G. & Read, A. J., "Where to catch a fish? The influence of foraging tactics on the ecology of bottlenose dolphins", Marine Mammal Science (2009).

A Cantor, M. et al., research on cooperative dolphin–human fishing in Laguna, Brazil, PNAS (2023).

THIS CONNECTS TO

Story 23: Penguins: Warmth by Taking Turns SAME MECHANISM

Both show cooperation producing an outcome — a capability, or survivable conditions — that no individual could achieve alone, through coordinated role-sharing rather than individual effort.

Story 11: Geese: Why the Lead Position Rotates SAME MECHANISM

Both describe genuine teamwork with differentiated roles, where individuals take on different parts of a shared task rather than each acting independently.

Story 41: Forests: Competition and Shelter LIMITATION

Dolphin cooperation is built on trust and coordinated roles among allies. Forests: Competition and Shelter shows that cooperation and competition can coexist within the very same system — a more complicated picture than pure teamwork.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Some dolphin populations hunt cooperatively using coordinated roles — certain individuals drive fish toward a barrier while others wait to catch them as they scatter — a division of labour documented in one dolphin population in Florida Bay.

What is the mechanism?

Individual dolphins consistently take on different functional roles within the same hunting group, and the roles depend on each other: the herders' technique only works because the catchers are positioned correctly, and vice versa, producing a capability neither role could achieve alone.

What organizational problem does it explain?

What capability does your team currently produce only because of genuine role differentiation and coordination — and would that capability survive if everyone were asked to do a bit of everything instead?

Level 2 — Make It Your Story

> *Tell me about a time a genuinely coordinated team produced a result that no individual, however skilled, could have produced alone.*

Use STAR to answer. Think about: a launch that required several distinct roles working in sync, a project where role clarity mattered more than individual talent.

TAKE THE IDEA FURTHER

> *Is there a capability your team currently has only because of how well-coordinated its distinct roles are — and do you protect that coordination, or treat each role as interchangeable?*

PART II

Leadership & Teams

How does nature organize groups?

10

Wolves: The Alpha That Never Existed

Evidence Level 1

Biological claim: Strong scientific evidence

Business application: Analogy

Almost everything the business world believes about wolves is wrong. The "alpha male" — the wolf who fights his way to the top and dominates the others — came from a 1940s study of unrelated captive wolves forced to live together in a zoo enclosure. It described animals under stress in an artificial situation.

When the biologist L. David Mech studied wild wolves on Ellesmere Island, he found something entirely different, and later publicly asked his own publisher to stop reprinting his earlier book. A wild wolf pack is a family. The "alpha pair" are simply the parents. They lead in the way parents lead — not by winning fights, but because they are older, more experienced and responsible for the young.

It is worth knowing where each version came from, because the difference is entirely methodological. The dominance model was built from a zoo population: unrelated adults, taken from different places, confined together with no possibility of leaving. Under those conditions they did fight for status, and the observations were accurate. The error was in the generalization. Mech's own work came from summers on Ellesmere Island, in the high Canadian Arctic, with a wild pack that gradually became tolerant of a researcher at close range — and there the same species behaved like a family, because that is what it was.

More interesting still: leadership within the pack shifts with the task. During a hunt, an experienced hunter may take the front. During travel, a different individual may set the pace — sometimes an older female, sometimes not the largest animal at all. Position at the front of a moving pack does not reliably indicate rank.

The submission that does occur in wild packs runs in an unexpected direction. Rather than being extracted by a dominant animal, greeting behaviour is typically initiated by the younger wolves — they approach the parents, muzzle-lick and crouch, in something closer to a food-begging display carried into adulthood than to a defeated animal yielding ground. The gesture is offered, not compelled. It is a small detail with an uncomfortable implication for the manager who reads deference in a meeting as evidence of authority rather than as something the room has decided to give.

There is a second feature of real packs that the corporate version never mentions. Young wolves leave. At one to three years old they disperse, sometimes travelling hundreds of kilometres, and either find a mate and found a pack of their own or fail in the attempt. The parents do not promote them and there is no contest to inherit. Succession happens by departure, and the pack a young wolf eventually leads is one it built rather than one it took. An organization in which ambitious people can only rise by displacing the person above them has reproduced the zoo, not the tundra — and should not be surprised by the behaviour that follows.

The business version of the alpha wolf — the leader who must dominate the room to hold authority — is therefore a myth built on captive animals in distress. What real wolves demonstrate is functional leadership: whoever has the relevant experience takes the lead for that situation, and gives it up again afterwards. The best leader for a crisis is often not the best leader for a negotiation, and pretending otherwise is how organizations end up with one exhausted person making every decision badly.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *For your current situation, who has the most relevant experience — and are they leading it?*

KEY TAKEAWAYS

1. The "alpha wolf" model comes from captive animals under stress and does not describe wild packs.
2. Real leadership in wolf packs is contextual: it moves according to the task at hand.
3. Dominance is not the same as leadership, and confusing the two damages teams.

SUMMARY

The famous alpha-wolf hierarchy was an artifact of studying unrelated wolves in captivity; wild packs are families led by parents. Leadership within them shifts depending on the situation and the relevant experience. For organizations, this replaces the idea of the dominant leader with functional leadership that changes according to context.

SCIENCE CHECK

ESTABLISHED

Wild wolf packs are family groups; leadership varies with the task, and dispersal rather than contest produces new packs.

INFERRED

That functional, task-based leadership is the right model for human teams.

UNCERTAIN

The dominance hierarchy was real in the captive study that produced it. The error was generalizing it to wild animals and then to workplaces.

VOCABULARY & PHRASAL VERBS

captive (adj.) kept in a cage or enclosure, not free. The study was done on captive animals, which changed everything.

pack (n.) a group of wolves or dogs living and hunting together. The pack travelled forty kilometres overnight.

rank (n.) a position in a hierarchy. Nobody could explain how rank was decided in the team.

dominance (n.) control or power over others. Dominance and leadership are not the same thing.

artifact (n.) here: a false result caused by the method of study.

The result was an artifact of how the survey was worded.

take the lead (phrase) to move into first position; to start directing. She took the lead when the client threatened to walk away.

fight one's way to the top (phrase) to succeed through aggressive competition. He fought his way to the top and arrived with no allies.

give up (phrasal verb) to surrender or hand over something. A good leader gives up control when someone else knows more.

end up with (phrasal verb) to finally have, often unintentionally. You end up with one exhausted person making every decision.

PRACTICE

Where the Analogy Breaks

Wolf packs are families with shared genetic interests. Companies are groups of strangers with separate interests and separate exits.

Use the Language

Make sentences using these three words: dominance, take the lead and give up.

Discuss

Who leads which situations in your team, and does the formal structure reflect it?

Write

Write 120 words on a situation where the best-placed person to lead was not the person who did.

SOURCES

A Mech, L. D., "Alpha status, dominance, and division of labour in wolf packs", Canadian Journal of Zoology 77, 1196–1203 (1999).

B Mech, L. D. & Boitani, L. (eds.), Wolves: Behaviour, Ecology, and Conservation, University of Chicago Press (2003).

THE CONTRAST

Story 10: Wolves: The Alpha That Never Existed vs. Story 11: Geese: Why the Lead Position Rotates

Story 10: Wolves: The Alpha That Never Existed: A popular leadership myth (the dominance-fighting alpha) turns out to be based

on captive animals under artificial stress, not how wild groups actually lead.

Story 11: Geese: Why the Lead Position Rotates: The real leadership mechanism in wild geese is need-based rotation, since the lead position is the most costly, not the most rewarded.

How much of what organizations believe about natural leadership hierarchies comes from studying animals under conditions that don't resemble their normal lives at all?

THIS CONNECTS TO

Story 12: Elephants: What the Oldest Female Knows LIMITATION
Wolves debunks a specific popularized leadership myth. Elephants offers the corrective it implicitly calls for: real natural leadership evidence, carefully documented, showing what actually predicts group-benefiting influence.

Story 11: Geese: Why the Lead Position Rotates SAME MECHANISM
Both correct a popular assumption about animal leadership by pointing to what the real mechanism costs and rewards, rather than what the folk version assumes.

Story 49: Predators: The Cost Is in the Search LIMITATION
Wolves corrects a myth about dominance and reward. Predators corrects a myth about effort and reward from a different angle — showing that the visible, dramatic moment (the chase, the fight for dominance) is usually not where the real cost sits.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

The popular "alpha wolf" concept — a wolf that fights its way to dominance and rules through aggression — does not describe how wild wolf packs actually function.

What is the mechanism?

The original alpha research was conducted on unrelated adult wolves forced into captivity together, a socially artificial and stressful situation that produced unusual dominance conflict; wild packs are typically family units, led by a breeding pair, with authority resembling parental structure rather than a hierarchy won through fighting.

What organizational problem does it explain?

How much of an organization's assumptions about strong, dominance-based leadership rest on evidence gathered under conditions that don't resemble the organization's actual, ordinary working environment?

Level 2 — Make It Your Story

> *Tell me about a time a widely accepted idea about leadership or hierarchy turned out, on closer inspection, not to hold up.*

Use STAR to answer. Think about: a leadership myth you were taught, a management practice you later learned was based on flawed evidence, a hierarchy you questioned.

TAKE THE IDEA FURTHER

> *What belief about leadership does your organization treat as settled fact, without anyone having checked recently where that belief actually came from?*

11

Geese: Why the Lead Position Rotates

Evidence Level 1

Biological claim: Strong scientific evidence

Business application: Analogy

The V-formation of migrating geese is one of the most-used images in management training, and unusually for such images, the underlying physics is real. A bird flying behind and slightly to the side of another can position itself in the upwash of air rolling off the leader's wingtip, gaining lift for free. Research measuring the heart rates and wingbeats of birds flying in formation confirmed that following birds do genuinely reduce their energy costs.

Measuring this required putting data loggers on birds. A study of northern bald ibises, hand-raised and following a microlight on a migration route, recorded position and wingbeat many times a second for every individual in the formation. The birds were not merely sitting behind a neighbour. They placed themselves to catch the rising air coming off the wingtip ahead, and — the detail that convinced people the effect was real — timed their own wingbeats so that the upstroke coincided with the moment the helpful air arrived. When a bird flew directly behind another, where the air is disturbed rather than rising, it did the opposite and flapped out of phase.

The bird at the front receives none of this benefit. It flies in undisturbed air and works hardest — which is exactly why the position rotates. When the leader tires, it drops back into the formation and another bird moves forward.

A second study followed the same birds through whole flights and compared how long each spent in front with how long it spent benefiting behind. The times matched closely, pair by pair: a bird that led for a stretch was repaid

with a comparable stretch of following. Nobody is keeping the ledger, and the honest position on the size of the energy saving is that estimates vary widely between species and studies. What is not in doubt is the shape of the arrangement — the cost is real, it falls on one position, and it circulates.

The organizational point is not the sentimental one about "supporting each other". It is a harder observation about sustainability. The lead position is not a reward. It is the position with the highest cost, and a group that never rotates it does not have a strong leader — it has a leader who is being consumed, and a team that is quietly losing the capability to fly at the front at all.

Rotating the front is harder in an organization than in a flock for one reason the birds do not face. In a V-formation the lead position carries cost and nothing else. In a company it carries cost and status and pay, so rotating it means rotating the rewards as well — and a rotation that moves the work without moving the recognition is not a rotation. It is a request for volunteers to be tired on someone else's behalf.

Ask a practical question of your own organization: if the person at the front dropped back tomorrow, is there anyone who has practised being there?

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *If the person at the front stopped tomorrow, who has practised being there?*

KEY TAKEAWAYS

1. The lead position carries the highest cost and no aerodynamic benefit.

2. Rotating leadership is what makes long-distance effort sustainable.

3. A team that never rotates loses the ability to lead, not just the willingness.

SUMMARY

Geese fly in a V because trailing birds gain lift from the leader's wake, while the leader gains nothing and works hardest — so the position is rotated. Studies of formation flight confirm both the energy saving and the reciprocal turn-taking. Organizations should likewise avoid making one person carry the front indefinitely.

SCIENCE CHECK

ESTABLISHED

Birds in formation position themselves to exploit upwash, time their wingbeats to it, and reciprocate time spent leading.

INFERRED

That leadership should rotate so that the cost of the front position is shared.

UNCERTAIN

The size of the energy saving varies widely between species and studies; the direction of the effect is not disputed, the magnitude is.

VOCABULARY & PHRASAL VERBS

formation (n.) an organized arrangement of moving things. The birds held formation for six hours without a break.

upwash / lift (n.) upward-moving air; the upward force that keeps a bird or plane flying. The follower catches the upwash and gains lift for free.

to rotate (v.) to take turns in a position or role. We rotate the on-call duty every two weeks.

reciprocal (adj.) given and received in return. The arrangement is reciprocal: each side repays the other.

sustainable (adj.) able to be continued over a long period. The pace is impressive but it is not sustainable.

drop back (phrasal verb) to move into a position further behind. He dropped back and let a colleague present the results.

take turns (phrase) to do something one after another. They take turns chairing the weekly review.

burn out (phrasal verb) to become exhausted from long-term overwork. Two of our best people burned out last winter.

fall behind (phrasal verb) to fail to keep up with others. A bird that falls behind loses the benefit entirely.

PRACTICE

Where the Analogy Breaks

Geese rotate a physical cost. Human leadership carries status and pay, so rotating it means rotating rewards, not just effort.

Use the Language

Make sentences using these three words: burn out, take turns and reciprocal.

Discuss

If the person carrying the most in your team stepped back for a month, who could take the front?

Write

Design a rotation for one demanding responsibility, and state what would have to move with it besides the work.

SOURCES

A Portugal, S. J. et al., "Upwash exploitation and downwash avoidance by flap phasing in ibis formation flight", Nature 505, 399–402 (2014).

A Voelkl, B. et al., "Matching times of leading and following suggest cooperation through direct reciprocity during V-formation flight in ibis", PNAS (2015).

THIS CONNECTS TO

Story 10: Wolves: The Alpha That Never Existed CONTRAST
See The Contrast in that chapter. In short: both correct a popular assumption about animal leadership, pointing to what the real mechanism actually costs and rewards rather than what the folk version assumes.

Story 9: Dolphins: Roles No One Plays Alone SAME MECHANISM
Both describe genuine teamwork with differentiated roles, where individuals take on different parts of a shared task rather than each acting independently.

Story 23: Penguins: Warmth by Taking Turns SAME MECHANISM
Both describe a costly position or condition being shared through rotation, spreading a burden that would exhaust or disadvantage any single individual permanently occupying it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

In a flying V formation, the lead goose position is rotated frequently among different birds, rather than being permanently held by one dominant individual.

What is the mechanism?

The lead position in a V formation gets no aerodynamic drafting benefit from birds ahead — it does all the work of breaking the air that following birds then draft off — so leading is the most physically costly position in the formation, and geese rotate through it based on who has energy to spare, not based on status.

What organizational problem does it explain?

Is your organization's leadership position structured as a reward for status, or as a genuinely costly role that should rotate based on who currently has the capacity to carry it?

Level 2 — Make It Your Story

> *Tell me about a time your team rotated a demanding role or responsibility based on capacity, rather than leaving it permanently with one person.*

Use STAR to answer. Think about: an on-call rotation, a project lead that changed hands, a costly task shared deliberately across a team.

TAKE THE IDEA FURTHER

> *What position in your organization is treated as a permanent reward for status, when it's actually the most costly role in the group and might function better as a rotation?*

12

Elephants: What the Oldest Female Knows

Evidence Level 1

Biological claim: Strong scientific evidence

Business application: Analogy

An elephant family is led by its oldest female, the matriarch, and researchers have been able to measure what her age is actually worth to the group.

In Amboseli National Park, researchers played recorded elephant calls to different families and watched how they reacted. Families with older matriarchs were significantly better at distinguishing the calls of strangers from those of familiar animals, and responded defensively when they should. Families with younger matriarchs were more likely to get it wrong — either bunching in alarm unnecessarily, or failing to react when a genuine stranger approached. A later study found that older matriarchs also responded more appropriately to the roars of male lions, which are far more dangerous to elephants than female lions.

The method is worth describing, because it turns a vague claim about wisdom into a measurement. Researchers concealed a loudspeaker in a vehicle and played recorded contact calls to a family — sometimes from a female the group knew well, sometimes from a stranger. Then they scored what the family did: whether they bunched defensively, whether they lifted their trunks to smell in the direction of the sound, how long they stayed alert. Families with older matriarchs bunched when a stranger called and ignored the familiar voices. Families led by younger females did both at the wrong times.

The knowledge is not written down anywhere. It exists only inside the head of an individual who has lived through enough events to recognize the patterns — including, in droughts, the location of water sources visited decades earlier. When a matriarch is killed, that entire library disappears in a moment, and the family's decision quality measurably declines.

There is a bleak corollary that makes the point sharper. Ivory poaching removes the oldest animals first, because they carry the largest tusks. The pressure therefore falls precisely on the individuals holding the accumulated knowledge, and heavily poached populations can end up containing plenty of elephants and very little experience — a group that looks healthy on a headcount and makes consistently worse decisions.

The knowledge is also spatial, and one drought made that measurable. During the severe East African drought of 1993, elephant families in Tarangire responded differently from one another: some left the park, some stayed. Analysis of calf mortality afterwards found that the groups that left lost fewer calves — and that those groups tended to contain females old enough to remember the catastrophic drought of the late 1950s, thirty-five years earlier. The decisive knowledge was not general competence. It was one specific event, remembered by one animal, and unavailable to any family whose matriarch was too young to have lived through it.

Organizations should read this carefully, because they routinely make the opposite calculation. Experienced staff are expensive, and on a spreadsheet an expensive employee looks like a cost. What the spreadsheet cannot show is the pattern recognition that leaves with them: which client behaves this way before cancelling, which supplier's excuse means real trouble, what happened the last time the company tried this. That knowledge was never in the documentation. It was in a person.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What does your longest-serving colleague know that exists nowhere in writing?*

KEY TAKEAWAYS

1. Older elephants hold social and ecological knowledge that measurably improves group decisions.
2. Critical experience is often undocumented and exists only in individuals.
3. Losing experienced people removes capability that no handover document captures.

SUMMARY

Research in Amboseli showed that elephant families with older matriarchs judged threats more accurately than families led by younger females. The advantage came from accumulated experience that exists nowhere except in the individual. Experienced employees are similarly repositories of institutional knowledge, not simply expensive headcount.

SCIENCE CHECK

ESTABLISHED

Families with older matriarchs discriminate threats more accurately, and drought survival tracked the age of the leading female.

INFERRED

That experienced employees hold institutional knowledge that no handover captures.

UNCERTAIN

How far a human organization can substitute documentation for experience — elephants have no such option, and companies rarely test whether they do.

VOCABULARY & PHRASAL VERBS

matriarch (n.) the female leader of a family or group. The matriarch decided when the family would move.

to distinguish (v.) to recognize the difference between two things. Older animals distinguish strangers from familiar neighbours.

pattern recognition (n.) the ability to notice a familiar situation repeating. Experience is largely pattern recognition you cannot explain.

repository (n.) a place where something is stored. She is a repository of everything the company learned the hard way.

handover (n.) the process of transferring a job to someone else. The handover document covered the process and none of the judgment.

headcount (n.) the number of employees in an organization. Headcount tells you how many people, not how much experience.

live through (phrasal verb) to experience something difficult and survive it. He lived through the last recession and recognized the signs.

write down (phrasal verb) to record something in writing. Nobody had written down why the rule existed.

pass on (phrasal verb) to give knowledge to someone else. She passed on what she knew before she retired.

look like (phrasal verb) to appear to be. On a spreadsheet, experience looks like cost.

PRACTICE

Where the Analogy Breaks

Elephant knowledge cannot be written down at all. Human institutional knowledge can be captured — organizations simply rarely try until it leaves.

Use the Language

Make sentences using these three words: repository, pass on and write down.

Discuss

What does your most experienced colleague know that exists nowhere in writing?

Write

Interview a long-serving colleague for ten minutes and write 150 words of what they know that no document holds.

SOURCES

A McComb, K. et al., "Matriarchs as repositories of social knowledge in African elephants", Science 292, 491–494 (2001).

A McComb, K. et al., "Leadership in elephants: the adaptive value of age", Proceedings of the Royal Society B (2011).

THIS CONNECTS TO

Story 10: Wolves: The Alpha That Never Existed LIMITATION

Wolves debunks a specific popularized leadership myth. Elephants offers the

corrective it implicitly calls for: real natural leadership evidence, carefully documented, showing what actually predicts group-benefiting influence.

Story 7: Octopuses: Thinking at the Edges CONTRAST
See The Contrast in that chapter. In short: octopus intelligence is pushed outward to the periphery, where information already is; elephant intelligence is concentrated in one individual who has accumulated decades of experience centrally.

Story 45: Scrub-Jays: Planning for a Future Need LIMITATION
Elephant matriarchs draw on accumulated past experience to guide present decisions. Scrub-jays draw on a different cognitive capacity entirely — anticipating a future need not currently being experienced — showing memory and foresight are separate mechanisms, not the same one.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Elephant herds led by older matriarchs make measurably better group decisions during rare, high-stakes events, such as correctly identifying and responding to unfamiliar predator threats or locating water during severe drought.

What is the mechanism?

Matriarchs accumulate decades of ecological and social memory — the location of distant water sources during past droughts, the specific danger posed by different predator species — and the herd's survival during rare, high-stakes events correlates with the matriarch's age and accumulated experience, not just her formal position.

What organizational problem does it explain?

Does your organization have a mechanism for identifying and deferring to accumulated experience during genuinely rare, high-stakes situations, or does authority default to current position regardless of who has actually seen a comparable situation before?

Level 2 — Make It Your Story

> *Tell me about a time someone's long accumulated experience, rather than their current title, was what got your team through a rare or unfamiliar situation.*

Use STAR to answer. Think about: a crisis where a veteran's memory mattered, a decision you deferred on because someone had seen it before.

TAKE THE IDEA FURTHER

> *Who in your organization holds the kind of long, specific memory that would matter most in a rare crisis — and would your organization's structure actually surface their input in time?*

13

Bees: Specialists Who Can Switch Back

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A worker honeybee does not choose a career. She moves through several of them, roughly in order of age. In her first days she cleans cells. Then she becomes a nurse, feeding larvae. Later she receives and stores nectar, builds comb, guards the entrance, and only in the final phase of her life does she leave the hive to forage. This progression is called temporal polyethism, and it is driven by hormones and by contact with other bees rather than by any allocation system.

The progression is not a schedule so much as a chemical conversation. A bee's rate of development is governed by hormones, and those hormones respond to contact with other bees: returning foragers carry a compound on their bodies that, when it reaches young workers in the hive, slows their maturation. A colony rich in foragers is therefore continuously telling its nurses to stay nurses. When foraging bees stop coming home, the signal weakens and the young ones develop faster.

The elegant part is what happens when the pattern is disrupted. If a hive loses its foragers, some nurse bees will accelerate their development and start foraging early. If it loses its nurses, some foragers can even revert. The specialization is real — a foraging bee is genuinely better at foraging — but the boundaries stay permeable, and the colony rebalances itself according to what is actually missing.

The flexibility has a price worth knowing about. Bees that take up foraging early, in response to a shortage, do not perform as well as bees that arrive at the role at the usual age — they complete fewer successful trips and their lives are shorter. The colony survives the emergency and pays for it afterwards. Anyone who has watched a capable junior colleague pushed into a senior role three years early, and watched what happened to them over the following year, will recognize the transaction exactly.

Companies tend to get one half of this. Either everyone is a generalist, in which case nothing is done to a high standard, or specialists are so rigidly separated that a gap in one function stops the whole operation while a hiring process runs for three months.

In practice that means two unglamorous things. Someone has to be able to do the neighbouring job well enough to hold it for a month, which requires deliberate rotation before there is an emergency rather than after one. And the shortage has to become visible to the people who could respond — a queue that is growing, a customer waiting, a signal that reaches the floor rather than a manager's weekly report. Bees do not redistribute effort because a supervisor noticed. They redistribute because the absence itself changed what each individual could detect.

The bee model suggests a third option: build genuine depth of skill, but design roles so that people can move across the boundary when the colony's needs change — and make sure the signals about what is missing actually reach the people who could respond.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where would your organization stop if one specialist were unavailable for a month?*

KEY TAKEAWAYS

1. Complex organizations need genuinely differentiated roles, not universal generalists.
2. Specialization should remain flexible enough to rebalance when needs change.
3. The system must signal what is missing so people can shift towards it.

SUMMARY

Honeybees specialize by age, moving through cleaning, nursing, storing and finally foraging, but they can accelerate or reverse this progression when the colony's needs change. Specialization is real yet permeable. Effective organizations combine complementary depth of skill with the flexibility to redistribute people when gaps appear.

SCIENCE CHECK

ESTABLISHED

Honeybee roles progress with age under hormonal control, respond to colony needs, and precocious foragers perform worse.

INFERRED

That organizations need deep specialization with permeable boundaries.

UNCERTAIN

Whether the human cost of early promotion follows the same curve; the parallel is suggestive, not measured.

VOCABULARY & PHRASAL VERBS

larva / larvae (n.) the young form of an insect before it becomes an adult. The nurse bees feed the larvae for several days.

to forage (v.) to search for food. Older bees leave the hive to forage.

comb (n.) the wax structure of cells bees build inside a hive. The workers build fresh comb along the top of the frame.

permeable (adj.) allowing things to pass through. The boundaries between the roles are permeable.

generalist / specialist (n.) someone with broad skills / deep skills in one area. A generalist covers gaps; a specialist goes deeper.

to revert (v.) to return to a previous state. Some foragers can revert to nursing when the colony needs it.

fill in for (phrasal verb) to temporarily do someone else's job. Can you fill in for her while she is on leave?

step up (phrasal verb) to take on more responsibility when needed. Two juniors stepped up when the lead developer left.

fall back on (phrasal verb) to use as an alternative when the first option fails. We had no one to fall back on when he resigned.

PRACTICE

Where the Analogy Breaks

Bees change role through hormones, not through applications, interviews and pay bands. Human role change is slow because it is contractual.

Use the Language

Make sentences using these three words: step up, fill in for and specialist.

Discuss

Which single specialist would stop your work if they were away for a month?

Write

Write a short plan for cross-training one critical role, including what would have to be given up to do it.

SOURCES

B Robinson, G. E., "Regulation of division of labour in insect societies", Annual Review of Entomology 37, 637–665 (1992).

A Huang, Z.-Y. & Robinson, G. E., "Regulation of honey bee division of labour by colony age demography", Behavioural Ecology and Sociobiology (1996).

THE CONTRAST

Story 13: Bees: Specialists Who Can Switch Back vs. Story 14: Cockroaches: A Mixture of Temperaments

Story 13: Bees: Specialists Who Can Switch Back: Effective organizations need genuinely differentiated roles, with individuals specializing rather than everyone doing a little of everything.

Story 14: Cockroaches: A Mixture of Temperaments: Effective groups need a genuine mixture of behavioural types, since uniform individuals decide worse collectively than a varied group.

Does organizational strength come more from role specialization, or from behavioural diversity within a group facing the same role?

THIS CONNECTS TO

Story 17: Ants: Thresholds, Not Assignments SAME MECHANISM
Both describe flexible task allocation within a colony, where individuals shift what they're doing based on need rather than sticking to a permanently fixed assignment.

Story 9: Dolphins: Roles No One Plays Alone SAME MECHANISM
Both show that genuine role specialization, rather than generalized effort from everyone, is what produces a complex organization's full range of capability.

Story 14: Cockroaches: A Mixture of Temperaments CONTRAST
See The Contrast in that chapter. In short: bees show the value of differentiated roles; cockroaches show the value of differentiated temperaments within the same role — two distinct, complementary kinds of diversity.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Worker bees specialize in different tasks — nursing, foraging, guarding, building — over the course of their lives, but individual bees can switch back to an earlier role if the colony's needs suddenly change.

What is the mechanism?

Age-based task specialization is the default, driven by physiological changes as a bee matures, but the assignment is flexible rather than fixed: if the colony loses too many foragers, younger bees that would normally still be nursing

can accelerate their development and switch to foraging early, restoring balance to the colony's overall capability.

What organizational problem does it explain?

Can your organization's specialists flex back into a different, previously-held role when a sudden shortage demands it, or does deep specialization make people brittle and unable to cover for each other?

Level 2 — Make It Your Story

> *Tell me about a time you or a colleague had to switch back into an earlier or different role because of a sudden gap in the team.*

Use STAR to answer. Think about: covering for an absent colleague, a specialist who stepped back into a generalist role temporarily, a skill you hadn't used in years that suddenly mattered again.

TAKE THE IDEA FURTHER

> *If your organization suddenly lost a third of the people in one specialized role, is there anyone positioned to flex back into it the way a bee colony's workers can?*

14

Cockroaches: A Mixture of Temperaments

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Cockroaches have personalities. This sounds like a joke, but it is a measurable finding: when individual cockroaches are tested repeatedly, some are consistently bolder — quicker to leave shelter and explore an open space — while others are consistently more cautious. The traits are stable over time, which is the technical definition of an animal personality.

Calling it a personality is a technical claim rather than a charming one, and it has to be earned. The test is repeatability: an individual is placed in the same situation many times, over days or weeks, and the question is whether its behaviour is consistent relative to others. A cockroach that leaves shelter quickly today must still be among the quicker ones two weeks later. When that holds — and for boldness in this species it does — the differences cannot be dismissed as random variation in the moment.

Planas-Sitjà's team, working with cockroaches, found that this variation is not noise in the system. Groups containing a mixture of bold and shy individuals reached collective decisions — such as which shelter to occupy — differently from groups of uniform temperament. Bold individuals explore and generate options; cautious individuals wait, and their reluctance slows the group's commitment until enough information has accumulated. Similar effects appear across many species: behavioural diversity within a group changes both the speed and the quality of collective choices.

The effect is not confined to insects. Groups of fish, birds and mammals containing a mixture of bold and cautious individuals have repeatedly been found to behave differently from uniform groups — not simply as the average of their members, but with different dynamics. They explore more of the available options, they commit more slowly, and they are less likely to follow a single confident individual into a bad outcome. The variation is not something the group puts up with. It is part of how the group computes.

The temptation in hiring is to recruit for a single ideal profile — usually a version of the people already succeeding at the company. It feels efficient. What it actually does is remove the tension that makes group decisions work. A team of bold people commits fast and often commits wrongly. A team of cautious people analyzes a problem until the opportunity has gone.

There is a practical way to check this without psychometrics. Look at the last five significant decisions your team made and ask who argued against each one, by name. If the same person always does it, they are functioning as the team's entire supply of caution and will eventually be labelled negative and managed out. If nobody did, the group has no variation left to compute with, and its confidence is a product of its composition rather than of the evidence.

You do not need everyone to be balanced. You need the group to be balanced — which means tolerating the colleague whose instinct is the opposite of yours, and recognizing that their reluctance is doing a job.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Whose instinct on your team is the opposite of yours — and do you listen to it?*

KEY TAKEAWAYS

1. Individual animals have stable behavioural traits, and this variation affects group outcomes.
2. Mixed groups make different — often better-calibrated — collective decisions than uniform ones.
3. Hiring for a single ideal profile removes the productive tension a team needs.

SUMMARY

Cockroaches show consistent individual differences in boldness, and groups containing a mixture of temperaments make collective decisions differently from uniform groups. Behavioural diversity functions as part of the decision mechanism rather than as noise. A good team is therefore not a collection of identical personalities.

SCIENCE CHECK

ESTABLISHED

Individual cockroaches show repeatable differences in boldness, and mixed groups decide differently from uniform ones.

INFERRED

That teams should be composed for behavioural variation rather than a single ideal profile.

UNCERTAIN

Human 'boldness' is performed as well as felt — people display confidence strategically, which insects do not.

VOCABULARY & PHRASAL VERBS

bold / cautious (adj.) willing to take risks / careful to avoid risk. The bold ones explore; the cautious ones slow the group down usefully.

trait (n.) a characteristic feature of a person or animal. Boldness is a trait, not a mood.

temperament (n.) a person's or animal's typical emotional character. The team has one temperament and no internal argument.

uniform (adj.) all the same. A uniform group commits quickly and often commits wrongly.

reluctance (n.) unwillingness to do something. Her reluctance was doing a job nobody had assigned her.

well-calibrated (adj.) correctly adjusted; neither too much nor too little. A well-calibrated team is neither reckless nor paralyzed.

hold back (phrasal verb) to hesitate; to prevent something from advancing. Somebody has to hold back when everyone else is enthusiastic.

rush into (phrasal verb) to do something too quickly without thinking. We rushed into the contract without reading the terms.

weigh up (phrasal verb) to consider the advantages and disadvantages. Weigh up the risk before you commit the budget.

PRACTICE

Where the Analogy Breaks

Cockroach boldness is fixed and unstrategic. Humans perform caution or confidence for social reasons, which distorts the signal.

Use the Language

Make sentences using these three words: trait, hold back and rush into.

Discuss

Who in your last five decisions argued against the majority? Was it always the same person?

Write

Describe a colleague whose instincts are opposite to yours, and one occasion when they were right.

SOURCES

A Planas-Sitjà, I. et al., "Group personality during collective decision-making: a multi-level approach", Proceedings of the Royal Society B 282 (2015).

A Jolles, J. W. et al., "Consistent individual differences drive collective behaviour and group functioning", Current Biology (2017).

THIS CONNECTS TO

Story 13: Bees: Specialists Who Can Switch Back CONTRAST

See The Contrast in that chapter. In short: bees show the value of differentiated roles; cockroaches show the value of differentiated temperaments within the same role — two distinct, complementary kinds of diversity.

Story 25: Forests: Spare Parts That Differ SAME MECHANISM

Both cases show that a mixture of different individual types — temperaments, or species performing the same function — outperforms a uniform group when conditions are unpredictable.

Story 2: Honeybees: Quorum, Not Consensus LIMITATION

Cockroach group decisions benefit from a mix of bold and cautious individuals deciding together. Honeybee quorum sensing achieves reliability

through a different route — independent assessment by specialized scout individuals, not a mixture of general temperaments.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Individual cockroaches show consistent, measurable differences in boldness and caution — some individuals reliably explore new areas faster, others hang back — and groups made up of a mixture of these temperaments make different collective decisions than uniform groups do.

What is the mechanism?

Research shows cockroach groups with a mix of bold and cautious individuals reach decisions — such as choosing a shelter — differently, and often more adaptively, than groups made up of only one temperament type, because bold individuals generate exploration while cautious individuals prevent premature or risky commitment.

What organizational problem does it explain?

Does your team's composition include a genuine mixture of risk temperaments, or has hiring or culture quietly selected for one type, leaving the group with a systematic blind spot?

Level 2 — Make It Your Story

> *Tell me about a time your team's mixture of cautious and bold personalities produced a better outcome than a more uniform group would have.*

Use STAR to answer. Think about: a decision that benefited from both a risk-taker and a skeptic, a team you noticed was missing one temperament type.

TAKE THE IDEA FURTHER

> *Does your team currently have a genuine mixture of bold and cautious temperaments, or has it quietly become uniform — and if so, in which direction?*

15

Termites: Building With No Blueprint

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

A termite mound can stand five metres high and take years to build. No termite has seen the finished structure, no termite has a plan of it, and the queen — despite the name — issues no instructions. The building is coordinated entirely through the material itself.

The mechanism is called stigmergy: an individual's action changes the environment in a way that triggers the next action. A termite deposits a pellet of soil mixed with saliva. That pellet is thought to carry a chemical trace (a 'cement pheromone', whose role is still debated) that makes other termites more likely to deposit nearby. Small piles become pillars; pillars attract deposits on their sides; opposing pillars grow towards each other and meet as arches. Nobody decided to build an arch. The arch is what happens when that simple rule runs for long enough in that particular geometry.

Engineers have taken this seriously. A Harvard team built TERMES, a group of small autonomous robots that construct structures using exactly this principle: each robot follows local rules and reads the state of the building instead of receiving instructions from a controller, so the swarm keeps working even if individual robots fail. Additive manufacturing researchers have used termite-inspired multi-agent methods for generative design.

The mechanism has a condition attached. The chemical trace in each pellet decays, so if too few termites are working, the signal fades before the next builder arrives and no structure forms at all. Stigmergy needs a critical density

of activity and a trace that outlives the gap between contributors. Shared work states in organizations fail in exactly this way: a board nobody updates for two weeks stops coordinating anything.

The robot version makes the requirement explicit. Each TERMES unit carries no map and receives no instructions; it climbs the structure, reads the local configuration of blocks beneath it, and places a brick where the rules permit. Because every robot reads the building rather than a plan, the swarm keeps working when individual robots fail, and finishes the same structure with three units or thirty. The engineering cost of that robustness is that the rules must be worked out in advance with real care — which is also the cost in an organization, and the part usually skipped.

What termites cannot do is hide the work. Every deposit is in the structure, visible to whoever arrives next.

The management insight is precise, and it is not "let people do whatever they want". Termites have very strict rules. What they lack is a manager. Give people clear rules, make the current state of the work visible to everyone, and let the work itself tell each person what to do next.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Can everyone see the current state of the work without being told?*

KEY TAKEAWAYS

1. Stigmergy means coordinating through the shared work rather than through instructions.

2. Sophisticated structures emerge from strict simple rules plus a visible shared state.
3. Systems built this way keep working when individual members fail or leave.

SUMMARY

Termites build enormously complex mounds with no plan and no supervision, because each deposit of soil changes the environment in a way that guides the next termite — a process called stigmergy. Robotics and manufacturing researchers have deliberately copied it. Organizations get the same effect by combining clear rules with a visible shared state of the work.

SCIENCE CHECK

ESTABLISHED

Termite construction proceeds by stigmergy, requires a critical density of builders, and has been reproduced in robot swarms.

INFERRED

That clear local rules plus a visible shared work state can replace supervision.

UNCERTAIN

Termites cannot misrepresent the structure. Shared human work states are interpreted differently by different readers, which is where the analogy strains.

VOCABULARY & PHRASAL VERBS

mound (n.) a raised mass of earth. The mound had stood for years before anyone measured it.

stigmergy (n.) coordination through traces left in a shared environment. Stigmergy means coordinating through the work itself.

pellet (n.) a small hard ball of material. Each termite adds a pellet of soil and saliva.

pillar / arch (n.) a vertical support / a curved structure between two supports. The pillars grow towards each other and meet as an arch.

autonomous (adj.) able to operate independently.

Each robot is autonomous and receives no instructions.

to trigger (v.) to cause something to start. One deposit triggers the next.

build up (phrasal verb) to accumulate gradually. The structure builds up without anyone directing it.

take on (phrasal verb) here: to begin to have a particular quality or form. The pile takes on a shape nobody designed.

keep going (phrasal verb) to continue despite difficulty. The swarm keeps going even when individual robots fail.

PRACTICE

Where the Analogy Breaks

Termites cannot misread the mound. Shared human work states are interpreted differently by different readers, which is where coordination fails.

Use the Language

Make sentences using these three words: trigger, autonomous and build up.

Discuss

Can everyone on your team see the current state of the work without asking anyone?

Write

Write 120 words describing how work is handed on in your team, using the idea of stigmergy.

SOURCES

A Werfel, J., Petersen, K. & Nagpal, R., "Designing collective behaviour in a termite-inspired robot construction team", Science 343, 754–758 (2014).

A "A generative multi-agent design methodology for additively manufactured parts inspired by termite nest building", ScienceDirect. http s://www.sciencedirect.com/science/article/abs/pii/S0007850617300392

THIS CONNECTS TO

Story 1: Ants: Order Without a Manager SAME MECHANISM

Both show sophisticated, coordinated outcomes arising from many local units acting on local information, with no central blueprint or controller directing the result.

Story 7: Octopuses: Thinking at the Edges SAME MECHANISM

Both show sophisticated, coordinated outcomes arising from many local units acting on local information, with no central plan or controller directing the result.

Story 37: Termite Mounds: Walls That Breathe SAME MECHANISM

Both chapters look at termite-built structures — this one at how they're built with no blueprint, the other at what the finished structure does once built — two halves of the same mechanism, construction and function.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Termites build enormous, structurally complex mounds without any individual termite holding a plan of the finished structure, and without any central coordinator directing construction.

What is the mechanism?

Termites coordinate through stigmergy: each termite responds to the current state of the physical structure itself — depositing material where previous deposits or pheromone cues indicate building is needed — rather than to instructions from other termites, so the work in progress becomes the coordination mechanism.

What organizational problem does it explain?

Could your organization coordinate complex work by making the current state of the work itself visible and actionable to everyone, rather than routing every next step through direct instruction?

Level 2 — Make It Your Story

> *Tell me about a time a project stayed coordinated because the shared state of the work was visible to everyone, rather than because people were explicitly told what to do next.*

Use STAR to answer. Think about: a shared document, a visible task board, a project that self-organized once progress was made transparent.

TAKE THE IDEA FURTHER

> *Is there a complex, ongoing piece of work in your organization currently coordinated through direct instruction that might coordinate itself better if the work's current state were simply made visible to everyone involved?*

16

Birds: The Nest Fits the Conditions

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

There is no single bird architecture. A sociable weaver in the Kalahari builds a permanent apartment block of dry grass that houses a hundred pairs and lasts for a century, with thick insulation that keeps interior chambers cool in the day and warm at night. A tailorbird sews leaves together with plant fibre. A swift builds a cup from its own hardened saliva. An emperor penguin builds nothing at all and carries the egg on its feet, because in Antarctica there is nothing to build with.

Nor is the skill purely inherited. When researchers filmed individual southern masked weavers building nest after nest, they found that birds differed from one another in technique and were consistent in their own — some worked predominantly from the left, others from the right — and that individuals improved with practice, dropping fewer pieces of grass as they went. The design is in the species. The competence is in the bird, and it is acquired.

Comparative work on animal nest construction stresses exactly this point: builders adapt their process to locally available materials and to local environmental conditions, and the "right" structure depends on the climate, the predators and the raw materials to hand. There is no universally optimal nest. A weaver's insulated fortress would be a fatal design in a wet tropical forest; a swift's saliva cup would be useless in a desert.

The material can also dictate the entire form. Ovenbirds build a domed chamber of clay and dung that can weigh several kilograms and outlast the

pair that made it; swifts, with no access to anything solid at altitude, build from hardened saliva; the tailorbird, surrounded by broad leaves, sews. Each solution is unrecognizable as a version of the others, and each is correct where it stands.

Companies with several markets often make the opposite assumption. A process that worked in the head office is documented, standardized and exported everywhere, and when it fails in a different country the local team is blamed for poor implementation. The design was never wrong at home. It was wrong for the environment it was moved into.

The constraint can also be absolute, which is worth remembering before admiring flexibility too much. Swallows and martins build with mud, and where there is no mud of the right consistency they do not improvise an alternative — they simply do not breed there. A company has an advantage no bird has: it can change the design. It routinely declines to use it, exports the head-office method unchanged, and then treats the resulting failure as a local performance problem.

Birds suggest a better rule. Standardize the function — keep the eggs warm, dry and safe from predators. Localize the method — because what materials and threats exist here is a genuinely different question from what exists there.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which head-office process fails in your local market, and is the local team being blamed for it?*

KEY TAKEAWAYS

1. Nest design varies enormously because it is optimized for local materials and conditions.
2. There is no universally best structure, only structures suited to an environment.
3. Standardize the purpose of a process; localize how it is carried out.

SUMMARY

Bird species build radically different nests — from century-old communal grass structures to cups of hardened saliva — because construction is adapted to local materials, climate and predators. No single design is universally best. Organizations should therefore standardize the function of a process while allowing the method to fit the local environment.

SCIENCE CHECK

ESTABLISHED

Nest design varies with local materials and conditions, and individual birds improve with practice.

INFERRED

That organizations should standardize function and localize method.

UNCERTAIN

Companies standardize partly for control and reporting rather than performance — a motive that has no equivalent in nest building.

VOCABULARY & PHRASAL VERBS

insulation (n.) material that stops heat passing through. The thick walls provide insulation against the midday heat.

chamber (n.) an enclosed space or room. Each pair occupies its own chamber inside the mass.

fibre (n.) a thin thread of material. The bird sews the leaves together with plant fibre.

predator (n.) an animal that hunts and eats other animals. The design depends on which predators live nearby.

to standardize / to localize (v.) to make uniform everywhere / to adapt to a specific place. Standardize the purpose; localize the method.

to hand (phrase) available nearby.

They build with whatever material is to hand.

put up (phrasal verb) to build or erect. They put up the same structure in every market and wondered why it failed.

fit in with (phrasal verb) to be suitable for a situation or environment. The process has to fit in with local conditions.

roll out (phrasal verb) to introduce something across an organization. Head office rolled out the system unchanged.

PRACTICE

Where the Analogy Breaks

Birds cannot export a design to a subsidiary. Human standardization exists partly for control and reporting, not only for function.

Use the Language

Make sentences using these three words: standardize, fit in with and predator.

Discuss

Which head-office process fails locally, and who gets blamed for it?

Write

Take one standard process and separate it into function and method. Rewrite the method for your own conditions.

SOURCES

B Hansell, M., Animal Architecture, Oxford University Press (2005).

A "Tools, behaviour, and materials: What should we learn from animal nest construction?", ScienceDirect. https://www.sciencedirect.com/science/article/pii/B978012821053600014X

THIS CONNECTS TO

Story 20: Trees: Wind Is What Makes Them Strong SAME MECHANISM

Both show a biological structure calibrated directly to local physical conditions — wind stress, or local materials and climate — rather than built to one fixed, universal design.

Story 43: Mangroves: Making Their Own Ground LIMITATION

Bird nests adapt their design to whatever local materials and conditions already exist. Mangroves go a step further — actively changing the physical conditions around them, building new ground rather than just adapting to existing terrain.

Story 15: Termites: Building With No Blueprint LIMITATION

Bird nest design varies by individual bird adapting to local materials, without any coordination between birds. Termite mound construction is coordinated

among thousands of individuals through stigmergy — a very different mechanism producing a similarly locally-adapted structure.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Bird nest design varies enormously, even within the same species, depending on locally available materials, climate and predation risk, rather than following one fixed blueprint.

What is the mechanism?

Nest-building behaviour combines an inherited general structure with substantial individual adaptation to the specific materials, temperature range and predator pressure present at each nest site, so two nests built by the same species in different environments can look markedly different while serving the identical function.

What organizational problem does it explain?

Is your organization applying one fixed process or template everywhere, when local conditions — team size, market, available resources — actually call for real structural adaptation rather than surface customization?

Level 2 — Make It Your Story

> *Tell me about a time you had to substantially adapt a standard process or template because local conditions made the standard version a poor fit.*

Use STAR to answer. Think about: a global process adapted for a local market, a template you had to rebuild for a smaller team, a one-size-fits-all policy that didn't fit.

TAKE THE IDEA FURTHER

> *What process in your organization is currently applied identically everywhere, when the underlying conditions it needs to work in actually vary enough to justify real structural adaptation, not just cosmetic changes?*

17

Ants: Thresholds, Not Assignments

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

In many ant species, work is allocated by a mechanism called the response threshold model. Each ant has an internal threshold for each type of task — one ant might start cleaning the nest when the mess reaches a low level, another only when it becomes severe. Nobody assigns anything. Individuals simply act when the stimulus around them crosses their personal threshold.

The consequence is a workforce that reallocates itself automatically. When the ants with low thresholds for brood care are removed, the level of unattended brood rises until it crosses the higher thresholds of other ants — who then start doing the work, even though they had never done it before. Experiments removing whole task groups show colonies restoring function without any signal being sent. The redistribution is a side-effect of the design.

Thresholds are not fixed, either, which is where specialists come from. Performing a task successfully tends to lower an individual's threshold for that task, so she becomes more likely to do it again; failing repeatedly raises it. Run that for a few weeks and a colony that started with broadly similar ants contains recognizable specialists — none of whom was appointed, and any of whom can drift back if the colony's needs change. Most organizations reproduce the first half of this by letting people gravitate towards what they are good at, then freeze it permanently with a job title.

The reserve makes this possible. At any given moment a substantial proportion of a colony's workers are doing nothing observable at all, and

researchers who removed the active workers found that these apparently idle individuals became active in their place. A colony running every worker at full capacity would have nothing left to reallocate — which is the ordinary condition of most understaffed teams, and the reason a single resignation can stop them.

Both points translate directly. Organizations become resilient not by documenting who does what, but by ensuring that unmet work is visible and that people are permitted to pick it up. And a team with no slack in it cannot absorb a single absence.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Is unfinished work visible to people who are free to pick it up?*

KEY TAKEAWAYS

1. Work is reallocated automatically when unmet needs are visible and people may act on them.
2. Colonies recover from losing whole task groups without any central reassignment.
3. Apparent slack in a system is what makes redistribution possible.

SUMMARY

Ants take on tasks when the surrounding stimulus crosses their individual response threshold, so removing one group of workers causes others to start doing that work automatically. Colonies also maintain a reserve of inactive

individuals who become active when needed. Organizations should build visible work and spare capacity rather than dependence on particular people.

SCIENCE CHECK

ESTABLISHED

Ants take on tasks when local stimulus crosses individual thresholds; colonies restore function after whole task groups are removed.

INFERRED

That visible unmet work plus spare capacity produces resilience without reassignment.

UNCERTAIN

Ants face no status cost for doing work classified as someone else's. This is the main reason the mechanism transfers imperfectly.

VOCABULARY & PHRASAL VERBS

threshold (n.) the level at which something starts to happen. Each ant has its own threshold for each kind of work.

stimulus (n.) something that causes a reaction. The stimulus accumulates until somebody responds to it.

brood (n.) the young of an animal, collectively. Unattended brood is a signal, not a crisis.

to reallocate (v.) to give out again in a different way. The colony reallocates effort without any instruction.

utilization (n.) the degree to which capacity is being used. A team at full utilization cannot absorb one absence.

slack (n.) spare capacity or unused time.

Slack looks like waste until the day you need it.

take over (phrasal verb) to begin doing something that someone else was doing. Somebody quietly took over the work when she left.

pick up (phrasal verb) here: to start doing a task that is waiting. Unfinished work should be visible so that others can pick it up.

cover for (phrasal verb) to do someone's work while they are absent. Two colleagues covered for him for two weeks.

PRACTICE

Where the Analogy Breaks

Ants have no job descriptions to defend. Humans face status costs for doing work classified as someone else's.

Use the Language

Make sentences using these three words: threshold, slack and cover for.

Discuss

Is unfinished work visible in your team, and are people free to pick it up?

Write

Write 120 words on where your team has no slack, and what would break first if someone were ill.

SOURCES

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THE CONTRAST

Story 17: Ants: Thresholds, Not Assignments vs. Story 13: Bees: Specialists Who Can Switch Back

Story 17: Ants: Thresholds, Not Assignments: Work reallocates automatically the instant unmet need becomes visible enough to cross an individual's response threshold.

Story 13: Bees: Specialists Who Can Switch Back: Roles are more stable and specialized by default, with switching happening only when a significant gap opens.

Should task allocation happen through constant, low-friction threshold-response, or through stable specialization with occasional deliberate switching?

THIS CONNECTS TO

Story 13: Bees: Specialists Who Can Switch Back SAME MECHANISM
Both describe flexible task allocation within a colony, where individuals shift what they're doing based on need rather than sticking to a permanently fixed assignment.

Story 1: Ants: Order Without a Manager LIMITATION
Order Without a Manager shows decisions emerging from reinforcement over time. Thresholds shows a faster, second mechanism in the same species — individuals responding instantly to visible unmet need — proving colonies use more than one coordination method depending on the problem.

Story 38: Ant Colonies: Growth Without Overhead SAME MECHANISM
Both describe how ant colonies coordinate work locally, without a manager

tracking overall colony state, letting the coordination mechanism itself scale with colony size.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Ants don't hold fixed job assignments; instead, individual ants have different response thresholds for various tasks, and take on whichever job currently has the most visible unmet need, shifting fluidly as conditions change.

What is the mechanism?

Each ant has a genetically and experientially shaped threshold for responding to a given task-related stimulus (such as brood needing care, or a breach in the nest wall); as unmet need in a task rises, more ants' thresholds are crossed and they join that task, and as the need is met, ants drift back to their default activity — producing continuous, self-correcting task allocation with no assignment step at all.

What organizational problem does it explain?

Is work in your organization allocated through fixed job assignments that lag behind actual need, or could a threshold-style system — where visible need itself pulls in help — respond faster?

Level 2 — Make It Your Story

> *Tell me about a time work got reallocated quickly because the need became visible to the right people, rather than through a formal reassignment process.*

Use STAR to answer. Think about: an informal all-hands-on-deck moment, a team that self-organized around a visible gap, a slow formal reassignment you wish had happened faster.

TAKE THE IDEA FURTHER

> *Where in your organization does visible unmet need currently sit unaddressed simply because reallocating people to it requires a formal process rather than a natural threshold response?*

18

Bees: Dissent That Improves the Choice

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Return to the swarm hanging on its branch. Several scouts have found different sites, and each is advertising hers. For a while, the colony holds several competing proposals at once, and the dances actively compete for recruits.

What stops a mediocre site winning early? Two things. First, a scout's dance decays: she advertises a site with fewer and fewer circuits on each return, so support for a weak site runs out unless new scouts independently confirm it is good. Enthusiasm has to be re-earned, and it is re-earned by evidence rather than by repetition. Second, scouts committed to one site can deliver "stop signals" — a brief vibration — to bees dancing for a rival site. Cross-inhibition of this kind prevents deadlock and helps the swarm break out of a tie.

The stop signal is a physical thing, not a metaphor. A bee delivers it by butting her head against a dancer while producing a short vibrational pulse of about a tenth of a second, and a dancer who receives enough of them stops advertising and leaves the dance floor. Scouts committed to one site direct these signals at bees dancing for rival sites, and the rate rises as support for the sender's own site grows. Two equally good options therefore do not deadlock the swarm: each side actively suppresses the other until one crosses the quorum, and the bees that were suppressed are then recruited to the winner.

So the colony is not a harmonious meeting. It contains real competition, active interference and open disagreement — and it converges anyway, because the disagreement is about the options, not about who is right.

The uncomfortable implication for meetings is that the smooth ones may be the dangerous ones. If a proposal arrives and everyone agrees within four minutes, the group has almost certainly not generated an alternative — and without an alternative, nobody has actually decided anything. They have only accepted something.

Notice also what the decay rule does to seniority. A scout's influence is not a standing quantity she carries into the next decision — it lasts precisely as long as her most recent inspection supports it, and she has to go back and look again to renew it. In a human meeting, credibility earned two years ago is spent freely on today's question, by people who have not visited the site in months.

The practical form is simple and slightly artificial, which is fine. Require that any proposal arrive with a named alternative that somebody is willing to argue for, and that the two be heard before either is discussed on its merits. This is not a debating exercise. It exists because a group that has considered only one option has not made a decision at all — it has accepted a suggestion, and the difference will only become visible later.

If you chair meetings, this gives you a concrete job: do not ask "does anyone disagree?", which invites silence. Require a second option to exist before the first can be chosen.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *In your last meeting, did a second real option exist before the first was chosen?*

KEY TAKEAWAYS

1. Healthy decision processes hold several competing options open before converging.
2. Support must decay unless it is renewed by independent evidence.
3. Immediate agreement usually means no alternative was generated, not that the choice is right.

SUMMARY

Bee swarms decide between nest sites through genuine competition between advertisers, with support decaying unless independently confirmed and stop signals used against rival options. Disagreement is built into the mechanism, yet the swarm still converges. A meeting in which everyone agrees at once has generated no alternative, and should be treated as a warning sign.

SCIENCE CHECK

ESTABLISHED

Scouts deliver vibrational stop signals to rivals, support decays without fresh confirmation, and swarms still converge.

INFERRED

That healthy decisions require competing options and that instant agreement is a warning sign.

UNCERTAIN

Bee dissent carries no career risk. Human silence may indicate agreement, calculation or fear, and the three are hard to tell apart.

VOCABULARY & PHRASAL VERBS

dissent (n.) disagreement with a majority opinion. Dissent is part of the mechanism, not a failure of it.

mediocre (adj.) of only average or poor quality. A mediocre option wins when nobody proposes another.

to decay (v.) to weaken and decline gradually. Support should decay unless fresh evidence renews it.

cross-inhibition (n.) one option actively suppressing support for another. Cross-inhibition stops two good options deadlocking the group.

deadlock (n.) a situation where no progress is possible. The committee reached deadlock and adjourned.

to chair (a meeting) (v.) to lead or preside over a meeting. Whoever chairs the meeting decides what gets heard.

run out (phrasal verb) to be used up completely. Enthusiasm runs out unless something confirms it.

break out of (phrasal verb) to escape from a situation. The group needed help to break out of the tie.

go along with (phrasal verb) to agree with something without real conviction. Everyone went along with it and nobody had checked.

PRACTICE

Where the Analogy Breaks

Bee dissent has no social consequences. Human disagreement carries career risk, which is why silence is not the same as agreement.

Use the Language

Make sentences using these three words: go along with, decay and mediocre.

Discuss

When did your team last choose between two genuine options rather than accepting one?

Write

Rewrite the agenda of a recent meeting so that a second option must exist before the first can be chosen.

SOURCES

A Seeley, T. D. et al., "Stop signals provide cross inhibition in collective decision-making by honeybee swarms", Science 335, 108–111 (2012).

B Seeley, T. D., Honeybee Democracy, Princeton University Press (2010).

THIS CONNECTS TO

Story 2: Honeybees: Quorum, Not Consensus SAME MECHANISM

Both chapters describe the same species using a decision process built to prevent premature agreement, whether through independent scouting or deliberately sustained disagreement.

Story 3: Locusts: Why Alignment Resists Change CONTRAST

See The Contrast in that chapter. In short: bees preserve productive dissent as a deliberate safeguard; locusts show what happens without that safeguard — alignment that becomes self-reinforcing and impossible to correct once established.

Story 6: Ants: Weighing Two Sources of Truth LIMITATION

This chapter is about preserving competing options long enough to be properly tested. Weighing Two Sources of Truth is about a different safeguard — weighting information by reliability — showing good group judgment needs more than one kind of protection.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

During honeybee swarm decision-making, scouts advertising a weaker candidate nest site don't simply abandon their position the moment a competing site is proposed; the competing options are held open and contested for a period before the swarm converges.

What is the mechanism?

Scouts advertising different sites use head-butting signals to inhibit each other's recruitment dances, effectively slowing down premature convergence around any one option; this contest continues until one option accumulates enough independent support to cross the quorum threshold, ensuring the swarm doesn't converge simply because one option was proposed first or loudest.

What organizational problem does it explain?

Does your organization's decision process actively protect a competing option from being abandoned too early, or does the first proposal to gain any momentum tend to steamroll genuine alternatives?

Level 2 — Make It Your Story

> *Tell me about a time you deliberately kept a dissenting or minority option alive in a group discussion, rather than letting the group converge early.*

Use STAR to answer. Think about: a devil's advocate role you played, a decision you slowed down deliberately, a minority opinion that turned out to matter.

TAKE THE IDEA FURTHER

> *In your organization's last major decision, was a competing option given a real, protected chance to be tested — or did it quietly disappear the moment the group started leaning one way?*

PART III

Resilience & Adaptation

How does nature survive disruption?

19

Axolotls: Regrowth Instead of Scar

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Cut off an axolotl's leg and it grows a new one — with bone, muscle, nerve, blood vessels and skin in the right arrangement, and no scar where the old limb was. It can do the same for parts of its heart, its spinal cord and portions of its brain, and it can do it repeatedly.

The mechanism is what makes this useful rather than merely astonishing. Mammals seal a wound fast. Cells rush to the site, lay down dense collagen, and close the gap with scar tissue — a repair that restores the barrier but not the function. The axolotl does something slower and stranger. Cells near the wound dedifferentiate: they partly abandon their specialized identity and form a mass of flexible cells called a blastema, which then rebuilds the missing structure according to positional information the remaining tissue still carries. The animal does not patch the hole. It regrows the part.

The blastema is the part that has no mammalian equivalent. Within days of an amputation, cells beneath the wound surface lose the markers that identified them as muscle, cartilage or connective tissue and revert to a flexible state, dividing rapidly to form a pale bud at the end of the stump. What makes this more than a mass of spare parts is that the cells retain a sense of where they are along the limb: tissue taken from a wrist position builds a wrist and a hand, not a whole arm. The blastema knows both what is missing and where it starts.

Both strategies work. They simply optimize for different things — speed of closure against restoration of capability.

Scarring is not a failure of the mammalian system either, which is the part worth being careful about. A wound closed quickly is a wound that does not become infected, and in an animal with a high metabolic rate and a long list of predators, speed of closure has usually been worth more than perfect restoration. The axolotl can afford the slower route partly because it is aquatic, cool-bodied and unhurried. When an organization chooses the fast patch under real pressure, it is making the same trade — and the mistake is rarely the patch itself. It is leaving the patch in place for a decade after the emergency has passed.

Organizations almost always choose the mammalian option, and rarely notice they have chosen. Something goes wrong, and within two weeks there is a new approval step, a new form, a new mandatory sign-off. The gap is closed. But the tissue that closed it is not functional tissue, and it never becomes functional again: three years later nobody remembers the incident and everybody is still filling in the form. That is organizational scar tissue, and enough of it will stiffen a company until it can no longer move.

There is a diagnostic that separates the two. Look at a control that was introduced after an incident and ask what would happen if you removed it tomorrow. If somebody can name the failure it prevents and describe how they would notice its absence, it is functioning tissue. If the answer is that nobody is quite sure but it seems unwise to touch, it is scar — and scar accumulates, because every incident adds a layer and nothing ever removes one.

The question worth asking after any failure is therefore not "have we fixed it?" but a harder one: have we restored the capability, or have we merely sealed the gap?

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *After your last serious failure, did you restore capability — or add a permanent approval step?*

KEY TAKEAWAYS

1. Mammals close wounds with scar tissue; axolotls rebuild the missing structure itself.
2. Fast closure and full restoration of function are different goals, and they trade off.
3. Procedures added after a failure often persist as permanent scar tissue that reduces capability.

SUMMARY

Axolotls regenerate whole limbs and organs by forming a blastema of dedifferentiated cells rather than sealing wounds with collagen scar as mammals do. The trade-off is speed of closure against restoration of function. Organizations that answer every failure with a new procedure are choosing scar tissue, and should ask whether capability was restored or merely covered.

SCIENCE CHECK

ESTABLISHED

Axolotls form a blastema of dedifferentiated cells that rebuilds the missing structure with correct positional identity.

INFERRED

That organizations should restore capability after failure rather than adding permanent controls.

UNCERTAIN

Why mammals scar instead is an active research question, and fast closure carries real survival advantages the analogy tends to skip.

VOCABULARY & PHRASAL VERBS

regeneration (n.) the regrowth of lost or damaged body parts. Regeneration rebuilds the part; repair merely closes the gap.

blastema (n.) a mass of unspecialized cells that rebuilds a lost structure. The blastema forms within days of the injury.

to dedifferentiate (v.) to lose a specialized role and become flexible again. The cells dedifferentiate and become flexible again.

collagen (n.) the fibrous protein that forms scar tissue. Mammals lay down collagen and close the wound quickly.

scar tissue (n.) fibrous material that closes a wound but does not restore full function. Every incident left another layer of procedural scar tissue.

sign-off (n.) official approval required before work can proceed. The new sign-off has outlived the problem that caused it.

grow back (phrasal verb) to regrow after loss. The limb grows back complete, with no scar.

patch up (phrasal verb) to repair quickly and imperfectly. We patched it up and never returned to the real cause.

cover up (phrasal verb) to hide a problem rather than solve it. Adding a form covered up the failure without fixing it.

PRACTICE

Where the Analogy Breaks

An axolotl regenerates the same limb it had. Organizations rarely need the identical structure back, so 'full restoration' is itself a judgment.

Use the Language

Make sentences using these three words: scar tissue, patch up and cover up.

Discuss

After your last serious failure, did you restore the capability or add a control?

Write

List every approval step in one of your processes. Beside each, name the failure it prevents. Mark the ones nobody can explain.

SOURCES

A Tanaka, E. M., "The molecular and cellular choreography of appendage regeneration", Cell 165, 1598–1608 (2016).

A Nowoshilow, S. et al., "The axolotl genome and the evolution of key tissue formation regulators", Nature 554, 50–55 (2018).

B Seifert, A. W. & Maden, M., "New insights into vertebrate skin regeneration", International Review of Cell and Molecular Biology (2014).

THE CONTRAST

Story 19: Axolotls: Regrowth Instead of Scar vs. Story 20: Trees: Wind Is What Makes Them Strong

Story 19: Axolotls: Regrowth Instead of Scar: Full recovery means rebuilding the original structure completely, rather than patching over the damage.

Story 20: Trees: Wind Is What Makes Them Strong: Full resilience is built proactively, in advance of damage, through repeated exposure to stress.

Is resilience better built by rebuilding fully after damage occurs, or by strengthening structure in advance of damage ever happening?

THIS CONNECTS TO

Story 27: Wildfires: Renewal That Needs Burning SAME MECHANISM
Both show a system whose recovery process doesn't just restore the previous state but depends on disruption itself as part of how renewal actually happens.

Story 21: Plants: Roots Before Visible Growth LIMITATION
Axolotl regeneration is a response to acute, sudden damage. Plants: Roots Before Visible Growth is about slow, invisible investment made in the absence of any damage at all — two different relationships between preparation and disruption.

Story 47: Tardigrades: Stopping Instead of Dying LIMITATION
Axolotls respond to damage by actively rebuilding the lost structure. Tardigrades respond to extreme conditions by doing close to the opposite — suspending activity almost entirely rather than actively resisting or repairing.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Where most mammals heal a serious wound by forming scar tissue — a functional but structurally different patch — axolotls can regenerate a fully functional, structurally complete replacement of an entire lost limb, including bone, muscle, nerves and skin.

What is the mechanism?

At the injury site, axolotl cells de-differentiate into a mass of progenitor cells called a blastema, which then regrows the missing structure following the same patterning information used in original limb development, rather than simply closing the wound with generalized connective tissue the way scarring does.

What organizational problem does it explain?

When your organization recovers from a serious setback, does it rebuild the actual lost capability, or does it patch over the gap with something that functions but isn't structurally the same thing that was lost?

Level 2 — Make It Your Story

> *Tell me about a time your team fully rebuilt a lost capability after a setback, rather than patching around the gap.*

Use STAR to answer. Think about: a key person leaving and being properly replaced versus just covered for, a system rebuilt rather than patched, a process restored to full strength after a disruption.

TAKE THE IDEA FURTHER

> *What capability did your organization lose recently that got a scar-tissue patch — functional, but not the same thing — rather than the deliberate investment a full rebuild would have required?*

20

Trees: Wind Is What Makes Them Strong

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A tree grown in a greenhouse, protected from all wind, often cannot stand up. This is not a metaphor. Plants respond to mechanical stress — wind, touch, bending — with a process called thigmomorphogenesis: they grow shorter, thicker and stronger in the direction the stress comes from. Remove the stress and the tree grows tall and slender and eventually falls over under its own weight. Biosphere 2, the sealed experimental habitat in Arizona, made this famous: trees inside grew fast in windless air and collapsed before maturity.

The mechanism is specific enough to be measured. Bending a stem repeatedly triggers changes in gene expression within minutes, and the resulting growth is redistributed rather than merely increased: less height, more diameter, thicker cell walls, and wood laid down preferentially on the side that took the load. A tree shaken in a wind tunnel from one direction becomes stronger in that direction and no stronger in the other. It is not becoming generally tougher. It is answering a specific question about where the force comes from.

But individual strength is only half the story. A tree standing alone in an open field faces the full force of every storm. In a forest, wind is slowed by the canopy as a whole, roots interlock, and neighbouring trees limit how far any one of them can sway. Foresters routinely observe that trees suddenly exposed when their neighbours are removed become far more vulnerable to being

blown down — they had been relying on a protection they never grew the wood to replace.

The forest effect has been quantified too, in a way foresters find painful. A stand thinned too heavily loses the mutual shelter that its trees grew up depending on, and the survivors — tall, slender, and adapted to a wind load they never carried alone — are markedly more likely to be thrown by the next storm. The trees did not become weaker when their neighbours were removed. They simply stopped being protected, and their trunks recorded the assumption that protection would continue.

The timing matters as much as the presence of stress. Seedlings need enough shelter to establish at all, and a young tree exposed to a gale it cannot yet withstand simply breaks — the response is adaptive only within a range the plant can survive. This is where the fashionable version of the lesson usually goes wrong. "What doesn't kill you" is not a mechanism; the useful statement is narrower and more demanding: repeated, survivable, direction-specific load produces strength, and everything outside that description produces damage.

Two lessons follow, and they pull in opposite directions in a useful way. First, protecting people from all difficulty produces exactly the fragility you were trying to prevent; controlled stress is how strength is built. Second, no amount of individual strength substitutes for the structure around you. A resilient organization is not a collection of exceptionally tough individuals. It is a set of relationships dense enough that no one takes the full force of the storm alone.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which people are you protecting from difficulty in a way that is preventing them from growing?*

KEY TAKEAWAYS

1. Plants build strength in direct response to mechanical stress; total protection creates fragility.
2. Trees in forests are protected by their neighbours, not only by their own trunks.
3. Removing surrounding support exposes individuals to forces they never adapted to.

SUMMARY

Trees grown without wind become tall, weak and prone to collapse, because mechanical stress is what triggers strengthening growth. Forest trees are also protected by their neighbours, and become far more vulnerable when suddenly isolated. Resilience therefore comes from both appropriate stress and the strength of surrounding relationships.

SCIENCE CHECK

ESTABLISHED

Mechanical stress alters gene expression and redistributes growth; sheltered trees are structurally weaker and thinning increases windthrow.

INFERRED

That protecting people from all difficulty produces fragility, and that support structures matter as much as individual strength.

UNCERTAIN

The range of survivable stress in a human career is not measurable the way a wind load is, which makes the lesson easy to abuse.

VOCABULARY & PHRASAL VERBS

thigmomorphogenesis (n.) growth changes caused by mechanical stimulation. Thigmomorphogenesis is growth altered by mechanical stress.

slender (adj.) thin and tall. Sheltered trees grow tall and slender and fall over.

canopy (n.) the upper layer of branches and leaves in a forest. The canopy slows the wind before it reaches the trunks.

to sway (v.) to move slowly from side to side. Neighbouring trees limit how far any one of them can sway.

fragility (n.) the quality of being easily broken.

Total protection produces fragility, not safety.

vulnerable (adj.) easily harmed. A tree suddenly exposed is vulnerable to the next storm.

stand up (phrasal verb) to remain upright; to resist pressure. A greenhouse tree often cannot stand up on its own.

blow down (phrasal verb) to be knocked over by wind. Half the stand was blown down after the thinning.

hold up (phrasal verb) to remain strong under pressure. The structure held up because the load was shared.

rely on (phrasal verb) to depend on. They relied on shelter they had not grown the wood to replace.

PRACTICE

Where the Analogy Breaks

Trees cannot be over-stressed by their manager's anxiety. Human stress is not automatically strengthening, and the wrong kind is simply damage.

Use the Language

Make sentences using these three words: hold up, rely on and fragility.

Discuss

Who on your team is being protected from difficulty in a way that is preventing them from growing?

Write

Write 120 words on a difficulty that made you more capable, and say precisely why it helped rather than harmed.

SOURCES

A Jaffe, M. J., "Thigmomorphogenesis: the response of plant growth and development to mechanical stimulation", Planta 114, 143–157 (1973).

A Telewski, F. W., "A unified hypothesis of mechanoperception in plants", American Journal of Botany (2006).

THE CONTRAST

Story 20: Trees: Wind Is What Makes Them Strong vs. Story 19: Axolotls: Regrowth Instead of Scar

Story 20: Trees: Wind Is What Makes Them Strong: Full resilience is built proactively, in advance of damage, through repeated exposure to stress.

Story 19: Axolotls: Regrowth Instead of Scar: Full recovery means rebuilding the original structure completely, rather than patching over the damage, after it occurs.

Is resilience better built by strengthening structure in advance of damage, or by rebuilding fully after damage occurs?

THIS CONNECTS TO

Story 16: Birds: The Nest Fits the Conditions SAME MECHANISM
Both show a biological structure calibrated directly to local physical conditions — wind stress, or local materials and climate — rather than built to one fixed, universal design.

Story 22: Camels: Tolerance, Not Reserves LIMITATION
Trees build resilience through repeated, moderate stress exposure over time. Camels achieve resilience through a completely different mechanism — tolerating a wider range of internal conditions rather than adapting structurally to repeated stress.

Story 50: Sunflowers: Turning Before the Sun LIMITATION
Trees respond to mechanical stress by building targeted structural strength. Sunflowers respond to a predictable environmental cycle with active, energy-costly tracking — two different strategies for dealing with a recurring environmental force.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A tree grown in a completely wind-sheltered environment develops thinner, taller, weaker trunk growth than a tree of the same species exposed to regular wind, and is more likely to fail catastrophically when eventually exposed to real wind stress.

What is the mechanism?

Mechanical stress from wind triggers a genuine growth response called thigmomorphogenesis, in which the tree allocates additional growth to trunk and root strength specifically in response to the stress it has actually experienced, meaning moderate, repeated stress functions as a signal the tree uses to calibrate its own structural investment.

What organizational problem does it explain?

Is your organization shielding a team or system from all stress in the name of stability, when moderate, repeated exposure to real difficulty is actually what would build the structural strength needed for a genuine crisis later?

Level 2 — Make It Your Story

> *Tell me about a time moderate, repeated exposure to a real difficulty made you or your team stronger than being protected from it would have.*

Use STAR to answer. Think about: a challenging assignment that built real capability, a team that got stronger from handling smaller crises before a big one hit.

TAKE THE IDEA FURTHER

> *Is there a team or system in your organization currently shielded from all stress, in a way that might leave it structurally weaker when real pressure eventually arrives?*

21

Plants: Roots Before Visible Growth

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Chinese bamboo is the standard story in motivational books, and it is worth treating carefully because the usual version exaggerates. The real biology is more interesting anyway.

Many perennial plants spend their early years investing almost everything below ground. Above the surface, growth looks negligible. Underground, the plant is constructing a root system and, in many species, storing carbohydrate reserves in rhizomes or bulbs. When conditions are right, that stored energy is released and shoots can extend with startling speed — certain bamboo species are among the fastest-growing plants measured, capable of extending remarkably quickly during their growth phase, precisely because the material was accumulated in advance.

The economics of this have been studied in their own right. Storing carbohydrate rather than spending it on immediate growth is a real cost — those reserves could have been leaves, competing for light this season — and the plants that do it are the ones facing predictable interruption: fire, drought, grazing, winter. Storage buys the ability to rebuild fast after a loss, and it is paid for by growing more slowly than a neighbour that keeps nothing back. Nobody watching the two plants in a good year would identify the cautious one as better adapted.

The visible speed is therefore not the growth. It is the release of growth that already happened somewhere invisible.

This has an obvious application to careers and businesses, and a less obvious one. The obvious version: research, capability-building and relationships often show nothing for a long time and then produce results suddenly. The less obvious version is a warning about measurement. If your metrics only capture what is above ground, you will systematically defund the phase where the actual work is being done — and then be surprised by competitors whose visible growth appears to come from nowhere.

The reverse error is worth naming as well. Not every project with nothing to show is building a root system; some are simply not working, and the vocabulary of patient investment is a comfortable place for a failing initiative to hide. The distinction is not the absence of visible output. It is whether the invisible phase is producing anything at all — and whether anybody has agreed in advance what that would look like.

There is a second reason invisible work gets defunded, and it is not stupidity. Underground investment produces no evidence of its own quality until it emerges, which means an honest manager genuinely cannot distinguish a root system from an excuse. The answer is not to demand faith. It is to agree in advance what a healthy invisible phase should produce — a working prototype, a named customer who took the call, a technical question answered — so that there is something to inspect that is not yet a result.

Ask of any slow-looking project: is nothing happening, or is something happening that this dashboard cannot see? These are entirely different situations, and only one of them justifies cancellation.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which slow-looking project is your dashboard unable to see the progress of?*

KEY TAKEAWAYS

1. Long invisible periods are often investment in structure that later enables rapid growth.
2. Sudden visible growth is usually the release of stored capacity, not its creation.
3. Metrics that measure only visible output will defund the work that produces results.

SUMMARY

Many plants spend years building roots and storing energy underground before producing rapid visible growth, so the dramatic phase merely releases what was accumulated invisibly. Speed above ground is a consequence of earlier unseen investment. Organizations must distinguish projects where nothing is happening from those whose progress their metrics cannot detect.

VOCABULARY & PHRASAL VERBS

perennial (adj.) living for several years. Perennial plants invest below ground for years.

rhizome / bulb (n.) underground plant structures that store energy. The energy is stored in a rhizome underground.

carbohydrate (n.) a substance in food that provides energy. Carbohydrate reserves pay for the growth you can see.

negligible (adj.) so small as to be unimportant. Progress above ground was negligible for two seasons.

to defund (v.) to stop providing money for something. Metrics that see only output will defund the real work.

dashboard (n.) a screen showing key business measurements. The dashboard cannot see what the roots are doing.

pay off (phrasal verb) to produce a good result after effort. The research paid off three years after it started.

build up (phrasal verb) to accumulate over time. Reserves build up long before anything appears.

write off (phrasal verb) to decide something is a failure and abandon it. Do not write off a project simply because it is quiet.

PRACTICE

Where the Analogy Breaks

A plant cannot be asked what it is doing underground. Human projects can be inspected — the failure is usually of measurement, not of visibility.

Use the Language

Make sentences using these three words: negligible, pay off and write off.

Discuss

Which slow-looking project can your reporting not see the progress of?

Write

For one quiet project, write down what a healthy invisible phase should produce in the next month.

SOURCES

B Chapin, F. S., Schulze, E.-D. & Mooney, H. A., "The ecology and economics of storage in plants", Annual Review of Ecology and Systematics 21, 423–447 (1990). · Note: the popular "five years then 90 feet in six weeks" bamboo story is a simplification; the underlying storage-then-release biology is real.

THIS CONNECTS TO

Story 52: Cicadas: Built for One Loud Season SAME MECHANISM
Both describe a long, invisible period of preparation whose value only becomes visible in a short, intense period afterward — and both warn against judging the invisible phase as wasted time.

Story 19: Axolotls: Regrowth Instead of Scar LIMITATION
Plants: Roots Before Visible Growth is about slow, invisible investment made in the absence of any damage at all. Axolotl regeneration is a response to acute, sudden damage — two different relationships between preparation and disruption.

Story 45: Scrub-Jays: Planning for a Future Need SAME MECHANISM
Both describe investing effort now — storing energy underground, caching food — against a future need that isn't currently being experienced, based on anticipation rather than immediate reward.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Many perennial plants, popularly exemplified by bamboo, appear to do

almost nothing above ground for years, then produce startlingly fast visible growth in a short period.

What is the mechanism?

During the apparently inactive years, the plant is investing heavily in root system development and underground carbohydrate storage; the dramatic visible growth spurt is simply the release of energy that was accumulated slowly and invisibly beforehand, not a sudden acquisition of new capability.

What organizational problem does it explain?

Does your organization's measurement system only register visible output, in a way that would systematically defund the invisible-but-real work — research, relationship-building, capability development — that later produces the visible result?

Level 2 — Make It Your Story

> *Tell me about a time a slow, invisible period of investment was later followed by a sudden, visible result that surprised people who hadn't been tracking the invisible work.*

Use STAR to answer. Think about: a research project, a relationship built over years before it paid off, a skill developed quietly before it suddenly mattered.

TAKE THE IDEA FURTHER

> *What slow-looking project in your organization is at risk of being defunded because your current metrics can only see above-ground growth, not the root system being built underneath it?*

22

Camels: Tolerance, Not Reserves

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Almost everything people believe about camels is slightly wrong, and the truth is a better lesson. The hump does not store water. It stores fat — a concentrated energy reserve deliberately kept in one place rather than spread under the skin, so the rest of the body can lose heat efficiently. Storage was located where it would not compromise cooling.

The water strategy is a series of tolerances rather than a reservoir. A camel allows its body temperature to swing by several degrees over a day, absorbing heat rather than sweating to remove it, then releasing it into the cold desert night. Its kidneys concentrate urine severely; its nostrils recapture moisture from exhaled breath. Its red blood cells have an unusual oval shape and can tolerate extreme swings in hydration, which is why a camel can rehydrate very rapidly without damage. It can lose a proportion of body water that would kill most mammals and keep functioning.

The measurements behind this are unusually elegant. Working in the Algerian Sahara in the 1950s, Knut Schmidt- Nielsen and colleagues recorded the body temperature of camels through the day and found that a dehydrated animal let itself swing by around six degrees, from roughly 34°C in the early morning to 40°C by afternoon. A camel that sweated to hold a constant temperature would have lost several litres of water doing it. Instead it stores the heat, carries it, and unloads it into the cold desert night for free.

Read as a business lesson, the interesting part is that none of this is "efficiency" in the usual sense of doing the same thing more cheaply. The camel changed what it tolerates. It accepted a fluctuating body temperature, which mammals normally treat as non-negotiable, in exchange for water.

There is one more piece of the design worth borrowing. The camel does not merely tolerate loss; it recovers from it at extraordinary speed, drinking on the order of a hundred litres in a matter of minutes without the cell damage that rapid rehydration causes in most mammals — a tolerance its unusual oval red blood cells appear to support. Endurance and recovery were engineered together. A system that can go without something for a long time, and then refill quickly when the resource reappears, is in a different position from one that has merely learned to be frugal.

The comparison that makes this concrete is a donkey. A donkey drinks more often and can also survive rough conditions, but it does so by finding water, not by tolerating its absence — which ties it to a route. The camel's tolerances buy range: it can cross ground where the resource simply is not available, and arrive. Constraint-driven strategy usually looks like this. The question is rarely how to consume less of something. It is what becoming able to go without it would let you reach.

When resources are scarce, most organizations look for savings inside their existing assumptions. The camel's approach is to ask which assumption itself is expensive — which comfort, standard or fixed expectation is quietly consuming everything — and then to renegotiate that instead.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which fixed expectation in your organization is quietly consuming the most resource?*

KEY TAKEAWAYS

1. The camel concentrates its reserves where storage does not interfere with cooling.
2. Its survival comes from raising tolerances, not from holding a large water reserve.
3. Under scarcity, examine which fixed assumptions are consuming resources.

SUMMARY

Camels survive deserts not by storing water but by tolerating large swings in body temperature and hydration, and by concentrating fat in a hump that leaves the rest of the body free to cool. Their adaptation changed what they tolerate rather than what they consume. Scarcity is best met by questioning expensive assumptions, not only by trimming costs.

SCIENCE CHECK

ESTABLISHED

Camels tolerate a body-temperature swing of around six degrees and severe dehydration, then rehydrate rapidly without cell damage.

INFERRED

That scarcity is better met by questioning expensive assumptions than by trimming consumption.

UNCERTAIN

Physiological tolerances are fixed by evolution; organizational 'tolerances' are social agreements, and lowering them affects people who did not consent.

VOCABULARY & PHRASAL VERBS

hump (n.) the raised part on a camel's back. The hump stores fat, not water.

reserve (n.) a supply kept for future use. The reserve is concentrated where it will not block cooling.

tolerance (n.) the ability to endure difficult conditions. Survival came from raising tolerances, not hoarding supplies.

to fluctuate (v.) to rise and fall irregularly. The animal lets its temperature fluctuate by six degrees.

to rehydrate (v.) to take in water again after losing it. A camel can rehydrate in minutes without damage.

scarcity (n.) a shortage of something needed.

Scarcity forces better questions than abundance does.

cut back on (phrasal verb) to reduce consumption of something. We cut back on travel and missed the real cost entirely.

get by on (phrasal verb) to survive with only a small amount. They got by on half the budget by dropping one assumption.

run out of (phrasal verb) to have no more of something left. The team ran out of cash before it ran out of ideas.

PRACTICE

Where the Analogy Breaks

A camel's tolerances are physiological. Human standards are social agreements, and lowering them affects people who did not consent.

Use the Language

Make sentences using these three words: tolerance, cut back on and get by on.

Discuss

Which fixed expectation in your organization consumes the most and is never questioned?

Write

Write 120 words on what your team could reach if it became able to operate without one thing it currently requires.

SOURCES

A Schmidt-Nielsen, K. et al., "Body temperature of the camel and its relation to water economy", American Journal of Physiology (1957).

B Schmidt-Nielsen, K., Desert Animals: Physiological Problems of Heat and Water, Oxford University Press (1964).

THE CONTRAST

Story 22: Camels: Tolerance, Not Reserves vs. Story 20: Trees: Wind Is What Makes Them Strong

Story 22: Camels: Tolerance, Not Reserves: Resilience comes from tolerating a wider range of internal conditions, not from building structural strength through repeated stress exposure.

Story 20: Trees: Wind Is What Makes Them Strong: Resilience is built proactively, through repeated exposure to stress that triggers targeted structural growth.

Is resilience better achieved by tolerating a wider range of conditions, or by building structural strength in response to repeated stress?

THIS CONNECTS TO

Story 33: Beetles: Drinking From Fog SAME MECHANISM

Both describe an organism thriving in a resource-scarce environment not through storage or brute-force acquisition, but through a specifically adapted mechanism for managing the scarce resource efficiently.

Story 47: Tardigrades: Stopping Instead of Dying LIMITATION

Camels tolerate extreme conditions by widening their internal operating range while staying active. Tardigrades tolerate extreme conditions through a different strategy entirely — suspending activity almost completely rather than continuing to function through it.

Story 20: Trees: Wind Is What Makes Them Strong LIMITATION

Trees build resilience through repeated, moderate stress exposure over time. Camels achieve resilience through a completely different mechanism — tolerating a wider range of internal conditions rather than adapting structurally to repeated stress.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Camels do not primarily survive desert conditions by storing large water reserves, as the popular account of the hump suggests; the hump is fat storage, not water storage, and camels instead tolerate physiological extremes — including body temperature swings and blood water loss — that would be dangerous or fatal for most mammals.

What is the mechanism?

A camel's body tolerates a much wider range of core body temperature and blood osmotic concentration than most mammals can safely withstand, allowing it to avoid the energy cost of constant thermoregulation and delay water loss through sweating, rather than relying on a large stored reserve to draw down during scarcity.

What organizational problem does it explain?

Is your organization trying to survive scarcity by building larger reserves, when the camel's approach — tolerating a wider range of operating conditions before intervention is needed — might be the more efficient strategy?

Level 2 — Make It Your Story

> *Tell me about a time your organization got through a period of scarcity by tolerating a wider range of conditions, rather than by drawing down a stored reserve.*

Use STAR to answer. Think about: operating with less buffer than usual, a team that adapted its normal operating range under constraint, a reserve you didn't actually need to use.

TAKE THE IDEA FURTHER

> *Is your organization defaulting to building bigger reserves against scarcity, when widening your tolerance for variation might be the cheaper and more resilient strategy?*

23

Penguins: Warmth by Taking Turns

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Emperor penguins breed in the Antarctic winter, in temperatures below −40°C and winds that can exceed 150 km/h. The males incubate for around two months, at the end of a fast lasting nearly four, each balancing a single egg on his feet. No individual physiology is sufficient for this. What makes it survivable is the huddle.

Thousands of birds pack together at very high density, and the temperature inside can rise above 20°C even while the air outside is lethal. Getting this on record took some ingenuity. Researchers filmed colonies from fixed cameras through the Antarctic winter and tracked individual birds frame by frame, which revealed that the huddle is never still: every thirty seconds to two minutes, a small movement by one bird triggers a wave that travels through the packed mass, and repeated waves gradually merge separate huddles and rotate individuals through the structure. The birds are not shuffling randomly. Each step is only five to ten centimetres, small enough to keep the packing tight while the whole arrangement slowly turns itself inside out.

So the birds on the outside are not sacrificing themselves. They are taking a turn. If the huddle did not rotate, its edge would freeze and the structure would fail. The mechanism only works because exposure is temporary for everyone.

The temperature gradient inside is what makes the arrangement worth the discomfort. Measurements taken within huddles have recorded temperatures

rising well above freezing — in places above twenty degrees — while the air a few metres away sits at forty below. The heat is not generated for the group; it is the birds' own metabolic waste, retained because the packing is tight enough to stop it escaping. Nobody is being generous. The structure simply refuses to let anything leave.

The energetics explain why they tolerate the crush at all. A male emperor fasts for around four months while incubating, losing a substantial share of his body mass, and huddling reduces his metabolic rate enough to make the difference between arriving at hatching with reserves and not arriving at all. This is not comfort-seeking. It is the only version of the winter that ends with a chick, which is why birds tolerate a density that would otherwise be intolerable.

The organizational version is uncomfortable in a specific way. Most companies do have people on the cold edge during a crisis — support staff facing angry customers, the night shift, the team absorbing the consequences of someone else's decision. What they usually lack is the rotation. Mutual support becomes more valuable as conditions get harsher, but only if the position on the edge is genuinely temporary and everyone eventually stands there.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Who has been on the cold edge of your organization for too long?*

KEY TAKEAWAYS

1. Cooperation makes survivable what no individual could endure alone.

2. The huddle works because exposure rotates; a static structure would fail at its edge.
3. Support systems must move people out of the hardest position, not merely thank them.

SUMMARY

Emperor penguins survive the Antarctic winter by huddling, raising internal temperatures far above the lethal outside air, with travelling waves slowly rotating individuals between the exposed edge and the warm centre. Nobody stays on the outside permanently. Mutual support in harsh conditions depends on that rotation being real.

SCIENCE CHECK

ESTABLISHED

Huddles raise internal temperature far above the ambient air, and travelling waves rotate individuals through the structure.

INFERRED

That mutual support becomes more valuable in harsh conditions, provided exposure genuinely rotates.

UNCERTAIN

Penguin rotation is a physical consequence of crowding. Human rotation requires somebody with authority to insist on it.

VOCABULARY & PHRASAL VERBS

huddle (n./v.) a tightly packed group; to crowd together for warmth. The huddle raises the internal temperature above freezing.

lethal (adj.) causing death. Outside the group the conditions are lethal.

windward (adj.) facing the direction the wind comes from. Birds on the windward edge take the full force.

to churn (v.) to move around continuously and irregularly. The mass churns slowly and everyone moves through the centre.

exposure (n.) the state of being unprotected from harsh conditions.

Exposure has to be temporary or the structure fails.

shift (n.) a period of work. Somebody has covered the night shift for eight months.

pack together (phrasal verb) to press closely into a small space. They pack together tightly enough to trap their own heat.

take a turn (phrase) to do something in rotation with others. Everyone takes a turn on the outside.

hold out (phrasal verb) to survive or resist for a period of time. The males hold out for four months without food.

PRACTICE

Where the Analogy Breaks

Penguins rotate automatically through physical crowding. Human rotation requires someone with authority to insist on it.

Use the Language

Make sentences using these three words: exposure, shift and hold out.

Discuss

Who has been on the exposed edge of your organization longest, and when do they come inside?

Write

Design a rotation for the hardest position in your team, and state what makes it hard to implement.

SOURCES

A Gilbert, C. et al., "Huddling behaviour in emperor penguins: dynamics of huddling", Physiology & Behaviour (2006).

A Zitterbart, D. P. et al., "Coordinated movements prevent jamming in an emperor penguin huddle", PLOS ONE (2011).

THIS CONNECTS TO

Story 9: Dolphins: Roles No One Plays Alone SAME MECHANISM

Both show cooperation producing an outcome — a capability, or survivable conditions — that no individual could achieve alone, through coordinated role-sharing rather than individual effort.

Story 11: Geese: Why the Lead Position Rotates SAME MECHANISM

Both describe a costly position or condition being shared through rotation, spreading a burden that would exhaust or disadvantage any single individual permanently occupying it.

Story 43: Mangroves: Making Their Own Ground LIMITATION

Penguins survive extreme cold by continuously rotating individuals through the harshest position. Mangroves survive a harsh environment through a structural, not behavioural, solution — a fixed physical design that absorbs stress rather than a rotating group behaviour.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Emperor penguins breeding through the Antarctic winter survive temperatures and wind chill that would be lethal to an isolated individual, by forming a huddle that continuously rotates its members.

What is the mechanism?

Penguins on the exposed outer edge of the huddle experience the most severe cold and wind exposure, and periodically move toward the interior as other individuals rotate outward to take their place, so no single penguin remains in the harshest position long enough to suffer disproportionate cold exposure, and the group's average survival is far higher than any individual's would be alone.

What organizational problem does it explain?

Does your organization have a mechanism for rotating people out of its most exposed, costly positions, or does the same person or team absorb disproportionate strain indefinitely because rotation was never built in?

Level 2 — Make It Your Story

> *Tell me about a time your team deliberately rotated people through a particularly difficult or exposed responsibility to prevent burnout.*

Use STAR to answer. Think about: a high-pressure client relationship shared across a team, an on-call rotation, a difficult project rotated rather than left with one person.

TAKE THE IDEA FURTHER

> *Who in your organization is currently in the huddle's outer edge — absorbing disproportionate strain — with no rotation mechanism to bring them back to the middle?*

24

Fungi: The Network Under the Forest

> **Evidence Level 2**
> **Biological claim:** Contested — the mechanism is real, its extent is debated
> **Business application:** Analogy

Beneath a forest floor lies a network of fungal threads called mycorrhizae, which connect to tree roots and to each other. The relationship is a trade: the fungus supplies water and mineral nutrients such as phosphorus that its enormous thread network can reach, and the tree supplies sugars produced by photosynthesis, which the fungus cannot make. Neither partner can be replaced by effort. Each simply has access to something the other does not.

Research using radioactive and stable isotope tracers has shown that carbon can move between trees through these fungal connections, including between different species, and that transfer can run towards seedlings growing in deep shade. The interpretation is actively debated — how much moves, whether it benefits the recipient tree, and whether the fungus is best understood as a cooperative highway or as an organism managing its own interests. It is a good example of a real finding whose meaning is still contested.

The scale of the thing is easy to underestimate. A single gram of forest soil can contain hundreds of metres of fungal filament, and the network's total surface area for absorbing water and phosphorus dwarfs what the tree's own roots could ever reach. That is the basis of the trade: the tree can photosynthesize and the fungus cannot; the fungus can explore soil at a resolution the root cannot approach. Neither partner is being generous. Each is buying access to a capability it has no way to build.

What is not in doubt is the structural point. The forest's resource distribution does not run through the largest tree. It runs through a network layer that belongs to neither party and connects everyone.

The relationship also has a policing problem, which is the part most business writing about networks omits. Not every fungal partner delivers, and not every tree pays. Experimental work has shown that plants can direct more carbon towards fungal partners that supply more phosphorus, and that fungi preferentially move nutrients towards the roots that reward them — a mutual discipline that keeps the trade honest. A network without that discipline does not stay cooperative for long.

It is also worth noting who pays. The tree hands over a substantial fraction of the sugar it fixes — estimates commonly put it at up to around a fifth of photosynthetic output — for a service it cannot perform itself. Organizations rarely account for their network layer at all, which is why the coordinator who keeps three departments talking is usually classified as overhead, and is the first role cut when overhead is under review.

Organizations have this layer too, and rarely acknowledge it. The people who actually move information between departments are frequently not senior. They are the long-serving coordinator, the person everyone asks, the one who knows who to call. Reorganize without seeing that layer and you will cut connections that no chart records.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Who actually moves information between departments here — and would a reorganization notice them?*

KEY TAKEAWAYS

1. Mycorrhizal networks trade resources between fungi and trees that neither could obtain alone.
2. Resources and information can move through a network layer that belongs to no single member.
3. Informal connectors carry organizational flow that formal charts do not show.

SUMMARY

Fungal networks in soil exchange nutrients with tree roots and can transfer carbon between trees, though the extent and interpretation remain scientifically debated. The forest's distribution system is a shared network rather than a hierarchy. Organizations similarly depend on informal connectors who move resources and information across formal boundaries.

VOCABULARY & PHRASAL VERBS

mycorrhiza / mycorrhizae (n.) a symbiotic association between fungus and plant roots. Mycorrhizae connect the roots of different trees.

nutrient (n.) a substance that provides nourishment. The fungus supplies nutrients the root cannot reach.

seedling (n.) a very young plant. Carbon can move towards seedlings growing in deep shade.

tracer (n.) a substance used to track movement through a system. Isotope tracers showed where the carbon went.

contested (adj.) disputed; not agreed upon. The interpretation is still contested among researchers.

symbiosis (n.) a close relationship between two different organisms. The symbiosis works because neither partner can do the other's job.

hook up (phrasal verb) to connect. New roots hook up to the existing network.

draw on (phrasal verb) to use a resource or supply. Young trees draw on a network they did not build.

cut off (phrasal verb) to separate or disconnect. A reorganization can cut off connections no chart records.

PRACTICE

Where the Analogy Breaks

Trees do not choose their fungal partners strategically. Human connectors decide who to help, and that discretion is itself a form of power.

Use the Language

Make sentences using these three words: symbiosis, draw on and cut off.

Discuss

Who moves information between departments here, and how are they classified on the org chart?

Write

Draw your organization's real network layer: who talks to whom. Compare it with the formal structure.

SOURCES

A Simard, S. W. et al., "Net transfer of carbon between ectomycorrhizal tree species in the field", Nature 388, 579–582 (1997).

A Karst, J., Jones, M. D. & Hoeksema, J. D., "Positive citation bias and overinterpreted results lead to misinformation on common mycorrhizal networks in forests", Nature Ecology & Evolution (2023).

THIS CONNECTS TO

Story 42: Coral Reefs: Abundance From Poor Water SAME MECHANISM
Both describe a partnership between two different organisms that trades complementary capabilities, producing a result — nutrient access, reef abundance — that neither partner could achieve alone.

Story 41: Forests: Competition and Shelter SAME MECHANISM
Both look at forests as systems that are simultaneously competitive and cooperative, possibly exchanging resources through shared fungal networks even while individual trees compete for light.

Story 8: Slime Mould: Build Wide, Then Prune LIMITATION
Mycorrhizal networks are a standing, persistent infrastructure that trees rely on continuously. Slime mould builds and prunes its network actively in response to a specific, temporary foraging problem — a different relationship between a network and the traffic it carries.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Beneath a forest floor, a network of fungal threads called mycorrhizae connects to tree roots and to each other, trading resources between the fungus and multiple trees, including transferring carbon between different tree species.

What is the mechanism?

The fungus supplies water and mineral nutrients its enormous thread network can reach far more efficiently than a tree's own roots, while the tree supplies sugars from photosynthesis the fungus cannot produce itself; research has shown that plants direct more resources toward fungal partners that deliver more in return, and fungi do the same toward more generous roots, keeping the exchange broadly reciprocal rather than purely altruistic.

What organizational problem does it explain?

Does your organization recognize and account for its informal network layer — the people who move information and resources between departments — or does it get invisibly cut whenever a reorganization only looks at the formal chart?

Level 2 — Make It Your Story

> *Tell me about a time an informal connector or network in your organization made cooperation possible in a way the formal structure didn't capture.*

Use STAR to answer. Think about: the person everyone actually asks, informal information-sharing that outperformed official channels, a reorganization that accidentally cut a valuable connection.

TAKE THE IDEA FURTHER

> *Who in your organization actually moves information and resources between departments — and would a reorganization notice and protect that role, or accidentally cut it?*

25

Forests: Spare Parts That Differ

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Ecologists use the term functional redundancy for a situation where several different species perform broadly the same job in an ecosystem — several pollinators visiting the same flowers, several decomposers breaking down the same litter. For a long time this looked wasteful. Why should an ecosystem support five species doing one job?

The answer is that they do not respond identically to disturbance. One pollinator may be sensitive to drought, another to cold, another to a particular disease. When conditions shift and one declines, the others compensate and the function continues. This is described as the insurance hypothesis in biodiversity research, and the difference in how species respond is precisely what provides the insurance. Systems that lost their redundancy — where a single species carries a whole function — collapse abruptly when that species is hit, and history offers hard examples: the Irish potato famine followed from near-total dependence on a narrow genetic base, and banana cultivation has twice faced catastrophe for the same reason.

The insurance is only real if the members respond differently, and that condition can quietly fail. Plant several varieties of the same crop and you have apparent redundancy; if all of them share a single resistance gene, one pathogen removes the entire set at once. This is precisely what happened to the Gros Michel banana, and it is why the Cavendish that replaced it — a clone, grown worldwide — is now facing a soil fungus with no genetic

variation to fall back on. Duplication that shares a hidden dependency is not redundancy. It is one thing wearing several names.

Redundancy is therefore not inefficiency. It is the price of continuity, and it is only visible as a cost until the day it is the only thing that works.

There is a version of this that organizations actively create. Standardization is usually justified as efficiency, and it does deliver efficiency — while quietly converting a set of different local practices into a single practice with one failure mode. Every team using the same tool, the same template and the same supplier is a system with excellent margins and no functional redundancy at all. The saving is real. So is what was sold to obtain it, and it is rarely written down anywhere.

The same hidden dependency shows up in companies as a shared assumption rather than a shared gene. Three suppliers who all ship through one port; four analysts trained by the same person, using the same model, on the same data. The test is not how many alternatives you have. It is whether any single event could remove all of them, and that question is answered by looking for what they have in common rather than by counting them.

Every organization has single points of failure, and they are usually people rather than machines: one person who can operate the legacy system, one who holds the key client relationship, one who understands why the process exists. Efficiency drives will find them, because they look exactly like duplication on a spreadsheet. The test is simple and worth running annually: if this person were unavailable for a month, what would stop?

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *If one person were unavailable for a month, what would stop?*

KEY TAKEAWAYS

1. Functional redundancy means several members can perform the same essential function.
2. Redundancy provides insurance because members respond differently to disturbance.
3. Single points of failure look like efficiency until the moment they fail.

SUMMARY

Ecosystems maintain several species capable of the same function, and because those species respond differently to stress, the function survives disturbances that remove any one of them. Systems without this redundancy collapse suddenly. Organizations should treat single points of knowledge or capability as vulnerabilities rather than as savings.

SCIENCE CHECK

ESTABLISHED

Ecosystems maintain several species performing similar functions, and the insurance depends on their responding differently to stress.

INFERRED

That single points of failure in organizations are vulnerabilities rather than savings.

UNCERTAIN

Whether apparent organizational redundancy is real: shared tools, training and suppliers can hide a single failure mode behind several names.

VOCABULARY & PHRASAL VERBS

redundancy (n.) here: duplication that provides backup capacity. Redundancy is the price of continuity.

pollinator / decomposer (n.) an organism that fertilizes plants / breaks down dead matter. Several pollinators do broadly the same job.

disturbance (n.) an event that disrupts a system. The species respond differently to the same disturbance.

to compensate (v.) to make up for a loss. When one declines, the others compensate.

single point of failure (n.) one element whose failure stops the whole system. One person who understands the system is a single point of failure.

legacy system (n.) old technology still in use. Only he can operate the legacy system.

break down (phrasal verb) to stop working; to decompose. The whole process breaks down when she is away.

make up for (phrasal verb) to compensate for something missing. Nothing else makes up for losing that knowledge.

carry out (phrasal verb) to perform a function or task. Several species can carry out the same function.

PRACTICE

Where the Analogy Breaks

Species redundancy costs an ecosystem nothing in salary. Spare human capacity has a visible price and an invisible benefit, which is why it loses budget arguments.

Use the Language

Make sentences using these three words: break down, make up for and redundancy.

Discuss

What do your alternatives have in common? Could one event remove all of them at once?

Write

Write 120 words identifying a hidden shared dependency in something your team treats as redundant.

SOURCES

A Yachi, S. & Loreau, M., "Biodiversity and ecosystem productivity in a fluctuating environment: the insurance hypothesis", PNAS 96, 1463–1468 (1999).

A Walker, B., "Biodiversity and ecological redundancy", Conservation Biology (1992).

THE CONTRAST

Story 25: Forests: Spare Parts That Differ vs. Story 8: Slime Mould: Build Wide, Then Prune

Story 25: Forests: Spare Parts That Differ: Redundancy — several different species capable of the same essential function — should be deliberately preserved as insurance against a future that hasn't happened yet.

Story 8: Slime Mould: Build Wide, Then Prune: Redundancy that isn't currently being used should be pruned away once it's confirmed unnecessary, in favour of efficiency.

Is unused redundancy waste to be pruned, or insurance to be deliberately preserved?

THIS CONNECTS TO

Story 14: Cockroaches: A Mixture of Temperaments SAME MECHANISM
Both cases show that a mixture of different individual types — temperaments, or species performing the same function — outperforms a uniform group when conditions are unpredictable.

Story 8: Slime Mould: Build Wide, Then Prune LIMITATION
Slime mould prunes away genuine redundancy once it's confirmed unnecessary. Forests: Spare Parts That Differ makes the opposite case — some redundancy should be deliberately preserved, because it protects against a future that hasn't happened yet.

Story 42: Coral Reefs: Abundance From Poor Water LIMITATION
Functional redundancy protects a forest against an unpredictable future stress. Coral reef abundance depends on a single, non-substitutable partnership, which means it has far less of this kind of protection when conditions change.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A resilient forest typically contains multiple different species capable of performing the same core ecological function — several species that can each

act as the dominant canopy tree, for instance — even though, at any given time, only one or two of them may be common.

What is the mechanism?

Ecologists call this functional redundancy: because different species performing the same function respond differently to disease, pests, drought or fire, a stress that devastates one species leaves the function intact, since another functionally similar species can expand to fill the gap; a forest of a single dominant species has no such backup if that species is hit.

What organizational problem does it explain?

Is your organization treating apparent duplication — multiple people or systems capable of the same function — as pure inefficiency to eliminate, without accounting for what happens if the one you keep is the one that fails?

Level 2 — Make It Your Story

> *Tell me about a time deliberately maintained redundancy — a backup system, a second person who could do a critical job — protected your organization when the primary option failed.*

Use STAR to answer. Think about: cross-training, a backup vendor you almost cut, a single point of failure you noticed too late or just in time.

TAKE THE IDEA FURTHER

> *Where has your organization recently eliminated apparent duplication in the name of efficiency, without asking what would happen if the single remaining option failed?*

26

Rivers: Around, Not Through

> **Evidence Level 3**
> **Biological claim:** Descriptive or established
> **Business application:** Metaphor / inference

A river has one non-negotiable goal — the sea — and no attachment whatsoever to the route. Put a boulder in its path and it goes around. Put a ridge of hard rock across it and it turns sideways and follows the softer stone until it finds a way through. Block it entirely and it fills the space, rises, and leaves over the lowest point available. Over long periods it will cut through the obstacle, but it never waits for that to happen before continuing.

Geologists distinguish the erosion that eventually cuts a gorge from the immediate rerouting that happens in the meantime — and the rerouting is what keeps the river a river. Water does not argue with granite. It applies constant pressure in every available direction and moves through whichever one yields.

The geology behind the metaphor is worth stating properly, because it contains the interesting part. A river's course is decided by the rock it is cutting through: it follows fault lines, joints and softer beds, so its path is not a compromise between water and stone but a map of where the stone is weakest. Where a channel does cut straight through hard rock, it is often because the river was there first and the ground rose slowly beneath it — the water kept its position and sawed downwards, producing gorges that look like brute force and are actually persistence at geological speed.

This is an analogy, not a scientific finding about business, and it is worth saying so plainly. But it captures something people persistently confuse: the

difference between a goal and a plan. When a strategy fails, the emotional reaction is to treat it as though the destination has been lost. Usually it hasn't. What has been lost is one route.

It is also worth noticing what a river does with the material it removes. Nothing is discarded: sediment eroded from an outer bend is deposited on the inner bend a little downstream, which is why meanders migrate slowly across a floodplain rather than simply deepening. The obstacle is not defeated and it is not avoided, exactly — it is disassembled and used to build the next stretch of the route.

The straightening example is worth carrying further, because engineers have run this experiment on a large scale. Channelizing a river shortens its path, steepens its gradient and speeds the flow, which moves water off the immediate site efficiently — and delivers it downstream faster, higher and with more erosive force than the banks there evolved to take. Many countries now spend considerable sums putting meanders back. The lesson is not that speed is bad. It is that removing the mechanism by which a system dissipates energy relocates the energy rather than removing it.

The practical discipline is to write down, before you begin, which part of an initiative is the destination and which is merely the path. Teams that have not made that distinction in advance tend to defend the path — because in the middle of a failure, the path is the thing that feels like the commitment.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What are you defending — the destination, or one route to it?*

KEY TAKEAWAYS

1. A river is committed to its destination and completely uncommitted to its route.
2. Constant pressure in many directions finds the yielding point faster than force in one.
3. Distinguish the goal from the plan before difficulty makes them feel identical.

SUMMARY

Rivers reach the sea by continuously rerouting around obstacles rather than confronting them, eroding hard rock only over much longer timescales. The destination is fixed; the path never is. As an analogy, it argues for changing the route when a strategy fails while keeping the objective intact.

SCIENCE CHECK

ESTABLISHED

River courses follow zones of weakness in the underlying rock, and channelization moves flood energy downstream rather than removing it.

INFERRED

That a failing strategy calls for a change of route rather than a change of destination.

UNCERTAIN

This is an analogy throughout. Rivers have no ego, no sunk costs and no need to explain the change of route to a board.

VOCABULARY & PHRASAL VERBS

boulder (n.) a very large rock. The river goes around the boulder without slowing much.

ridge (n.) a long, narrow raised area of land. A ridge of hard rock turned the channel sideways.

erosion (n.) the gradual wearing away of rock or soil. Erosion cuts the gorge over geological time.

gorge (n.) a deep, narrow valley with steep sides. The gorge looks like force and is actually persistence.

to yield (v.) to give way under pressure. Water finds whichever direction yields.

non-negotiable (adj.) not open to discussion or change. The destination is non-negotiable; the route never is.

go around (phrasal verb) to avoid by taking a different route. When the strategy failed, we went around rather than stopping.

give way (phrasal verb) to collapse or stop resisting. The softer bank gave way first.

stick to (phrasal verb) to continue with something without changing. They stuck to the plan long after the goal had moved.

back off (phrasal verb) to retreat or withdraw. Backing off a route is not abandoning the objective.

PRACTICE

Where the Analogy Breaks

A river has one goal and no ego. Humans confuse the plan with the commitment because abandoning a route looks like admitting error.

Use the Language

Make sentences using these three words: yield, go around and stick to.

Discuss

What are you defending — the destination, or one route to it?

Write

Take a current initiative and write two lists: what is the goal, and what is merely the plan.

SOURCES

B Whipple, K. X., "Bedrock rivers and the geomorphology of active orogens", Annual Review of Earth and Planetary Sciences (2004). · Presented here as a Level 3 analogy: the geology is factual; the business application is metaphorical.

THE CONTRAST

Story 26: Rivers: Around, Not Through vs. Story 27: Wildfires: Renewal That Needs Burning

Story 26: Rivers: Around, Not Through: Persistence toward a goal means total flexibility about the specific route, going around any obstacle rather than confronting it.

Story 27: Wildfires: Renewal That Needs Burning: Persistence toward renewal means depending on disruption directly — periodic destruction is a required part of the system, not something to route around.

Is resilience better achieved by flexibly avoiding every obstacle, or by depending on periodic disruption as part of the system's normal function?

THIS CONNECTS TO

Story 43: Mangroves: Making Their Own Ground LIMITATION
A river avoids obstacles by finding the path of least resistance around them. Mangroves take the opposite approach to a hostile environment — building a permeable structure that absorbs and dissipates force directly, rather than avoiding it.

Story 3: Locusts: Why Alignment Resists Change LIMITATION
A river is completely committed to its destination while remaining totally flexible about its route. Locust swarms show the opposite failure mode — a route or direction becoming so fixed that it can't be abandoned even when it should be.

Story 49: Predators: The Cost Is in the Search SAME MECHANISM
Both describe a system optimizing for the goal itself rather than for any particular fixed method of reaching it, abandoning a specific approach the moment it stops being the cheapest path forward.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A river has one fixed commitment — reaching lower ground, ultimately the

sea — and no attachment whatsoever to any particular route; when it meets an obstacle, it simply finds the path of least resistance around it rather than expending energy trying to move through or over it.

What is the mechanism?

Water always follows the path of least resistance available to it at any given point, meaning a river's specific course is entirely a product of local terrain rather than any commitment to a predetermined path; over time this produces meandering, braided, or diverted channels that look inefficient locally but represent the genuinely lowest-energy route to the fixed destination.

What organizational problem does it explain?

Is your organization staying committed to a specific method or route toward a goal, when total commitment to the destination and total flexibility about the path would actually get there with less wasted effort?

Level 2 — Make It Your Story

> *Tell me about a time you stayed committed to a goal while being willing to completely abandon your original planned route to get there.*

Use STAR to answer. Think about: a strategy pivot that kept the same underlying objective, an obstacle you routed around rather than pushed through.

TAKE THE IDEA FURTHER

> *Is your organization currently expending energy pushing directly through an obstacle, when going around it would reach the same destination with far less resistance?*

27

Wildfires: Renewal That Needs Burning

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Some plants do not merely survive fire. They require it. The cones of certain pines are sealed with resin that melts only at high temperature, so seeds are released precisely when fire has cleared the competing canopy and left a bed of nutrient-rich ash — a strategy called serotiny. Several Australian and South African species need smoke or heat to trigger germination at all. Fire, for these plants, is not a disaster. It is the reproductive signal.

The complication is important, and it is where the business analogy is usually abused. These systems are adapted to a particular fire regime — a characteristic frequency and intensity. Suppressing fire for decades allows fuel to accumulate, and the eventual fire burns hotter than anything the ecosystem evolved with, killing seed banks and mature trees that low-intensity fire would have spared. Too much fire is equally destructive. The adaptation is to a range, not to fire in general.

That nuance is the actual lesson. It is fashionable to say organizations should "embrace disruption", and mostly this is said by people not being disrupted. The honest version is narrower: regular small disruptions prevent catastrophic large ones. Ecosystems where nothing ever burns are not stable; they are accumulating fuel.

The management history is instructive. Fire suppression was pursued in North America and Australia for most of the twentieth century with real conviction, and it worked in the short term — which is exactly how the fuel

accumulated. The fires that eventually arrived burned hotter than the systems had evolved with, killing mature trees and seed banks that low-intensity fire would have spared. Agencies now deliberately light fires under controlled conditions, and the political difficulty of that policy is entirely familiar: the cost of the small fire is visible, immediate and attributable, and the cost of the large one is hypothetical until it is not.

Practically, that argues for deliberately introduced small shocks — retiring a product before it dies, restructuring a team before it calcifies, running the failure drill — on the grounds that the alternative is not calm. The alternative is a much larger fire later, at a time you do not choose.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What small disruption could you introduce now to prevent a large one later?*

KEY TAKEAWAYS

1. Some ecosystems depend on fire for regeneration, with seeds released or triggered by heat.
2. Adaptation is to a specific fire regime; suppression creates catastrophic fires later.
3. Regular small disruptions prevent the accumulation that makes large ones destructive.

SUMMARY

Serotinous plants release or germinate seeds only after fire, making periodic burning essential rather than merely survivable — but only within the frequency and intensity they evolved with. Suppressing fire builds fuel for a far more destructive event. Organizations similarly benefit from deliberate small disruptions rather than the illusion of permanent stability.

SCIENCE CHECK

ESTABLISHED

Serotinous species require fire to release or germinate seed, and suppression produces fires more severe than the system evolved with.

INFERRED

That regular small disruptions prevent catastrophic ones.

UNCERTAIN

Systems are adapted to a specific frequency and intensity; too-frequent fire is as destructive as suppression, and 'embrace disruption' states neither.

VOCABULARY & PHRASAL VERBS

serotiny (n.) the retention of seeds until an environmental trigger, usually fire. Serotiny holds the seeds until fire releases them.

resin (n.) a sticky substance produced by some trees. The resin melts only at temperatures reached in fire.

germination (n.) the beginning of growth from a seed. Smoke chemicals trigger germination in some species.

fire regime (n.) the typical pattern of fire frequency and intensity in an area. Each system is adapted to a particular fire regime.

to suppress (v.) to prevent or stop something happening. Suppressing every small fire builds fuel for a large one.

to calcify (v.) to become rigid and unable to change. Teams calcify when nothing is ever disturbed.

build up (phrasal verb) to accumulate. Fuel builds up in the years when nothing burns.

wipe out (phrasal verb) to destroy completely. An intense fire can wipe out the seed bank entirely.

clear out (phrasal verb) to remove everything from a space. The fire clears out the competing canopy.

PRACTICE

Where the Analogy Breaks

Fire regimes are measurable and repeatable. 'Embracing disruption' in business is usually recommended by people who will not absorb it.

Use the Language

Make sentences using these three words: suppress, calcify and build up.

Discuss

What small disruption could you introduce deliberately, before a large one arrives?

Write

Write 120 words arguing for retiring something that still works, and be honest about who would object.

SOURCES

B Keeley, J. E. et al., Fire in Mediterranean Ecosystems: Ecology, Evolution and Management, Cambridge University Press (2012).

C The Biomimicry Institute, Research & Resources (including wildfire adaptation work). https://biomimicry.org/research-resources/

THE CONTRAST

Story 27: Wildfires: Renewal That Needs Burning vs. Story 26: Rivers: Around, Not Through

Story 27: Wildfires: Renewal That Needs Burning: Persistence toward renewal means depending on disruption directly — periodic destruction is a required part of the system, not something to route around.

Story 26: Rivers: Around, Not Through: Persistence toward a goal means total flexibility about the specific route, going around any obstacle rather than confronting it.

Is resilience better achieved by depending on periodic disruption as part of the system's normal function, or by flexibly avoiding every obstacle?

THIS CONNECTS TO

Story 19: Axolotls: Regrowth Instead of Scar SAME MECHANISM

Both show a system whose recovery process doesn't just restore the previous

state but depends on disruption itself as part of how renewal actually happens.

Story 26: Rivers: Around, Not Through LIMITATION
Wildfire-adapted ecosystems depend directly on periodic destruction as part of their normal function. Rivers show the opposite relationship to obstacles — flexibly avoiding them entirely rather than depending on confronting them.

Story 52: Cicadas: Built for One Loud Season LIMITATION
Wildfire ecosystems depend on periodic, externally-triggered disruption for renewal. Cicadas depend on an internally-timed cycle of preparation and release — both involve a long buildup followed by a triggering event, but one is imposed from outside and the other is scheduled from within.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Some forest ecosystems, and specific tree species within them, do not merely survive wildfire — they structurally depend on periodic fire for regeneration, and suppressing fire entirely can leave the ecosystem worse off in the long run.

What is the mechanism?

Certain pine species produce serotinous cones, sealed with resin that only melts and releases seeds at fire temperatures, timing seed release precisely to moments when fire has cleared competing vegetation and deposited nutrient-rich ash — meaning the destruction is not merely tolerated but functions as the actual trigger and nutrient source for the next generation.

What organizational problem does it explain?

Is your organization suppressing all disruption in the name of stability, when a certain amount of periodic, controlled disruption might actually be the mechanism that clears out what's no longer working and enables genuine renewal?

Level 2 — Make It Your Story

> *Tell me about a time a period of real disruption in your organization turned out to be necessary for genuine renewal, not just something to be endured and minimized.*

Use STAR to answer. Think about: a reorganization that cleared out accumulated dead weight, a difficult reset that enabled real change, disruption you initially resisted but later saw the value of.

TAKE THE IDEA FURTHER

> *Is your organization suppressing all disruption in the name of stability, in a way that might be preventing the periodic renewal that a healthy system actually needs?*

PART IV

Innovation & Design

How does nature solve problems?

28

Humpback Whales: The Bumpy Edge

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

A humpback whale is about fifteen metres long and weighs thirty tonnes, yet it can turn inside a circle a few metres across to trap fish. For a long time nobody asked why the leading edge of its flipper is covered in bumps. Every engineering principle said a leading edge should be smooth.

The biologist Frank Fish noticed the bumps — called tubercles — on a sculpture in a gift shop, assumed the artist had made a mistake, and discovered that the whale really is built that way. Wind tunnel testing of model flippers with and without tubercles showed that the bumped version delayed stall to a steeper angle of attack and improved lift characteristics. The tubercles channel the flow into streams that stay attached to the surface at angles where a smooth wing would lose lift entirely.

The wind tunnel work is worth describing because it shows what "tested" means. Two model flippers were built, identical except that one carried the bumps, and both were run through a range of angles while lift and drag were measured. The smooth model behaved as textbooks predict: lift rises with angle until the airflow separates from the upper surface, at which point it collapses. The bumped model kept generating lift several degrees past that point. The tubercles split the oncoming flow into streams that stay attached where a smooth surface has already given up.

The application became commercial. Tubercle-inspired leading edges have been applied to fan blades and turbine designs, and the World Economic

Forum cites humpback tubercles as an example of nature-inspired engineering.

There is a limit worth naming honestly. Tubercles improve performance at high angles of attack and can slightly reduce efficiency at low ones, which is exactly why they have not appeared on aircraft wings, which spend most of their life cruising rather than manoeuvring at the edge of stall. Copying a feature without copying the operating regime it was built for produces a worse design, not a better one — a caution that applies equally to a company adopting a competitor's practice without asking what problem that practice was actually solving for them.

The business lesson is about where innovation lives. The whale did not need a new material, a new energy source or a new category of machine. It needed a modified edge. Companies searching for transformation frequently overlook the equivalent — the small geometric change at the point where their system meets the world, which alters performance far more than another strategic review would.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where does your product or service actually meet the customer — and when did you last change that edge?*

KEY TAKEAWAYS

1. Bumps on humpback flippers delay stall and improve performance at steep angles.

2. The insight came from questioning an assumption everyone treated as settled.
3. Large performance gains can come from small changes at the right point in a system.

SUMMARY

Humpback whales have tubercles along their flippers that improve lift and delay stall, contradicting the engineering assumption that leading edges must be smooth. Wind tunnel work confirmed the effect and it has been applied to fans and turbines. Innovation often means modifying one critical edge rather than redesigning the whole system.

SCIENCE CHECK

ESTABLISHED

Tubercles measurably delay stall and improve lift at steep angles in wind-tunnel testing.

INFERRED

That large performance gains can come from a small change at the point of contact with the environment.

UNCERTAIN

The benefit is specific to high-angle, low-speed manoeuvring — which is why the design has not transferred to aircraft, which mostly cruise.

VOCABULARY & PHRASAL VERBS

tubercle (n.) a small rounded bump on a surface. The tubercles delay stall at steep angles.

flipper (n.) the flat limb of a sea mammal used for swimming. The flipper carries bumps along its leading edge.

leading edge (n.) the front edge of a wing or blade. The leading edge is the part that meets the airflow first.

to stall (v.) to lose lift because airflow separates from the wing. A smooth wing stalls sooner than a bumped one.

angle of attack (n.) the angle between a wing and the oncoming airflow.

Lift holds several degrees past the usual angle of attack.

turbine (n.) a machine that converts moving fluid into rotation. The design was applied to fan and turbine blades.

come across (phrasal verb) to find something by chance. He came across the bumps on a gift-shop sculpture.

look into (phrasal verb) to investigate. Engineers looked into why the shape worked at all.

overlook (v.) to fail to notice something. The obvious small change is easy to overlook.

PRACTICE

Where the Analogy Breaks

A flipper has one function. Business 'edges' serve several purposes at once, so a change that helps one may damage another.

Use the Language

Make sentences using these three words: leading edge, come across and look into.

Discuss

Where does your product or service meet the customer, and when did you last examine that edge closely?

Write

Describe a small design change that had a disproportionate effect, and name the specific condition it worked under.

SOURCES

A Fish, F. E. & Battle, J. M., "Hydrodynamic design of the humpback whale flipper", Journal of Morphology 225, 51–60 (1995).

A Miklosovic, D. S. et al., "Leading-edge tubercles delay stall on humpback whale flippers", Physics of Fluids (2004).

C World Economic Forum (2026). https://www.weforum.org/stories/2026/04/nature-leadership-resilience-fractured-world/

THIS CONNECTS TO

Story 29: Kingfishers: The Beak and the Train SAME MECHANISM

Both are direct biomimicry cases where a specific biological structure was measured, then reproduced as an engineered solution to an unrelated technical problem.

Story 36: Fish: Riding the Vortex Behind SAME MECHANISM

Both show an aquatic animal exploiting fluid dynamics — turbulence around a structure, or a vortex left behind an obstacle — to gain a performance advantage a smooth or naive design wouldn't achieve.

Story 30: Geckos: Grip That Releases on Command SAME MECHANISM

Both are engineering solutions borrowed directly from a specific biological surface structure — whale flipper bumps, gecko footpads — that outperforms the smooth or simple version most designers would default to.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Humpback whale flippers have large, irregular bumps called tubercles along their leading edge, rather than the smooth edge that conventional aerodynamic and hydrodynamic design would predict as optimal.

What is the mechanism?

Wind tunnel and water tank testing has shown that the tubercles delay flow separation (stall) and allow the flipper to maintain lift at steeper angles of attack than a smooth-edged equivalent, by channelling flow into organized vortices along the bumps rather than allowing it to separate chaotically — this discovery has since been applied to some wind turbine blades and fan designs.

What organizational problem does it explain?

Is your organization designing toward a conventionally 'clean' or 'smooth' solution by default, when a structured, deliberately irregular approach — bumps, not a smooth edge — might actually outperform it under real conditions?

Level 2 — Make It Your Story

> *Tell me about a time a deliberately imperfect or irregular approach outperformed the conventionally 'clean' solution most people would default to.*

Use STAR to answer. Think about: a messy process that worked better than the tidy one, an unconventional design choice that improved performance.

TAKE THE IDEA FURTHER

> *Where in your organization is a smooth, conventional-looking solution actually underperforming a rougher, more structured alternative nobody has seriously tested?*

29

Kingfishers: The Beak and the Train

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

When Japan's Shinkansen trains entered tunnels at high speed, they pushed a wall of air ahead of them. The pressure wave compressed, reached the far end and burst out as a loud boom that could be heard hundreds of metres away, disturbing residents and constraining operating speed.

The physics of the original problem is well documented independently of the bird. A train entering a tunnel compresses the air ahead of it; that compression steepens into a pressure wave which travels the length of the tunnel at the speed of sound and radiates from the far portal as a low-frequency boom. The effect scales steeply with speed, which is why it constrained the Shinkansen and not conventional railways, and the engineering fix is well understood: lengthen the nose so that the cross-section increases gradually and the air is displaced over a longer interval rather than all at once.

Eiji Nakatsu, an engineer on the project and a keen birdwatcher, had been thinking about how a kingfisher enters water. The bird dives from air into a much denser medium at speed and barely makes a splash, because its beak is long, narrow and wedge-shaped, so the pressure change is spread gradually along its length instead of arriving all at once. The train's blunt nose was doing exactly the opposite.

The redesigned 500 Series nose was substantially longer and shaped on the same principle. Reported results included reduced tunnel boom, and the design was also associated with lower energy use and higher speeds.

The story is worth one further correction, since this book asks readers to be careful about exactly this kind of claim. The aerodynamic benefit of the redesigned nose is documented in the peer-reviewed literature on high-speed rail; the specific attribution to the kingfisher rests substantially on Nakatsu's own account of his thinking, told after the fact. Both things can be true together — the physics is real, and the origin story is one engineer's memory of how he arrived at it — and a careful reader holds them apart rather than letting the more vivid one stand in for the more rigorous one.

The train borrowed from more than one animal, which is usually left out of the retelling. The pantograph — the arm that collects current from the overhead wire — was a major noise source, and its supporting structure was reshaped after the body of a penguin, while serrations modelled on the leading edge of owl feathers were added to break up the vortices that generate noise. None of this was a single flash of inspiration. It was an engineering team with a list of specific problems, going looking for organisms that had solved each one.

The transferable lesson is about the shape of the analogy. The kingfisher had nothing to do with railways. What it shared with the train was an abstract problem: passing between two media of different density without generating a shock. Solutions transfer across industries when you describe your problem abstractly enough to be recognizable somewhere else.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Can you state your current problem abstractly enough that another industry would recognize it?*

KEY TAKEAWAYS

1. The Shinkansen's tunnel boom was solved by copying the kingfisher's beak geometry.
2. The transfer worked because the underlying physical problem was the same.
3. Describing a problem abstractly lets you find solutions outside your own industry.

SUMMARY

Japanese engineers reduced the Shinkansen's tunnel boom by reshaping the train's nose after the beak of a kingfisher, which enters water with minimal splash. The design also brought efficiency and speed benefits. Cross-industry solutions become findable when a problem is described in abstract physical terms.

SCIENCE CHECK

ESTABLISHED

The redesigned nose measurably reduced the Shinkansen's tunnel micro-pressure wave.

INFERRED

That describing a problem abstractly enables solutions to transfer across industries.

UNCERTAIN

The kingfisher attribution rests mainly on the engineer's own account; the aerodynamics are independently documented, the origin story is not.

VOCABULARY & PHRASAL VERBS

pressure wave (n.) a moving region of compressed air. The pressure wave compressed and burst out as a boom.

blunt (adj.) not sharp or pointed. The original nose was too blunt for the speed.

wedge-shaped (adj.) thick at one end and narrowing to a thin edge. A wedge-shaped beak spreads the pressure change gradually.

medium / media (n.) a substance through which something travels. The kingfisher moves between two media of different density.

serration (n.) a row of small teeth-like points on an edge. Serrations on the pantograph reduced the noise.

splash (n.) the sound and water thrown up when something enters liquid. The kingfisher enters the water with barely a splash.

burst out (phrasal verb) to emerge suddenly and violently. The pressure wave burst out of the far tunnel portal.

draw on (phrasal verb) to use as a source of inspiration. The design drew on more than one bird.

come up against (phrasal verb) to encounter a problem or obstacle. Engineers came up against the same physical limit twice.

PRACTICE

Where the Analogy Breaks

Physics transfers across domains; management practice often does not, because context and culture are part of the mechanism.

Use the Language

Make sentences using these three words: medium, draw on and come up against.

Discuss

Can you describe your current problem abstractly enough that another industry would recognize it?

Write

Rewrite a work problem without any industry vocabulary, then note which fields might already have solved it.

SOURCES

B Raghunathan, R. S., Kim, H.-D. & Setoguchi, T., "Aerodynamics of high-speed railway train", Progress in Aerospace Sciences 38, 469–514 (2002) — peer-reviewed treatment of the tunnel micro-pressure wave and nose geometry.

C Nakatsu, E., engineering accounts of the 500 Series design, published by JR West and reproduced widely in the biomimicry literature. Note: the kingfisher attribution rests on the lead engineer's own testimony rather than on a controlled study. The aerodynamics are well established; the story of where the idea came from is a first-person account, and this book labels it as such.

THIS CONNECTS TO

Story 28: Humpback Whales: The Bumpy Edge SAME MECHANISM

Both are direct biomimicry cases where a specific biological structure was measured, then reproduced as an engineered solution to an unrelated technical problem.

Story 30: Geckos: Grip That Releases on Command SAME MECHANISM
Both show an engineer studying a specific biological structure closely enough to reproduce its underlying physical mechanism directly, rather than settling for a loose analogy.

Story 33: Beetles: Drinking From Fog SAME MECHANISM
Both are precise engineering solutions copied directly from a specific, measured biological surface or structure, applied to solve a real, unrelated technical problem.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Japan's Shinkansen high-speed trains originally produced a loud tunnel boom when exiting tunnels at speed, caused by a sudden pressure wave — a problem engineers solved by studying an unrelated animal entirely.

What is the mechanism?

Engineer Eiji Nakatsu, an avid birdwatcher, proposed redesigning the train's nose based on the kingfisher's beak, which allows the bird to enter water at high speed from air — a much denser medium transition — with minimal splash or resistance; the resulting elongated, tapered nose design smoothed the train's transition between the tunnel and open air, sharply reducing the tunnel boom while also reducing drag and cutting energy consumption.

What organizational problem does it explain?

Has your organization ever solved a difficult technical or process problem by

looking at a completely unrelated field or discipline for a structural solution that already exists there?

Level 2 — Make It Your Story

> *Tell me about a time a solution to a technical or process problem came from a completely unrelated field or discipline.*

Use STAR to answer. Think about: an idea borrowed from an unrelated industry, a hobby or outside interest that solved a work problem, a cross-disciplinary insight.

TAKE THE IDEA FURTHER

> *What unsolved technical or process problem in your organization might already have a working solution in a completely unrelated field, if someone thought to look there?*

30

Geckos: Grip That Releases on Command

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

A gecko can hang from a polished glass ceiling by one toe. It secretes no glue, and its feet are dry and leave no residue. For over two thousand years — Aristotle mentions the problem — nobody could explain it.

The answer is scale. Each gecko foot carries around half a million hair-like structures called setae, and each seta splits at its tip into hundreds of even smaller spatula-shaped endings a few hundred nanometres wide. At that scale, van der Waals forces — extremely weak attractions between molecules that are close enough — become significant. Any single contact is negligible. A billion of them together produce enough adhesion to hold many times the animal's weight. Direct measurements of the force of a single seta confirmed the mechanism.

Establishing the mechanism required measuring a single hair. Autumn's team mounted one seta on a sensor and found that its adhesion depended on direction — pulled at a shallow angle along the surface it held with surprising force, and lifted at a steeper angle it released almost entirely. That directionality is what rules out most competing explanations. Suction would not care about angle; friction alone would not produce those magnitudes; and glue would not switch off. What remains is intermolecular attraction across an enormous number of contacts, engaged and disengaged geometrically.

Just as important is the release. The gecko detaches simply by changing the angle of its toe, which peels the setae off. The adhesion is strong in one direction and almost absent in another, which is why the animal can run up a wall at speed rather than getting stuck to it. Gecko-inspired dry adhesives have since been developed for climbing robots and grippers, including work at NASA's Jet Propulsion Laboratory on gripping in space.

The scaling problem is worth knowing about too, because it explains why synthetic versions took decades to catch up. A gecko's weight is supported by roughly two billion contact points across its four feet, and reproducing that density in a manufactured material — with each contact soft enough to make real molecular contact and stiff enough not to collapse into its neighbours — turned out to be a harder engineering problem than describing the mechanism had been. Understanding how something works and being able to build it remain two separate achievements.

Gecko adhesion also has a limitation that the engineering literature respects and the motivational version ignores. It needs a reasonably smooth surface. On dusty or rough ground the setae cannot make enough contact and the animal's grip fails — which is why a gecko can hold a polished ceiling and struggles on loose sand. The mechanism is not universally strong. It is extraordinarily strong within conditions it requires, and knowing those conditions is part of knowing the mechanism.

The design lesson is worth stating carefully: the gecko does not achieve strong adhesion and then solve release as an afterthought. Attachment and detachment were designed together. Any commitment you cannot exit — a contract, a platform, a partnership — is only half-engineered.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which commitment could you not exit if you needed to?*

KEY TAKEAWAYS

1. Gecko adhesion comes from billions of tiny contacts using van der Waals forces, not glue.
2. Enormous aggregate effects can be built from individually negligible interactions.
3. Attachment is only well designed if detachment is designed at the same time.

SUMMARY

Geckos climb using millions of microscopic setae whose combined van der Waals interactions produce powerful dry adhesion, released instantly by changing toe angle. The mechanism has inspired reusable dry adhesives and robotic grippers. Its deeper lesson is that strong attachment must be engineered together with easy release.

SCIENCE CHECK

ESTABLISHED

Gecko adhesion depends on van der Waals forces across billions of directional contacts, confirmed by single-seta measurement.

INFERRED

That attachment and detachment should be designed together, and that commitments need a built-in exit.

UNCERTAIN

The mechanism requires a smooth surface and fails on dust or rough ground — a limit the popular retelling usually omits.

VOCABULARY & PHRASAL VERBS

adhesion (n.) the ability to stick to a surface. Adhesion depends on the angle of pull.

seta / setae (n.) a stiff hair-like structure. Each foot carries half a million setae.

residue (n.) a substance left behind. The gecko leaves no residue behind it.

negligible (adj.) too small to be significant. One contact is negligible; a billion are not.

aggregate (adj./n.) the total formed by combining many parts. The aggregate force holds many times the animal's weight.

to detach (v.) to separate from something.

A change of angle lets the gecko detach instantly.

hold on to (phrasal verb) to keep gripping something. The mechanism holds on to smooth surfaces only.

peel off (phrasal verb) to remove by pulling gradually from one edge. Changing the angle peels the setae off cleanly.

get stuck (phrase) to become unable to move. A gecko does not get stuck to what it climbs.

PRACTICE

Where the Analogy Breaks

A gecko's release costs nothing. Human exits carry penalties, reputations and relationships, so detachment must be negotiated, not just designed.

Use the Language

Make sentences using these three words: adhesion, detach and get stuck.

Discuss

Which commitment could you not exit cleanly if you needed to?

Write

Describe a relationship or contract and design its exit at the same time as its entry.

SOURCES

A Autumn, K. et al., "Adhesive force of a single gecko foot-hair", Nature 405, 681–685 (2000).

A Autumn, K. et al., "Evidence for van der Waals adhesion in gecko setae", PNAS (2002).

THE CONTRAST

Story 30: Geckos: Grip That Releases on Command vs. Story 32: Mussels: Holding On to Wet Rock

Story 30: Geckos: Grip That Releases on Command: Adhesion works through billions of microscopic contact points using weak physical forces, and releases instantly and cleanly on command.

Story 32: Mussels: Holding On to Wet Rock: Adhesion works through a specialized chemical bond that displaces water, and is meant to be permanent, not easily released.

Does an adhesion problem call for a reversible, mechanical solution or a permanent, chemical one — and how do you tell which kind of problem you actually have?

THIS CONNECTS TO

Story 28: Humpback Whales: The Bumpy Edge SAME MECHANISM
Both are engineering solutions borrowed directly from a specific biological surface structure — whale flipper bumps, gecko footpads — that outperforms the smooth or simple version most designers would default to.

Story 29: Kingfishers: The Beak and the Train SAME MECHANISM
Both show an engineer studying a specific biological structure closely enough to reproduce its underlying physical mechanism directly, rather than settling for a loose analogy.

Story 32: Mussels: Holding On to Wet Rock CONTRAST
See The Contrast in that chapter. In short: gecko adhesion is reversible and mechanical; mussel adhesion is permanent and chemical — two different answers to sticking to a surface.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Geckos can support their entire body weight on a single toe against a smooth vertical surface, and can attach and release that grip almost instantly and repeatedly, thousands of times, without any adhesive residue or wear.

What is the mechanism?

Gecko toe pads are covered in millions of microscopic hair-like structures called setae, each splitting further into hundreds of even smaller spatula-shaped tips; the sheer number of these tiny contact points generates enough cumulative van der Waals force — a weak molecular-level attraction requiring no chemical bond — to hold body weight, and changing the angle of contact instantly releases the grip.

What organizational problem does it explain?

Does your organization's approach to a 'stickiness' problem — a partnership, a customer relationship, a process dependency — actually need to be permanent, or would a reversible, easily-released mechanism serve the goal better?

Level 2 — Make It Your Story

> *Tell me about a time you designed a relationship, process or system to be easily reversible rather than permanent, and that flexibility turned out to matter.*

Use STAR to answer. Think about: a trial partnership, a modular system you could disconnect easily, a commitment you deliberately kept reversible.

TAKE THE IDEA FURTHER

> *Where has your organization built a permanent-feeling commitment into something that would actually work better as an easily reversible one?*

31

Spider Webs: Damage That Stays Local

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

Spider dragline silk is frequently described as stronger than steel. The claim needs care: by weight, its tensile strength is comparable to high-grade steel, but its real advantage is toughness — the total energy it absorbs before breaking, which comes from combining strength with high extensibility. Silk stretches substantially before failing, and that combination is where it outperforms most engineered materials.

The web exploits this at the structural level. When a bee hits an orb web, the impact energy is absorbed by deformation across the whole structure. Research on web mechanics found something counter-intuitive: individual threads are designed to fail locally under extreme load, sacrificing a small region so that the rest of the web survives. Damage is deliberately localized rather than propagated.

The behaviour under load is unusual enough to have surprised materials scientists. Most engineering materials stiffen and then break; silk does something more useful — it softens as it is stretched, then stiffens again before failure. Modelling and testing showed that this sequence is what localizes damage: an overloaded thread yields and fails on its own rather than transmitting the load outward, so an impact takes out a small region and leaves the remaining structure intact. A web made of a stiffer, stronger fibre would tear across a much larger area.

There is a design choice hidden in the silk itself that deserves attention. Dragline silk is not one material behaving differently under different loads; a spider produces several distinct silks from different glands, each with its own protein structure and mechanical role — a stiffer thread for the frame and spokes, a stretchier and stickier one for the capture spiral. The web's overall toughness comes partly from combining materials with different properties in the positions where each is needed, which is a more deliberate strategy than the phrase "spider silk" usually suggests.

It is also worth noting the trade the spider is making. A web is expensive — silk is protein, and many orb weavers eat their own web each night to recover the material before rebuilding. The structure is designed to be sacrificed and reconstituted, not defended, which is why localized failure is acceptable: repair is routine and already budgeted for.

Interestingly, the web solved a completely different problem too. Some orb weavers include silk that reflects ultraviolet light, apparently making webs visible to birds and reducing costly collisions. That observation inspired Ornilux bird-protection glass, which incorporates a UV-reflective pattern visible to birds but nearly invisible to humans — one of the biomimicry cases documented by the Biomimicry Institute.

Both lessons are structural rather than material. Strength does not require mass; it requires geometry, and the ability to absorb energy rather than resist it rigidly. And a system that can lose a small part locally without transmitting the damage everywhere is more survivable than one that is uniformly rigid.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Can a failure in one part of your organization stay local, or does it spread?*

KEY TAKEAWAYS

1. Silk's advantage is toughness — strength combined with extensibility — not raw strength alone.
2. Webs localize damage deliberately so failure does not propagate through the structure.
3. UV-reflective silk inspired bird-safe architectural glass.

SUMMARY

Spider silk absorbs impact through a combination of strength and stretch, and orb webs are structured so that damage stays local instead of spreading. Web properties have also inspired UV-patterned bird-safe glass. Smart structure and controlled local failure outperform brute rigidity.

SCIENCE CHECK

ESTABLISHED

Spider silk's toughness comes from a combination of strength and extensibility, and webs localize damage by design.

INFERRED

That structural or architectural solutions can outperform simply adding material.

UNCERTAIN

Different silks serve different structural roles; treating 'spider silk' as one uniform material overstates the design's simplicity.

VOCABULARY & PHRASAL VERBS

dragline silk (n.) the strong silk a spider uses for its safety line and web frame. Dragline silk forms the frame and safety line of the web.

tensile strength (n.) resistance to being pulled apart. Its tensile strength rivals some steels by weight.

toughness (n.) the energy a material absorbs before breaking. Toughness combines strength with the ability to stretch.

extensibility (n.) the ability to stretch. Extensibility lets the thread absorb energy before breaking.

to propagate (v.) to spread from one area to another.

A failure should not propagate through the whole structure.

counter-intuitive (adj.) the opposite of what you would expect. The finding is counter-intuitive: local failure is the design.

give way (phrasal verb) to break or collapse under pressure. One thread gives way so the rest of the web survives.

soak up (phrasal verb) to absorb. The structure soaks up the impact across many threads.

hold together (phrasal verb) to remain intact as one piece. Different silks hold together for different jobs.

PRACTICE

Where the Analogy Breaks

A web sheds a thread without argument. Human units resist being the part that is sacrificed, however sound the structural logic.

Use the Language

Make sentences using these three words: propagate, give way and soak up.

Discuss

Can a small failure in your organization stay contained, or does it spread to everything connected to it?

Write

Identify one place where a single failure currently cascades, and propose where a deliberate break should sit.

SOURCES

A Cranford, S. W. et al., "Nonlinear material behaviour of spider silk yields robust webs", Nature 482, 72–76 (2012).

C Biomimicry Toolbox, Case Studies (Ornilux). https://toolbox.biomimicry.org/references/case-studies/

THIS CONNECTS TO

Story 34: Seashells: Tough From Brittle Parts SAME MECHANISM

Both show that a material's real strength advantage comes from a structural property — toughness, or the ability to contain damage — rather than from raw material strength alone.

Story 42: Coral Reefs: Abundance From Poor Water LIMITATION

Spider silk contains damage locally through a deliberate structural weak point. Coral reefs achieve resilience through a different structural principle — distributed partnership across the whole system rather than a single engineered failure point.

Story 35: Feathers: Strength at Almost No Weight SAME MECHANISM

Both are cases where a natural material achieves an unusual performance

advantage — contained damage, or strength at minimal weight — through structural architecture, not through the raw properties of the base material.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A spider web can sustain significant local damage — a strand breaking under the impact of a struggling insect or in strong wind — without the whole structure collapsing, because the damage tends to stay contained near where it occurred.

What is the mechanism?

Silk's real advantage is toughness, meaning it combines strength with extensibility — the ability to stretch substantially before breaking — and orb webs are engineered with deliberately sacrificial radial and spiral threads whose different tension properties mean a sudden overload breaks a local connection first, dissipating energy at that point rather than transmitting catastrophic stress through the entire structure.

What organizational problem does it explain?

Is your organization's system designed so that a local failure stays local, or is it built tightly enough that one broken connection can propagate stress through the whole structure?

Level 2 — Make It Your Story

> *Tell me about a time a system was deliberately designed so that a local failure stayed contained, rather than cascading through the whole structure.*

Use STAR to answer. Think about: a circuit breaker, a modular architecture, a process with a deliberate failure point that protected the rest of the system.

TAKE THE IDEA FURTHER

> *Where in your organization would a single local failure currently cascade through the whole system, when a deliberately engineered weak point could contain the damage instead?*

32

Mussels: Holding On to Wet Rock

> **Evidence Level 2**
>
> **Biological claim:** Documented biomimicry
>
> **Business application:** Technology transfer, then analogy

A mussel attaches itself to a rock in the intertidal zone — wet, salty, and hit by waves — and stays there. This is a serious engineering achievement, because the fundamental problem of adhesion is water. Almost every glue humans make fails when the surface is wet, since water molecules occupy the surface and prevent bonding.

The mussel produces a bundle of threads called a byssus, tipped with adhesive plaques. The proteins in these plaques are unusually rich in an amino acid called DOPA, which can displace water from a surface and form strong bonds with a wide range of materials — rock, wood, metal, plastic. The byssal threads themselves are graded in stiffness, soft where they meet the mussel and stiffer at the rock, so wave energy is absorbed rather than concentrated at a single joint.

The commercial line from this science is direct. Mussel-adhesion research inspired Kaichang Li at Oregon State University to develop a formaldehyde-free, soy-based wood adhesive (commercialized by Columbia Forest Products), allowing manufacturers to bond plywood without conventional formaldehyde-based resins — a health benefit as well as a technical one.

It is worth being precise about what "holding on" costs the mussel, since the picture of effortless grip is not quite right. Producing byssal threads is metabolically expensive, and animals in high-energy wave environments spend a measurably larger share of their energy budget on thread production

than those in sheltered water. The mussel is not indifferent to the sea. It is paying, continuously, for the right to stay where the waves are — which is a cost most business writing about resilience quietly leaves out of the account.

The threads are also disposable in a way the animal exploits. A mussel can release its byssus and lay down new threads to reposition itself, and it adds more threads when wave action increases — attachment strength is adjusted to conditions rather than fixed at maximum. Permanent maximum grip would be both expensive and immobilizing. What the animal maintains is the ability to hold hard when it must and let go when relocating is worth more.

Two things are worth taking from this. First, the mussel did not wait for calm water; the glue was designed for the wet, moving conditions it would actually face. Systems designed for ideal conditions fail in real ones. Second, the graded stiffness matters: the transition between two very different things — a soft animal and a hard rock — needed its own engineering, and most organizational failures happen at exactly such interfaces.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where do two very different parts of your organization meet — and who engineered that joint?*

KEY TAKEAWAYS

1. Mussel adhesive works underwater because DOPA proteins displace water and bond directly.
2. Byssal threads grade from soft to stiff so wave energy is spread, not concentrated.

3. The science led to formaldehyde-free wood adhesives used commercially.

SUMMARY

Mussels attach to wet rock using DOPA-rich proteins that displace water, with graded threads that absorb wave energy instead of concentrating it. The mechanism inspired safer, formaldehyde-free wood glues. Good systems are engineered for the conditions they will actually meet — especially at the interfaces between very different components.

SCIENCE CHECK

ESTABLISHED

DOPA-rich mussel proteins displace surface water and enable adhesion underwater, inspiring real adhesive technologies.

INFERRED

That systems should be engineered for the actual conditions they will meet, not for ideal ones.

UNCERTAIN

Byssus production is metabolically expensive; the appearance of effortless grip omits a continuous energy cost.

VOCABULARY & PHRASAL VERBS

byssus (n.) the bundle of threads a mussel uses to attach itself. The byssus is a bundle of threads anchoring the mussel.

plaque (n.) here: a flat adhesive pad at the end of a thread. Each thread ends in an adhesive plaque.

intertidal zone (n.) the shore area covered and uncovered by tides. The intertidal zone is wet, salty, and hit by waves.

to displace (v.) to force something out of its position. DOPA displaces water bound to the surface.

graded (adj.) changing gradually from one state to another. The threads are graded from soft to stiff.

interface (n.) the point where two systems meet. The interface between animal and rock is where the design lives.

hold on (phrasal verb) to keep gripping something firmly. The mussel holds on through repeated wave impact.

come off (phrasal verb) to become detached. A thread can come off and be replaced with a new one.

stand up to (phrasal verb) to resist damage from something. The adhesive stands up to constant immersion.

PRACTICE

Where the Analogy Breaks

Mussel adhesive faces constant, predictable conditions. Human interfaces face changing people, which no chemistry can hold.

Use the Language

Make sentences using these three words: interface, graded and stand up to.

Discuss

Where do two very different parts of your organization meet, and who is responsible for that join?

Write

Describe an interface in your work that needs graded, not uniform, strength — and why.

SOURCES

A Waite, J. H., "Mussel adhesion — essential footwork", Journal of Experimental Biology 220, 517–530 (2017).

C Biomimicry Toolbox, Case Studies (bio-based wood adhesives). https://toolbox.biomimicry.org/references/case-studies/

THE CONTRAST

Story 32: Mussels: Holding On to Wet Rock vs. Story 30: Geckos: Grip That Releases on Command

Story 32: Mussels: Holding On to Wet Rock: Adhesion works through a specialized chemical bond that displaces water, and is meant to be permanent, not easily released.

Story 30: Geckos: Grip That Releases on Command: Adhesion works through billions of microscopic contact points using weak physical forces, and releases instantly and cleanly on command.

Does an adhesion problem call for a permanent, chemical solution or a reversible, mechanical one — and how do you tell which kind of problem you actually have?

THIS CONNECTS TO

Story 30: Geckos: Grip That Releases on Command CONTRAST
See The Contrast in that chapter. In short: mussel adhesion is permanent and chemical; gecko adhesion is reversible and mechanical — two different answers to sticking to a surface.

Story 33: Beetles: Drinking From Fog LIMITATION
Mussels solve an adhesion problem in a wet, hostile environment through a specialized chemical mechanism. Beetles solve a resource-collection problem in a hostile environment through a specialized surface-physics mechanism — both are precise engineering answers evolved for one specific hostile condition.

Story 34: Seashells: Tough From Brittle Parts LIMITATION
Both mussels and their shells face the same wet, wave-battered environment, but solve different problems: this chapter is about sticking to the rock in the first place; Seashells is about surviving impact once attached.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Mussels attach themselves permanently and with remarkable strength to wet, often wave-battered rock surfaces — a genuinely difficult adhesion problem, since most conventional adhesives fail to bond properly to a wet surface at all.

What is the mechanism?

Mussels secrete threads tipped with adhesive plaques containing a modified amino acid called DOPA, which has a specific chemical property allowing it

to displace the thin layer of water molecules that normally coats a wet surface and prevents conventional adhesives from bonding, forming a direct, strong chemical bond to the rock beneath that water layer rather than trying to bond on top of it.

What organizational problem does it explain?

Where your organization is struggling to make something 'stick' — a change, a habit, a partnership — is the real obstacle the strength of the adhesive itself, or is it a hidden layer, like the water film on the rock, that needs to be displaced first?

Level 2 — Make It Your Story

> *Tell me about a time an initiative failed to 'stick' not because the effort was weak, but because something else was in the way that needed to be addressed first.*

Use STAR to answer. Think about: a change management effort, a habit that wouldn't form, a partnership that needed a hidden obstacle removed before it could take hold.

TAKE THE IDEA FURTHER

> *What is currently failing to 'stick' in your organization not because the effort behind it is weak, but because there's a hidden layer — like water on rock — that hasn't been displaced first?*

33

Beetles: Drinking From Fog

> **Evidence Level 2**
> **Biological claim:** Documented biomimicry
> **Business application:** Technology transfer, then analogy

The Namib Desert receives almost no rain, but fog rolls in from the Atlantic on many mornings. The darkling beetles that live there — the fog-basking Onymacris unguicularis is the famous case — climb to the crest of a dune, face into the wind and tilt their bodies forward. Fog droplets collect on the beetle's back, run down its surface and reach its mouth.

The physics of the shell attracted serious attention after a 2001 Nature paper proposed that the beetle's surface combines water-attracting bumps with a water-repelling waxed background, so droplets nucleate on the peaks and then roll off rather than evaporating. Later work has debated the details and the precise wettability pattern, which is a good illustration for students that biomimicry is a live science rather than a settled catalogue. What is not disputed is the behaviour and the outcome: the beetle drinks fog.

The engineering response was immediate and continues — fog-harvesting meshes and surfaces with patterned wettability have been developed to collect water in arid regions, an approach documented in biomimicry case collections.

It is worth adding what the engineering response actually looked like, because "inspired by" covers a wide range of rigour. Some fog-harvesting projects reproduce the bumpy, patterned surface fairly directly; others simply use fine mesh netting, which collects fog through an entirely different and much older mechanism and would work with or without the beetle. Not every

technology described as biomimetic in press coverage owes its function to the organism credited for it, which is worth remembering whenever a case study arrives without a citation attached.

The strategic point is easy to state and hard to practise. The beetle's environment contains almost no rain, which is the resource everyone else needs. Rather than competing for scarce water, it converted a feature of the environment that nobody was using into its water supply. Fog was not a resource until the beetle's morphology made it one. Ask what your environment contains in abundance that your industry currently treats as irrelevant.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What does your environment contain in abundance that your industry treats as worthless?*

KEY TAKEAWAYS

1. Namib beetles harvest drinking water directly from fog using surface properties and posture.
2. The mechanism inspired fog-harvesting meshes and patterned surfaces for arid regions.
3. A resource exists only once something is designed to capture it.

SUMMARY

Fog-basking beetles in the Namib collect water from morning fog on their bodies, exploiting surface wettability and a characteristic posture, though the exact surface mechanism remains debated. The principle has been applied in fog-harvesting technology. Strategically, it shows how an overlooked feature of an environment can become a resource.

SCIENCE CHECK

ESTABLISHED

Namib beetles collect drinking water from fog using posture and surface properties.

INFERRED

That an overlooked feature of an environment can become a resource once something is designed to capture it.

UNCERTAIN

The precise surface mechanism proposed in 2001 remains disputed, and some 'inspired' devices use unrelated mesh technology instead.

VOCABULARY & PHRASAL VERBS

fog (n.) thick cloud at ground level. Fog rolls in from the Atlantic most mornings.

crest (n.) the top ridge of a hill or dune. The beetle climbs to the crest of the dune.

to tilt (v.) to lean at an angle. It tilts its body to collect droplets on its back.

droplet (n.) a very small drop of liquid. Each droplet runs down towards the mouth.

wettability (n.) how easily a surface is covered by liquid. Wettability patterns may explain how droplets collect.

arid (adj.) extremely dry. The mechanism evolved in one of the most arid places on Earth.

roll in (phrasal verb) to arrive gradually in large amounts. Fog rolls in for perhaps an hour each morning.

run down (phrasal verb) to flow downwards along a surface. Water runs down the shell to the mouthparts.

roll off (phrasal verb) to move off a surface by rolling. Excess water rolls off rather than evaporating.

PRACTICE

Where the Analogy Breaks

A beetle's morphology is fixed by evolution; a business chooses its capture mechanism and can be wrong about it repeatedly.

Use the Language

Make sentences using these three words: arid, roll in and wettability.

Discuss

What does your environment contain in abundance that your industry currently treats as worthless?

Write

Propose one way to capture a resource nobody in your industry currently collects.

SOURCES

A Parker, A. R. & Lawrence, C. R., "Water capture by a desert beetle", Nature 414, 33–34 (2001).

A Nørgaard, T. & Dacke, M., "Fog-basking behaviour and water collection efficiency in Namib Desert Darkling beetles", Frontiers in Zoology (2010).

THE CONTRAST

Story 33: Beetles: Drinking From Fog vs. Story 22: Camels: Tolerance, Not Reserves

Story 33: Beetles: Drinking From Fog: Survival in a resource-scarce environment comes from a specialized mechanism for harvesting a resource others can't access at all.

Story 22: Camels: Tolerance, Not Reserves: Survival in a resource-scarce environment comes from tolerating a wider range of internal conditions, not acquiring more of the resource.

Is scarcity better solved by finding a way to access a resource nobody else can reach, or by needing less of it in the first place?

THIS CONNECTS TO

Story 22: Camels: Tolerance, Not Reserves SAME MECHANISM

Both describe an organism thriving in a resource-scarce environment not through storage or brute-force acquisition, but through a specifically adapted mechanism for managing the scarce resource efficiently.

Story 43: Mangroves: Making Their Own Ground LIMITATION

Namib beetles harvest an existing but hard-to-access resource (fog) directly. Mangroves take a different approach to a hostile environment — actively creating new usable ground, rather than more efficiently extracting from what's already there.

Story 29: Kingfishers: The Beak and the Train SAME MECHANISM

Both are precise engineering solutions copied directly from a specific, measured biological surface or structure, applied to solve a real, unrelated technical problem.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

The Namib desert beetle survives in one of the driest environments on Earth by harvesting drinking water directly from fog, using a combination of body surface structure and behaviour rather than depending on rainfall or standing water at all.

What is the mechanism?

One proposed explanation is that the beetle's back is covered in a pattern of water-attracting (hydrophilic) bumps surrounded by water-repelling (hydrophobic), wax-coated troughs; tiny fog droplets condense and stick to the bumps, grow larger, and eventually roll down the water-repelling channels directly to the beetle's mouth, and the beetle further improves this by tilting its body into the wind at dawn when fog is present.

What organizational problem does it explain?

Is there a resource your organization currently treats as inaccessible or scarce, when a more specific, purpose-built mechanism for harvesting the small amount that is available might make it viable after all?

Level 2 — Make It Your Story

> *Tell me about a time you found a way to access or harvest a resource that most people assumed simply wasn't available.*

Use STAR to answer. Think about: an underused data source, an overlooked talent pool, a resource everyone assumed was scarce that was actually just hard to access with the usual method.

TAKE THE IDEA FURTHER

> *What resource does your organization currently treat as unavailable or scarce, when a more specifically designed mechanism might let you harvest what small, usable amount actually is present?*

34

Seashells: Tough From Brittle Parts

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

The inner layer of an abalone shell, nacre, is roughly three thousand times tougher than the mineral it is mostly made of. That is a startling number, because the mineral in question — aragonite, a form of calcium carbonate — is chemically similar to chalk. Chalk is brittle. Something in the arrangement is doing the work.

Nacre is built like a microscopic brick wall: flat aragonite tablets, each around half a micrometre thick, stacked in layers and separated by very thin sheets of protein and polysaccharide. When a crack tries to travel through it, it cannot run straight. It is deflected around each tablet, tablets slide slightly against one another, and the organic layers stretch and absorb energy. The crack has to take a long, obstructed path and it runs out of energy before it gets anywhere.

The composition is about 95% brittle mineral and 5% organic material. Almost all the toughness comes from that 5% and from the architecture, not from the substance itself. Nacre-inspired layered composites are an active field in materials engineering for exactly this reason.

It is also worth being honest about what nacre is not good at, since a fair account names weaknesses as well as strengths. The structure resists a crack propagating through the shell; it does not make the shell hard in the everyday sense, and a sufficiently sharp, concentrated blow — exactly the kind a predator's claw or a determined human tool delivers — can still defeat it. Toughness against distributed stress and hardness against a sharp point are

different properties, and nacre was optimized for the threat the animal actually faces.

Engineers have found this much easier to admire than to reproduce. Nacre grows cell by cell, at ambient temperature, in seawater; a factory has to build the same architecture by freeze-casting, layer deposition or printing, and the resulting composites capture part of the toughness rather than all of it. The gap is instructive. The design is not a secret — it has been described in detail for decades — and the difficulty lies entirely in the manufacture, which is a fair description of most organizational designs that everybody admires and few reproduce.

For organizations, this is the most precise version of an idea that appears repeatedly in this course. Robustness is not primarily about the quality of your components. It is about whether a failure can travel in a straight line. If one problem in your organization can propagate cleanly from one department to the next without being deflected, absorbed or slowed, the strength of the individual parts will not save you.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where in your organization is quality being achieved by effort that structure could achieve instead?*

KEY TAKEAWAYS

1. Nacre's toughness comes from its layered brick-and-mortar architecture, not its minerals.
2. A tiny fraction of organic material produces most of the crack resistance.

3. Systems survive when failures are deflected and absorbed rather than allowed to run straight.

SUMMARY

Mother-of-pearl is thousands of times tougher than the brittle mineral it is made from because microscopic tablets and thin organic layers deflect and absorb cracks. Architecture, not material quality, produces the strength. Organizations similarly resist damage by ensuring problems cannot propagate directly through the structure.

SCIENCE CHECK

ESTABLISHED

Nacre's toughness comes from a layered tablet architecture that deflects cracks, not from the mineral itself.

INFERRED

That architecture, not raw material strength, produces resilience against distributed stress.

UNCERTAIN

Nacre resists distributed stress but not concentrated sharp impact; toughness and hardness are different properties, easily conflated.

VOCABULARY & PHRASAL VERBS

nacre (n.) mother-of-pearl, the iridescent inner layer of some shells. Nacre is thousands of times tougher than its own mineral.

brittle (adj.) hard but easily broken. Aragonite alone is brittle, like chalk.

crack (n.) a line of breakage in a material. A crack cannot travel straight through the structure.

to deflect (v.) to cause something to change direction. Each tablet deflects the crack slightly.

composite (n.) a material made from two or more different substances. The composite is 95% mineral and 5% organic material.

architecture (n.) here: the structural arrangement of parts. The toughness comes from architecture, not material.

run out of (phrasal verb) to exhaust a supply of something. The crack runs out of energy before crossing the shell.

give way (phrasal verb) to break under pressure. No single tablet has to give way for the shell to hold.

hold up (phrasal verb) to remain strong. The structure holds up against distributed stress.

PRACTICE

Where the Analogy Breaks

Nacre's architecture never changes. Organizational architecture is rebuilt constantly by people who did not design it.

Use the Language

Make sentences using these three words: brittle, deflect and run out of.

Discuss

Where is quality currently being achieved through effort that structure could achieve instead?

Write

Describe a process where a stronger design would remove the need for someone to be careful.

SOURCES

A Barthelat, F. et al., "On the mechanics of mother-of-pearl: a key feature in the material hierarchical structure", Journal of the Mechanics and Physics of Solids (2007).

A Meyers, M. A. et al., "Biological materials: Structure and mechanical properties", Progress in Materials Science (2008).

THIS CONNECTS TO

Story 31: Spider Webs: Damage That Stays Local SAME MECHANISM
Both are cases where a natural material achieves an unusual performance advantage — toughness, or contained damage — through structural architecture, not through the raw properties of the base material.

Story 35: Feathers: Strength at Almost No Weight SAME MECHANISM
Both show that a material's real performance comes from hierarchical structural architecture — layered brick-and-mortar, or hollow branching shafts — rather than from the intrinsic strength of the raw material itself.

Story 32: Mussels: Holding On to Wet Rock LIMITATION
Both mussels and their shells face the same wet, wave-battered environment, but solve different problems: Mussels is about sticking to the rock; this chapter is about surviving impact once attached.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Nacre, the iridescent inner layer of many mollusc shells (commonly called mother-of-pearl), is roughly three thousand times tougher than the pure mineral it's made from, despite that mineral itself being relatively brittle on its own.

What is the mechanism?

Nacre is built from microscopic tablets of a brittle mineral (aragonite) arranged in overlapping layers with a thin layer of elastic protein between each tablet, in a structure often described as brick-and-mortar; when a crack starts to form, this layered architecture forces it to travel around each individual tablet rather than straight through the material, and the protein layers absorb and dissipate energy, giving the composite dramatically more toughness than either component has alone.

What organizational problem does it explain?

Is your organization trying to build resilience by making a single component individually stronger, when a layered, composite architecture — combining a rigid element with a flexible one — might produce far greater toughness overall?

Level 2 — Make It Your Story

> *Tell me about a time combining a rigid, structured element with a more flexible one produced a result that was tougher than either component alone.*

Use STAR to answer. Think about: a process that combined strict rules with flexible exceptions, a team that combined structure with adaptability.

TAKE THE IDEA FURTHER

Is your organization trying to build resilience into a single component by making it individually stronger, when a layered combination of rigid and flexible elements might actually produce more toughness overall?

35

Feathers: Strength at Almost No Weight

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A feather weighs almost nothing and survives forces that would destroy an equivalent mass of most engineered materials. It is made of keratin — the same protein as fingernails — and gets its performance entirely from structure.

The central shaft, the rachis, is a hollow tube with a foam-like internal core. A hollow tube resists bending far better than a solid rod of the same mass, because material at the outside of a beam does most of the work; the foam core prevents the walls from buckling inward. Branching off the shaft are barbs, and off the barbs, thousands of barbules with microscopic hooks that interlock with the barbules of the neighbouring barb. This creates a surface that holds air pressure yet separates and re-zips when disturbed, which is why a bird repairs its wing surface by preening rather than by regrowing anything.

The internal structure is the part that does the work. The rachis is not a simple tube: its wall thickness varies along the length, and the interior is filled with a foam of keratin whose density changes from base to tip, so material sits where bending stress is highest and is absent where it is not. This is why a feather resists bending far better than its weight suggests, and why engineers interested in lightweight structures keep returning to it. The shaft is a graded structure rather than a uniform one.

Notice the second property, because it is the one engineers admire most: the feather is not merely light and strong, it is self-repairing at the point of failure, and the repair costs almost nothing.

Feathers also wear out, and how they are replaced is the part engineers find hardest to copy. Most birds moult gradually and symmetrically, shedding a few matched feathers from each wing at a time, so the surface is renewed continuously while the animal keeps flying. Replacement is scheduled around the requirement to stay airborne — the opposite of taking the system down for maintenance.

The lightness itself is worth grounding in a number, because "almost nothing" undersells the achievement. A flight feather can outperform its own weight in bending stiffness by a wide margin compared with a solid rod of the same material, which is precisely why aerospace engineers studying cellular and sandwich structures keep returning to feather geometry as a reference case rather than a curiosity.

There is a cost that the admiring version leaves out. Feathers are dead structures — once grown they cannot repair themselves at the material level, so damage accumulates through the year until the whole surface is replaced at moult. That replacement is expensive: it consumes protein and energy at a level that competes with breeding, and in many species it is scheduled to avoid overlapping with it. A system that renews itself continuously has paid for the capability somewhere, and in birds the bill arrives once a year.

Efficiency in organizations is usually treated as a resource problem — do the same thing with fewer people, less budget, less time. The feather suggests treating it as a structural problem instead. Where is the mass in your process actually doing work, and where is it just present? And when your process gets disturbed, does it need a manager to reassemble it, or can it re-zip by itself?

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which part of your process is doing real work, and which part is simply present?*

KEY TAKEAWAYS

1. Feather performance comes from hollow shafts and hierarchical branching, not special material.
2. Material placed where stress is highest does the work; the rest is unnecessary mass.
3. Interlocking barbules let a feather repair itself cheaply after damage.

SUMMARY

Feathers achieve high strength at almost no weight through a hollow foam-filled shaft and interlocking barbs and barbules made of ordinary keratin. They also reassemble themselves when disturbed. Efficiency is often a question of structure and self-repair rather than of consuming fewer resources.

SCIENCE CHECK

ESTABLISHED

Feathers achieve high stiffness at low weight through a graded, foam-filled shaft and self-repairing interlocking barbs.

INFERRED

That efficiency is often a structural rather than a resource problem.

UNCERTAIN

Feathers cannot self-repair at the material level; continuous renewal is achieved by costly annual replacement, not by the feather itself.

VOCABULARY & PHRASAL VERBS

keratin (n.) the protein in hair, nails and feathers. Feathers are built from keratin, the same protein as nails.

rachis (n.) the central shaft of a feather. The rachis is a hollow, foam-filled shaft.

barb / barbule (n.) the branches and sub-branches of a feather. Barbules interlock to hold the vane together.

to buckle (v.) to bend or collapse under pressure. The foam core stops the walls buckling inward.

to interlock (v.) to fit together and hold. Hooked barbules interlock with their neighbours.

to preen (v.) (of a bird) to clean and arrange its feathers. A bird preens to re-zip a disturbed feather.

branch off (phrasal verb) to divide and go in another direction. Barbs branch off the central shaft.

hold together (phrasal verb) to stay joined as one unit. Interlocking barbules hold the surface together.

come apart (phrasal verb) to separate into pieces. The vane comes apart easily and reassembles just as easily.

PRACTICE

Where the Analogy Breaks

A feather re-zips identically every time. Human self-repair depends on relationships, which do not reassemble on demand.

Use the Language

Make sentences using these three words: buckle, interlock and come apart.

Discuss

Which part of your process is doing real work, and which part is simply present out of habit?

Write

List the 'weight' in one process and identify what could be removed without loss of function.

SOURCES

A Sullivan, T. N. et al., "Extreme lightweight structures: avian feathers and bones", Materials Today (2017).

A Bachmann, T. et al., research on feather shaft structure and mechanics, Journal of the Royal Society Interface.

THIS CONNECTS TO

Story 34: Seashells: Tough From Brittle Parts SAME MECHANISM
Both show that a material's real performance comes from hierarchical structural architecture — layered brick-and-mortar, or hollow branching shafts — rather than from the intrinsic strength of the raw material itself.

Story 31: Spider Webs: Damage That Stays Local SAME MECHANISM
Both are cases where a natural material achieves an unusual performance advantage — strength at minimal weight, or contained damage — through structural architecture, not through the raw properties of the base material.

Story 28: Humpback Whales: The Bumpy Edge LIMITATION

Feather architecture achieves strength through internal hierarchical structure. Whale flipper tubercles achieve performance through external surface geometry — two different levels at which biological structure, rather than raw material, does the work.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A flight feather achieves an exceptional strength-to-weight ratio — rigid enough to generate lift and withstand aerodynamic force, yet remarkably light — using keratin, a protein no stronger, weight for weight, than the keratin in a fingernail or a hoof.

What is the mechanism?

The feather's central shaft is hollow rather than solid, with an internal foam-like structure providing support without the weight of solid material, and the shaft further tapers and branches hierarchically into barbs and barbules at progressively smaller scales, distributing structural support exactly where aerodynamic load requires it rather than uniformly throughout the material.

What organizational problem does it explain?

Is your organization assuming that more strength or more capability requires more material or more headcount, when a hollow, hierarchically-distributed structure — support concentrated only where load actually is — might achieve the same result with far less?

Level 2 — Make It Your Story

> *Tell me about a time a leaner, more precisely structured process or team outperformed a heavier one that had more resources thrown at it.*

Use STAR to answer. Think about: a small team that outperformed a larger one, a streamlined process that worked better than an over-resourced one.

TAKE THE IDEA FURTHER

> *Where in your organization has 'more resources' become the default answer to a strength or capability problem, when a more precisely structured, leaner approach might achieve the same result?*

36

Fish: Riding the Vortex Behind

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A trout holding station in a fast river sometimes beats its tail more slowly than the current would seem to require. Experiments in which fish were filmed swimming behind a cylinder in a flow tank showed that they adopt a distinctive "Kármán gait", slaloming between the vortices shed by the obstacle and extracting energy from the swirling water. Measurements of muscle activity confirmed that the fish were using less muscular effort than they would in uniform flow at the same speed.

The turbulence is not something the fish overcomes. It is fuel. And to use it the fish must give up control of its own rhythm and synchronize with a pattern created by something else.

The experiment that established this is unusually clean. A trout was placed in a flow tank downstream of a cylinder, which sheds vortices in a regular alternating pattern, and the fish's body movement was filmed while electrodes recorded activity in its swimming muscles. Behind the cylinder the animal adopted a distinctive slalom — larger, slower body undulations, timed to the passing vortices — and muscle activity dropped markedly compared with swimming in smooth flow at the same speed. Some fish held position with almost no measurable muscular effort at all.

The same theme appears in schooling. Fish positioned appropriately relative to their neighbours can exploit the wakes those neighbours generate, and research on schooling energetics indicates measurable savings compared with

swimming alone. Position within a group is not a social matter only; it is a hydrodynamic one.

It is worth adding the honest limit of the finding, since the original paper is more measured than its popular retelling. The energy saving was demonstrated for fish swimming behind a fixed cylinder shedding a regular vortex street — a controlled, predictable pattern. Real turbulence in open water is far less regular, and how much of the effect survives outside the laboratory, and across species, remains an active area of research rather than a settled fact.

One further detail sharpens the point. The fish does not choose its own rhythm behind the cylinder; the vortex street sets the tempo and the animal matches it. The energy is available only to a body prepared to move at a frequency it did not select. That is the uncomfortable part of the analogy, and the reason organizations so often fail to exploit forces already moving in their market: taking advantage of someone else's momentum requires abandoning your own schedule.

There is a real business analogy here, and it is more specific than "work smarter". A great deal of effort in organizations is spent fighting turbulence created by the market, by regulation, by a larger competitor's moves. The trout's strategy is neither to fight the current nor to leave the river. It is to change its own timing so that forces already in motion do part of the work. That requires letting go of a rhythm you chose, which is usually the difficult part.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which force already moving in your market could do some of your work if you changed your timing?*

KEY TAKEAWAYS

1. Trout extract energy from vortices behind obstacles, reducing muscular effort.
2. Exploiting external forces requires synchronizing with a rhythm you did not set.
3. Position relative to others changes the energetic cost of moving.

SUMMARY

Fish swimming behind obstacles adopt a Kármán gait that harvests energy from vortices, using less muscle than swimming in uniform flow, and schooling fish can similarly exploit their neighbours' wakes. Turbulence becomes a resource rather than an obstacle. Efficiency can come from changing how and when you move rather than from working harder.

SCIENCE CHECK

ESTABLISHED

Fish behind an obstacle adopt a distinct gait and show measurably reduced muscle activity, confirmed by EMG.

INFERRED

That aligning with external forces already in motion can reduce effort dramatically.

UNCERTAIN

The finding comes from a controlled, regular vortex street; how well it generalizes to irregular real-world turbulence is still studied.

VOCABULARY & PHRASAL VERBS

vortex / vortices (n.) a spinning mass of fluid. Vortices form in the wake of any obstacle in a current.

wake (n.) the disturbed water or air left behind a moving body. Position in a neighbour's wake can reduce effort.

turbulence (n.) irregular, chaotic movement in a fluid. Turbulence is treated as fuel rather than a problem.

gait (n.) a manner of moving. The fish adopts a distinctive gait behind the cylinder.

to synchronize (v.) to match timing with something else. Muscle activity synchronizes with the passing vortices.

to hold station (phrase) to stay in one position against a current. A trout can hold station using very little muscle.

give up (phrasal verb) to surrender or stop holding. Exploiting the current means giving up your own rhythm.

let go of (phrasal verb) to release. Efficiency required letting go of a chosen tempo.

go with the flow (phrase) to accept and move with the current situation. Going with the flow was a measurable energy saving.

PRACTICE

Where the Analogy Breaks

A trout can read the flow directly. Market turbulence is only visible in delayed, noisy data, so synchronizing is a bet.

Use the Language

Make sentences using these three words: wake, synchronize and give up.

Discuss

What force is already moving in your market that you are currently working against rather than with?

Write

Describe what changing your own timing, rather than your effort, would let you use.

SOURCES

A Liao, J. C. et al., "Fish exploiting vortices decrease muscle activity", Science 302, 1566–1569 (2003).

A Ashraf, I. et al., research on schooling and hydrodynamic interactions, PNAS / Journal of the Royal Society Interface.

THIS CONNECTS TO

Story 28: Humpback Whales: The Bumpy Edge SAME MECHANISM

Both show an aquatic animal exploiting fluid dynamics — turbulence around a structure, or a vortex left behind an obstacle — to gain a performance advantage a smooth or naive design wouldn't achieve.

Story 11: Geese: Why the Lead Position Rotates LIMITATION

Trout exploit turbulence left behind an obstacle to reduce their own effort. Geese exploit a related aerodynamic effect from a neighbouring bird's wingtip — both reduce individual energy cost by positioning relative to a disturbance another body creates, but one is from terrain, the other from a teammate.

Story 39: Beehives: Waiting Time as a Signal LIMITATION

Trout position themselves to extract energy from an existing environmental pattern rather than fighting it directly. Beehives regulate flow through an internal signal (waiting time) rather than an external physical structure — different mechanisms, similar principle of working with a system's natural dynamics rather than against them.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Trout swimming behind an obstacle in a stream — a rock, a bridge piling, even another fish — use substantially less muscular effort than swimming in open water, despite appearing to work against disturbed, turbulent water rather than smooth flow.

What is the mechanism?

An obstacle in flowing water sheds a regular, alternating pattern of vortices behind it (a von Kármán vortex street); a trout positioned in this wake can synchronize its body movements with the vortices, extracting kinetic energy from them in a way that reduces the muscular effort needed to hold position or move forward, effectively surfing the turbulence rather than fighting it.

What organizational problem does it explain?

Is your organization treating a disruptive, turbulent condition in its environment purely as an obstacle to expend energy fighting, when there might be a way to position against it that actually reduces the effort required?

Level 2 — Make It Your Story

> *Tell me about a time you positioned your team or strategy to work with a disruptive market or environmental condition, rather than spending energy fighting directly against it.*

Use STAR to answer. Think about: a downturn you positioned around rather than resisted, a disruption you rode rather than fought.

TAKE THE IDEA FURTHER

> *What turbulent condition in your organization's environment is currently being fought head-on, when a different position relative to it might actually reduce the effort required?*

PART V

Organizational Design & Systems

How does nature build sustainable systems?

37

Termite Mounds: Walls That Breathe

Evidence Level 2

Biological claim: Documented biomimicry

Business application: Technology transfer, then analogy

A termite mound in the African savannah stands in air that swings between roughly 3°C at night and 40°C by day. Inside, conditions stay far more stable — which matters because the colony farms a fungus that requires steady conditions and produces a great deal of carbon dioxide that must be exchanged for oxygen.

The early explanation was a simple chimney drawing hot air upward. Turner tested this directly by filling mounds with inert tracer gas and measuring how fast it left through the walls under different conditions. The exchange rate tracked surface temperature fluctuations rather than any steady internal draft, which ruled out the simple chimney model and pointed to the walls themselves — porous, thin-walled and full of interconnected channels — breathing in response to the daily heating and cooling cycle. The mound is closer to a lung than a flue.

There is a workplace version of the same swing worth naming, because most offices have one and ignore it. Occupancy, noise and demand for meeting rooms rise and fall predictably through the day, yet the physical office is usually built for peak load at every hour. A building — or a team's schedule — designed around the swing rather than against it would flex the way the mound does: opening capacity when it is needed and closing it down when it is not, instead of holding one expensive configuration permanently.

The Eastgate building translates the principle rather than copying the material. It has no conventional air conditioning: thick concrete floors and walls absorb heat during the day and release it at night, fans move air through voids in the structure at low cost, and the whole system is tuned to Harare's daily temperature swing rather than fighting it. Reported energy savings compared with a conventional design of the same size have been substantial, though the building relies on a climate with a wide day-night range — the same design would perform differently somewhere with a flatter temperature curve, which is worth remembering before treating any single biomimetic building as a universal template.

The workplace lesson runs beyond energy bills. Termites regulate their environment structurally, whereas most organizations try to regulate theirs behaviourally — a policy about focus time, a request that people be collaborative, a poster about wellbeing. If the physical and procedural environment works against the behaviour you want, no amount of encouragement will overcome it. Change the building, not the memo.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What in your physical or procedural environment works against the behaviour you keep asking for?*

KEY TAKEAWAYS

1. Termite mounds regulate gas exchange and temperature through their structure itself.
2. The Eastgate Centre applied passive termite-inspired cooling with large energy savings.

3. Environments shape behaviour more reliably than instructions about behaviour do.

SUMMARY

Termite mounds maintain stable internal conditions through porous walls driven by daily temperature cycles rather than through a simple chimney effect, and this principle informed passively cooled buildings such as Harare's Eastgate Centre. Structure performs the regulation. Workplaces likewise shape performance physically, not through policy alone.

SCIENCE CHECK

ESTABLISHED

Termite mound walls exchange gas in response to daily temperature cycles, not a steady chimney draft.

INFERRED

That physical and procedural environments shape behaviour more reliably than instructions do.

UNCERTAIN

The Eastgate strategy depends on a wide day-night temperature swing; the same approach would achieve less somewhere with a flatter climate.

VOCABULARY & PHRASAL VERBS

savannah (n.) a grassy plain in a hot region. The mound stands in open savannah with huge temperature swings.

porous (adj.) having small holes that allow air or liquid through. The porous walls act like a lung.

ventilation (n.) the movement of fresh air through a space. Passive ventilation replaces mechanical air conditioning.

thermal mass (n.) a material's ability to absorb and store heat. Thick concrete provides thermal mass for the building.

passive (cooling) (adj.) working without powered equipment.

Passive cooling only works where the climate provides a swing.

memo (n.) a short official written message in a company. A memo about focus time cannot fix a noisy office.

draw up / draw in (phrasal verb) to pull upward / to pull inward. Air is drawn in through the porous walls.

take up (phrasal verb) to adopt an idea or activity. The architect took up the termite principle directly.

work against (phrasal verb) to have the opposite effect to what is wanted. A noisy layout works against everything the memo asks for.

PRACTICE

Where the Analogy Breaks

A mound regulates one physical variable. Workplaces regulate attention, status and mood, which architecture influences but cannot determine.

Use the Language

Make sentences using these three words: passive, ventilation and work against.

Discuss

Does your physical or procedural environment work against the behaviour you keep asking for?

Write

Redesign one workplace routine so it flexes with a predictable daily swing instead of fighting it.

SOURCES

A Turner, J. S. & Soar, R. C., "Beyond biomimicry: What termites can tell us about realizing the living building", Proceedings of the First International Conference on Industrialized, Intelligent Construction (2008).

A "The role of mound functions and local environment in the diversity of termite mound structures", ScienceDirect. https://www.sciencedirect.com/science/article/pii/S0022519321002423

THIS CONNECTS TO

Story 15: Termites: Building With No Blueprint SAME MECHANISM
Both chapters look at termite-built structures — the other at how they're built with no blueprint, this one at what the finished structure does once built — two halves of the same mechanism, construction and function.

Story 43: Mangroves: Making Their Own Ground SAME MECHANISM
Both show a structure that manages a harsh environmental force — extreme temperature swings, or wave energy — passively, through its own physical architecture, without any active regulation.

Story 39: Beehives: Waiting Time as a Signal LIMITATION
Termite mounds regulate their internal environment through fixed passive structure. Beehives regulate their internal state through an active, ongoing behavioural signal (waiting time) — a static versus dynamic approach to the same underlying problem of internal regulation.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A termite mound in a savannah exposed to daily temperature swings between roughly 3°C at night and 40°C during the day maintains a far more stable internal temperature and steady gas exchange, without any active heating, cooling or ventilation mechanism.

What is the mechanism?

The mound's porous walls and porous walls exchange air in response to daily surface heating and cooling, drawing stale air out and fresh air in through the structure's own geometry, meaning the building itself performs the climate-control function rather than the termites actively managing conditions moment to moment.

What organizational problem does it explain?

Is your organization solving a regulation or stability problem with active, ongoing management effort, when a structural redesign might handle the same problem passively, with no ongoing effort required at all?

Level 2 — Make It Your Story

> *Tell me about a time redesigning a system's structure eliminated the need for active, ongoing management of a recurring problem.*

Use STAR to answer. Think about: a process redesigned to self-regulate, an automated system that replaced manual monitoring, a structural fix rather than a procedural one.

TAKE THE IDEA FURTHER

> *What recurring problem in your organization is currently managed through active, ongoing effort that a one-time structural redesign might handle passively instead?*

38

Ant Colonies: Growth Without Overhead

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A colony of two hundred ants and a colony of two hundred thousand ants are the same species running the same rules. Nothing is redesigned in between. This is worth pausing on, because in human organizations almost everything has to be redesigned at each stage of growth.

The reason ants scale is that their coordination cost does not rise with size. Each ant interacts with the ants immediately around it and with chemical traces in its local environment. Add a hundred thousand more and no individual's workload changes. Human coordination is different: if everyone must be aware of everyone, the number of possible connections grows roughly with the square of the number of people, which is why a process that worked beautifully at twelve people collapses at sixty.

Research on ant colonies also shows that behaviour changes with colony size in ways that are themselves part of the design — larger colonies show more pronounced division of labour and different collective dynamics, without any change in the underlying rules each ant follows. Scale-dependent behaviour emerges rather than being decreed.

There is a further wrinkle worth including for balance. Coordination cost in ant colonies is not literally zero at every scale; extremely large colonies of some species show measurable declines in certain kinds of collective efficiency, and very large 'supercolonies' formed by merging separate nests behave differently again. The claim that ants scale without cost is a strong first approximation,

not a law with no exceptions — which is itself a useful thing to model for students learning to read a confident biological claim carefully.

It is also worth stating what the model does not explain, since coordination is not the only limit on organizational size. Coordination cost is one constraint among several; capital, regulation and market size all cap growth too, and a process that scales perfectly in the ant sense can still fail for reasons that have nothing to do with coordination at all. The lesson is narrower than 'grow without limit'. It is that coordination need not be the constraint that stops you first.

The practical test for a growing company is therefore specific. Do not ask whether your process works. Ask what happens to it at ten times the size. Any process that depends on everyone knowing each other, on one person seeing everything, or on a single weekly meeting where all decisions are made is not a process. It is a temporary condition of being small — and it will fail exactly when the company is succeeding.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which of your processes only works because you are still small?*

KEY TAKEAWAYS

1. Ant coordination is local, so individual workload does not increase as the colony grows.
2. Human processes that require global awareness scale badly and fail as headcount rises.
3. Test processes by asking what happens at ten times the current size.

SUMMARY

Ant colonies function identically across enormous size differences because each ant interacts only locally, so coordination cost stays flat. Human processes requiring everyone to know everything fail as connections multiply. Businesses should design processes that do not depend on being small.

SCIENCE CHECK

ESTABLISHED

Ant coordination is local, so per-individual workload stays flat as colony size grows; behaviour still shifts with scale.

INFERRED

That processes requiring global awareness fail as organizations grow, while local coordination scales.

UNCERTAIN

Coordination cost is one constraint among several — capital, regulation and market size also cap growth, regardless of coordination design.

VOCABULARY & PHRASAL VERBS

scalability (n.) the ability to grow larger without breaking. Scalability means the process still works ten times larger.

coordination cost (n.) the effort required to keep people working together. Coordination cost rises with the square of headcount.

workload (n.) the amount of work a person must do. No individual ant's workload changes as the colony grows.

to decree (v.) to order officially. Nobody decreed the change in behaviour; size produced it.

headcount (n.) total number of employees. Headcount alone does not predict coordination cost.

bottleneck (n.) a point where flow is restricted.

Coordination, not capital, was the real bottleneck.

scale up (phrasal verb) to increase in size or capacity. The process broke the moment we tried to scale up.

break down (phrasal verb) to stop functioning. Weekly all-hands meetings break down past a certain size.

keep track of (phrasal verb) to maintain awareness of something. Nobody can keep track of sixty relationships at once.

PRACTICE

Where the Analogy Breaks

Ant coordination cost is genuinely flat. Human organizations add coordination cost with every layer, including the layers added to manage coordination.

Use the Language

Make sentences using these three words: scale up, bottleneck and keep track of.

Discuss

Which of your processes only works because your organization is still small?

Write

Write 120 words on what would break first at ten times your current size, and why.

SOURCES

B Gordon, D. M., Ant Encounters: Interaction Networks and Colony Behaviour, Princeton University Press (2010).

A Holbrook, C. T. et al., "Colony-size effects on task organization in the harvester ant", Insectes Sociaux (2011).

THIS CONNECTS TO

Story 17: Ants: Thresholds, Not Assignments SAME MECHANISM
Both describe how ant colonies coordinate work locally, without a manager tracking overall colony state, letting the coordination mechanism itself scale with colony size.

Story 1: Ants: Order Without a Manager SAME MECHANISM
Both show ant colonies achieving coordinated, efficient outcomes without any individual holding a view of the whole system, a mechanism that scales without adding coordination overhead.

Story 39: Beehives: Waiting Time as a Signal SAME MECHANISM
Both show a colony regulating its own collective behaviour through a simple local signal — waiting time, or local task density — rather than through any form of central tracking or reporting.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

As an ant colony grows larger, the coordination burden on any individual ant does not increase — an ant in a colony of a hundred thousand behaves

according to essentially the same local rules as an ant in a colony of a thousand.

What is the mechanism?

Because coordination happens entirely through local interactions — encountering other ants, detecting pheromone concentration, responding to task-specific thresholds — rather than through any mechanism that requires tracking colony-wide state, the coordination overhead per individual stays constant as the colony scales, unlike a hierarchical system where each additional layer of growth typically adds proportional management overhead.

What organizational problem does it explain?

Does your organization's coordination overhead grow faster than the work itself as you scale, and is that because coordination depends on someone tracking the whole system rather than on a mechanism built entirely from local interactions?

Level 2 — Make It Your Story

> *Tell me about a time your team kept a process running smoothly as it scaled, without a proportional increase in management or coordination overhead.*

Use STAR to answer. Think about: a process that scaled cleanly, local decision rules that didn't require more central tracking as headcount grew.

TAKE THE IDEA FURTHER

> *As your organization has grown, has coordination overhead grown roughly in proportion to headcount — and if so, is that because*

coordination still depends on someone tracking the whole system rather than on local rules that scale by themselves?

39

Beehives: Waiting Time as a Signal

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A foraging bee returns to the hive with nectar and does not store it herself. She hands it to a receiver bee. How long she has to wait to find one turns out to be a control signal for the entire colony.

If receivers are immediately available, the forager infers that the hive has spare storage capacity and advertises her food source energetically with a waggle dance, recruiting more foragers. If she has to search for a long time, the hive is saturated, and she may dance less or perform a tremble dance — which recruits more bees to become receivers instead. The colony balances collection against processing capacity using nothing but the delay experienced at the handover point.

There is a broader point about feedback design worth drawing out here. The bees' signal works because it is a direct physical consequence of the imbalance it reports — a forager cannot fake a short wait — whereas most human status reports are written by someone with an interest in how the number reads, and are one, sometimes several, steps removed from the event they describe. A feedback loop is only as trustworthy as its distance from the thing it measures, and the shortest loops are usually the ones nobody had to write down.

The mechanism has been probed by deliberately unbalancing it. When researchers raised nectar intake beyond what the receivers could handle, foragers' search times lengthened and more of them switched to the tremble dance that recruits new receivers; when receivers were plentiful, search times

were short and waggle dancing intensified. The colony's information was never inspected directly. It was inferred entirely from how long an individual had to wait at one handover point.

There is no report, no meeting and no manager comparing intake with storage. The information is generated exactly where the imbalance occurs, and it reaches the people who can respond within minutes.

Compare this with the typical corporate feedback loop. Sales sells more than operations can deliver. Operations struggles quietly. The mismatch surfaces in a monthly review, four to six weeks after it began, by which point customers have already been affected and the discussion is about blame rather than adjustment.

It is worth adding the failure mode this design has, because every mechanism has one. If foragers are delayed by congestion rather than by a genuine shortage of storage — for instance, if the colony over-recruits following an unusually good source — the wait-based signal can be swamped by simple congestion rather than accurately reflecting storage capacity. The system is elegant within its normal operating range and, like most local feedback mechanisms, can be misled by conditions its designers never had to face.

The design question is not "do we measure this?" but how long the loop takes and who receives the signal. A weekly number that reaches a manager is far less useful than an immediate signal that reaches the person creating the imbalance. Bees do not measure their problem. They feel it, in the moment, at the point of handover.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *How long does it take a problem here to reach someone who can change it?*

KEY TAKEAWAYS

1. Bees regulate collection and processing using the delay at the handover point.
2. The signal is generated where the imbalance occurs and reaches those who can act.
3. Loop speed and signal recipient matter more than whether something is measured.

SUMMARY

Honeybee foragers judge the hive's storage capacity by how long they wait for a receiver bee, adjusting recruitment or triggering tremble dances accordingly. This produces fast, local, self-correcting balance with no central oversight. Organizations need rapid feedback that reaches the people creating the imbalance, not delayed reports to managers.

SCIENCE CHECK

ESTABLISHED

Foragers infer storage capacity from wait time at the handover point, confirmed by deliberately unbalancing hives.

INFERRED

That fast, local feedback outperforms delayed reporting.

UNCERTAIN

The signal can be swamped by simple congestion under unusual conditions, and is elegant within, not beyond, its normal operating range.

VOCABULARY & PHRASAL VERBS

nectar (n.) the sweet liquid bees collect from flowers. Foragers bring nectar back and hand it to a receiver.

forager / receiver (n.) a bee that collects food / one that takes and stores it. The wait between forager and receiver carries the signal.

saturated (adj.) completely full; unable to take more. A saturated hive produces longer waits and less dancing.

handover (n.) the moment when one person passes work to another. The handover point is where the real information lives.

feedback loop (n.) a cycle in which results influence future action.

The shortest feedback loops are the most trustworthy.

mismatch (n.) a failure of two things to fit together. The mismatch surfaced weeks later, in a report nobody trusted.

hand over (phrasal verb) to give something to someone else. She hands over the number without anyone checking it.

build up (phrasal verb) to accumulate. Backlog builds up long before anyone official notices.

catch up with (phrasal verb) to reach the same level as something ahead. The report took a month to catch up with reality.

PRACTICE

Where the Analogy Breaks

A bee's wait is an unfakeable signal. Human status reports are written by people with an interest in how they read.

Use the Language

Make sentences using these three words: saturated, handover and catch up with.

Discuss

How long does it take a problem in your organization to reach someone who can act on it?

Write

Describe your slowest feedback loop and propose a shorter one that does not require a report.

SOURCES

B Seeley, T. D., The Wisdom of the Hive, Harvard University Press (1995).

A Seeley, T. D., "The tremble dance of the honey bee: message and meanings", Behavioural Ecology and Sociobiology (1992).

THIS CONNECTS TO

Story 38: Ant Colonies: Growth Without Overhead SAME MECHANISM

Both show a colony regulating its own collective behaviour through a simple local signal — waiting time, or local task density — rather than through any form of central tracking or reporting.

Story 40: Bacteria: Acting Only at Quorum SAME MECHANISM

Both describe a colony-level behaviour regulated by a simple, locally-sensed signal — a wait time, or a chemical concentration — that indirectly reflects a system-wide condition without anyone needing to measure that condition directly.

Story 36: Fish: Riding the Vortex Behind LIMITATION

Beehives regulate flow through an internal signal (waiting time) rather than an external physical structure. Trout position themselves relative to an existing physical pattern in their environment — different mechanisms, similar principle of working with a system's natural dynamics rather than against them.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Honeybee colonies regulate the balance between nectar collection and nectar processing without any bee directly measuring or communicating the colony's overall nectar supply or processing capacity.

What is the mechanism?

Foragers returning with nectar must wait for a processing bee (a receiver) to become available before unloading; the length of this wait time itself functions as the regulating signal — a long wait indicates the colony is over-supplied relative to processing capacity, which causes foragers to reduce recruitment for more foraging, while a short wait encourages more foraging — a self-regulating feedback loop built entirely from one simple, locally experienced delay.

What organizational problem does it explain?

Does your organization have a simple, naturally occurring signal — like a wait time or a queue length — that could regulate a process automatically, instead

of relying on someone actively measuring and reporting the underlying condition?

Level 2 — Make It Your Story

> *Tell me about a time a simple signal, like a wait time or a backlog length, ended up regulating a process automatically without anyone needing to actively manage the balance.*

Use STAR to answer. Think about: a queue that self-regulated intake, a wait time that naturally throttled a process, a backlog that signalled when to slow down.

TAKE THE IDEA FURTHER

> *Is there a simple, already-occurring signal in your organization's operations — a wait time, a queue length — that could regulate a process automatically, if anyone paid attention to it as a signal rather than just a delay?*

40

Bacteria: Acting Only at Quorum

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A single bacterium producing a toxin achieves nothing except alerting the host's immune system. A billion bacteria producing it simultaneously can overwhelm a host. The problem for the bacterium is knowing which situation it is in, and the solution is called quorum sensing.

Each cell continuously releases a small signalling molecule and continuously detects the concentration of that molecule around it. When the population is sparse, the signal diffuses away and stays dilute. As the population grows, concentration rises, and above a threshold the cells switch on genes they had been keeping silent — producing toxins, forming biofilms, or in Vibrio fischeri, generating light.

The clearest demonstration involves an organism that glows. Vibrio fischeri lives at low density freely in seawater, producing negligible light, and at high density inside the light organs of certain squid and fish, where it luminesces strongly enough to be useful to its host. Grown in a flask and diluted repeatedly, the same bacteria stay dark; concentrated past the threshold, the whole culture switches on within about an hour. Nothing about individual cells changed between the two states. Only the density did.

Bonnie Bassler's work showed that bacteria use several such languages, including a shared signal that lets different species estimate the total population rather than only their own kind.

The elegance is that no cell counts anything. Each measures one local chemical concentration, and population size is inferred from it. The collective decision is a by-product of a purely local measurement.

There is a strategic implication worth stating directly, because it runs against the instinct to persuade one skeptic at a time. If a desired behaviour only becomes normal above a density threshold, the efficient intervention is not maximum individual persuasion spread thinly across the whole organization. It is concentrating enough visible adopters in one team or one location that the local concentration crosses the threshold on its own — after which the behaviour looks normal to everyone who encounters that group, without anyone there having been individually convinced.

It is worth being honest about the limit of the human analogy too. Bacterial thresholds are chemical and identical across a population; human 'thresholds' for adopting a new behaviour vary enormously between individuals and are shaped by status, risk tolerance and trust in whoever is asking. Concentrating visible adopters helps for the same structural reason it helps a bacterial culture — local density is what crosses a threshold — but the threshold itself is not a fixed number waiting to be discovered.

For organizations this reframes what a "culture" is. Culture is not the values statement; it is the ambient signal concentration — what people around you actually do, and therefore what appears normal. Below a threshold, a new behaviour stays invisible and individuals adopting it feel exposed. Above it, the same behaviour becomes the obvious thing to do. If you are trying to change how an organization works, you are not persuading individuals one at a time. You are trying to reach a quorum.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What behaviour in your organization is invisible only because too few people do it yet?*

KEY TAKEAWAYS

1. Bacteria estimate population size from a local chemical concentration, not by counting.
2. Behaviour switches on only above a threshold, so timing is collective, not individual.
3. Cultural change works by reaching a visible threshold rather than by isolated persuasion.

SUMMARY

Quorum sensing lets bacteria detect population density through signalling molecules and activate group behaviours only once a threshold is passed. Sophisticated coordination emerges from each cell measuring one local quantity. Organizational culture changes similarly: behaviours spread once they become visibly normal rather than through individual persuasion alone.

SCIENCE CHECK

ESTABLISHED

Bacteria coordinate collective behaviour through chemical signalling molecules that reflect population density.

INFERRED

That visible behaviour spreads once local density crosses a threshold, which argues for concentrated rather than diffuse adoption efforts.

UNCERTAIN

Signalling molecules can be diluted in flowing or mixed environments, so detected density is not the same as true density.

VOCABULARY & PHRASAL VERBS

quorum sensing (n.) bacterial detection of population density via chemical signals. Quorum sensing lets bacteria detect their own density.

to diffuse (v.) to spread out through a medium. The signal diffuses away in a sparse population.

sparse / dilute (adj.) thinly spread / weak in concentration. In a sparse population the signal stays dilute.

biofilm (n.) a layer of microorganisms attached to a surface. Dense populations can switch on biofilm formation together.

threshold (n.) the point at which something begins to happen.

Behaviour switches on only above a threshold.

ambient (adj.) present in the surrounding environment. Culture is the ambient signal, not the values statement.

switch on / switch off (phrasal verb) to activate / to deactivate. Genes switch on collectively above the threshold.

catch on (phrasal verb) to become widely adopted. A behaviour catches on once local density is high enough.

stand out (phrasal verb) to be conspicuously different. Below the threshold, new behaviour stands out and feels risky.

PRACTICE

Where the Analogy Breaks

Bacterial thresholds are chemical and identical for all cells. Human norms are contested, and different groups reach different quorums.

Use the Language

Make sentences using these three words: threshold, catch on and stand out.

Discuss

What behaviour in your workplace is invisible only because too few people do it yet?

Write

Propose concentrating a new behaviour in one team first rather than spreading persuasion thinly.

SOURCES

A Bassler, B. L., "Small talk: cell-to-cell communication in bacteria", Cell 109, 421–424 (2002).

B Waters, C. M. & Bassler, B. L., "Quorum sensing: cell-to-cell communication in bacteria", Annual Review of Cell and Developmental Biology (2005).

THIS CONNECTS TO

Story 39: Beehives: Waiting Time as a Signal SAME MECHANISM

Both describe a colony-level behaviour regulated by a simple, locally-sensed signal — a wait time, or a chemical concentration — that indirectly reflects a system-wide condition without anyone needing to measure that condition directly.

Story 2: Honeybees: Quorum, Not Consensus SAME MECHANISM

Both use a quorum-style mechanism — action only once a threshold of independent local signals is reached — rather than either individual action or centrally coordinated agreement.

Story 1: Ants: Order Without a Manager SAME MECHANISM

Both show sophisticated colony-level behaviour emerging from individuals responding to simple, local signals, with no central controller directing when or how the group acts.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Individual bacteria typically do not launch a coordinated, resource-intensive collective action — such as producing a toxin, forming a biofilm, or triggering bioluminescence — until the local bacterial population reaches a sufficient density, even though no bacterium can count the population directly.

What is the mechanism?

Bacteria continuously release a signalling molecule at a low, constant per-cell rate; as population density increases, the local concentration of that molecule rises proportionally, and once it crosses a threshold concentration, it triggers a coordinated genetic response across the whole local population simultaneously — a mechanism called quorum sensing, which lets bacteria estimate population size indirectly through chemical concentration rather than needing any direct communication or counting.

What organizational problem does it explain?

Does your organization wait for a genuine, sufficient signal of readiness before committing resources to a coordinated action, or does it act on isolated individual initiative before the real conditions for success are actually in place?

Level 2 — Make It Your Story

> *Tell me about a time your organization correctly waited for a genuine threshold of readiness — enough demand, enough capacity — before committing to a coordinated, resource-intensive action.*

Use STAR to answer. Think about: a launch timed to real market signals, a scale-up that waited for genuine demand evidence rather than isolated enthusiasm.

TAKE THE IDEA FURTHER

> *Is your organization currently committing resources to a coordinated action based on isolated individual enthusiasm, without a genuine, broadly-sensed signal that the real conditions for success are actually present?*

41

Forests: Competition and Shelter

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Two facts about a forest are both true, and holding them at the same time is the whole lesson. Trees compete ferociously — for light above all, and for water and nutrients below. Height itself is an arms race: a tree grows tall mainly because its neighbours are growing tall. And simultaneously, those same trees depend on shared fungal networks, on the microclimate the canopy creates, on the soil that all of them build, and on the wind protection they give each other.

Business writers call this combination 'coopetition'; ecologists recognize that the same interaction can be competitive or facilitative depending on conditions. Research on the stress-gradient hypothesis found that facilitation between plants becomes more common in harsh environments, while competition dominates where conditions are benign. Whether your neighbour is your rival or your protector depends on how hard the environment is.

The stress-gradient work behind this finding compared plant interactions across many published studies spanning a range of environmental harshness, and found the balance between competition and facilitation shifting systematically with conditions — neighbours helping more often in physically demanding environments and competing more often in benign ones. The pattern is a tendency documented across many systems, not a fixed

law holding in every single case, and ecologists studying particular sites still find exceptions worth explaining.

That finding is directly usable. Internal competition between teams is not automatically toxic; in comfortable conditions it can sharpen performance. But organizations often make the mistake of maintaining the same aggressive internal competition when conditions become severe — during a downturn, a crisis, a regulatory shock — precisely when the ecological evidence suggests cooperation becomes more valuable.

There is a timing detail worth taking seriously, because it is where organizations most often get the balance wrong in practice. The shift from competition to facilitation in plant communities tends to happen at the point where conditions become survivable rather than comfortable — well before a crisis is obvious to a casual observer. Waiting for a downturn to be undeniable before easing internal competition means making the change later than the biology suggests is optimal.

There is a related asymmetry worth naming, because it explains why the wrong call is so common. Increasing internal competition is politically easy — it can be announced, it looks decisive, and it rewards whoever proposes it. Reducing it is politically harder, because it looks like giving something up, even when the underlying conditions justify exactly that. Organizations therefore drift towards more internal competition over time by default, regardless of what the environment currently calls for.

The mature view is not to choose. A company can compete internally for budget, headcount and attention while remaining completely dependent on cooperation to survive. Trees manage both at once. The skill is knowing which mode the environment currently rewards.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Are your internal teams competing at a moment when conditions call for cooperation?*

KEY TAKEAWAYS

1. Forests are simultaneously competitive and cooperative systems.
2. Facilitation between organisms increases as environmental conditions become harsher.
3. Internal competition suits benign conditions; crisis conditions reward cooperation.

SUMMARY

Trees compete intensely for light while depending on shared networks, soil and mutual shelter, and ecological research shows facilitation becomes more common as conditions get harsher. Competition and cooperation are modes rather than opposites. Organizations should shift the balance according to the severity of the environment rather than choosing one permanently.

SCIENCE CHECK

ESTABLISHED

Facilitation between plants becomes more common in harsher conditions across many studies; competition dominates in benign ones.

INFERRED

That internal competition and cooperation should shift with how harsh conditions actually are.

UNCERTAIN

The pattern is a documented tendency across many systems, not an exceptionless law; particular sites can and do behave differently.

VOCABULARY & PHRASAL VERBS

interdependence (n.) mutual dependence between parties. Interdependence and competition coexist in the same forest.

arms race (n.) a competition where each side escalates in response to the other. Height is partly an arms race between neighbours.

facilitation (n.) one organism making conditions better for another. Facilitation becomes more common as conditions worsen.

benign (adj.) gentle, favourable, not harmful. Competition dominates in benign conditions.

microclimate (n.) the climate of a very small local area.

The canopy creates its own microclimate below it.

downturn (n.) a period of reduced economic activity. Internal competition is easy to raise and hard to lower before a downturn.

crowd out (phrasal verb) to take space or resources others need. Internal rivalry can crowd out necessary cooperation.

look out for (phrasal verb) to take care of or protect. In hard conditions, neighbours look out for each other.

team up (phrasal verb) to join with others to work together. Teams team up when the environment gets harsher.

PRACTICE

Where the Analogy Breaks

Trees do not choose their mode. Humans can switch between competition and cooperation deliberately — and can get the timing badly wrong.

Use the Language

Make sentences using these three words: facilitation, benign and team up.

Discuss

Are your teams competing internally at a moment when conditions call for cooperation?

Write

Write 120 words on when your organization last eased internal competition, and what made that hard.

SOURCES

B Bertness, M. D. & Callaway, R., "Positive interactions in communities", Trends in Ecology & Evolution 9, 191–193 (1994).

A Brooker, R. W. et al., "Facilitation in plant communities: the past, the present, and the future", Journal of Ecology (2008).

THE CONTRAST

Story 41: Forests: Competition and Shelter vs. Story 9: Dolphins: Roles No One Plays Alone

Story 41: Forests: Competition and Shelter: A single system can be simultaneously and genuinely competitive and cooperative, with the same individuals doing both at once.

Story 9: Dolphins: Roles No One Plays Alone: Effective teamwork depends on trust and coordinated roles among cooperating allies, without the same kind of direct competition present.

Can genuine cooperation and genuine competition coexist within the same relationship, or does real teamwork require setting competition aside entirely?

THIS CONNECTS TO

Story 24: Fungi: The Network Under the Forest SAME MECHANISM
Both look at forests as systems that are simultaneously competitive and cooperative, possibly exchanging resources through shared fungal networks even while individual trees compete for light.

Story 9: Dolphins: Roles No One Plays Alone LIMITATION
Dolphin cooperation is built on trust and coordinated roles among allies. Forests: Competition and Shelter shows that cooperation and competition can coexist within the very same system — a more complicated picture than pure teamwork.

Story 42: Coral Reefs: Abundance From Poor Water SAME MECHANISM
Both describe an ecosystem whose overall abundance depends on a mix of cooperative resource-sharing and ordinary competitive pressure operating simultaneously, rather than one replacing the other.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Trees in a forest compete directly and constantly for light, water and nutrients, and yet the same forest also demonstrably benefits individual trees through shelter from wind, shared humidity, and possibly, though this is debated, resource exchange through mycorrhizal networks.

What is the mechanism?

Individual trees are simultaneously competing for above-ground resources like light, which drives them to grow taller and shade out neighbours, while also benefiting from the collective structure the forest provides — reduced wind exposure, moderated temperature and humidity, and, some studies suggest, limited transfer through fungal networks — meaning competition and cooperation are not sequential phases but two things happening in the same system at the same time.

What organizational problem does it explain?

Is your organization assuming that internal competition and genuine cooperation are mutually exclusive, when a forest suggests they can operate simultaneously within the same relationships, benefiting the whole system even as individuals compete?

Level 2 — Make It Your Story

> *Tell me about a time competition and cooperation coexisted within the same team or relationship, rather than one having to be eliminated for the other to work.*

Use STAR to answer. Think about: competing teams that still shared resources, colleagues who competed for the same promotion while genuinely helping each other.

TAKE THE IDEA FURTHER

> *Is your organization trying to eliminate all internal competition in the name of cooperation, when a forest suggests the two might be able to coexist productively within the same structure?*

42

Coral Reefs: Abundance From Poor Water

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A coral reef occupies well under one percent of the ocean floor and supports a very large share of marine species — a concentration of life that appears, at first glance, impossible in the nutrient-poor tropical water where reefs are found. The explanation is a partnership. Coral polyps host photosynthetic algae called zooxanthellae inside their own tissue. The algae supply the coral with a large majority of its energy; the coral supplies shelter and nutrients. Neither could build a reef alone.

That partnership creates the physical structure, and the structure creates the ecosystem: crevices for small fish, surfaces for sponges, hunting grounds for predators, nurseries for species that spend their adult lives elsewhere. The coral is not merely a resident of the reef. It is the platform provider — and thousands of species build businesses on top of it.

Bleaching itself is a controlled response before it is a catastrophe, which is a distinction worth making precisely. When water temperature rises even slightly above the coral's normal range, the algae's photosynthetic machinery becomes damaging to the host, and the coral expels its symbionts as a defensive measure — a functioning stress response, not a passive death. Survival after that point depends entirely on temperature returning to normal quickly enough for new algae to be taken up; prolonged heat converts a defensive measure into starvation.

There is a further point worth making about dependency itself, since the coral–algae relationship is often described as though only the coral needs saving. The algae, expelled during bleaching, do not simply vanish into a healthier existence outside the animal; free-living success for these particular symbiont lineages away from a host is limited, so the partnership's fragility runs in both directions. Neither party has a good alternative to the other, which is the sharpest version of the platform-dependency lesson this chapter is making.

There is a scale point worth adding before drawing a business lesson from any single reef. Coral–algae partnerships vary considerably between species and regions in their heat tolerance, and some populations have shown measurably greater resilience to warming than others nearby — variation that matters for conservation planning and complicates any single, universal statement about how fragile the arrangement is. The dependency this chapter describes is real. Its severity is not uniform.

The business ecosystem analogy is now familiar, but the reef adds two details usually left out. First, the platform's own survival rests on a partnership at a scale far below the visible structure. Second, being the platform is not safe. Everything depends on you, which means every stress in the system eventually arrives at your door — and if you go, nothing built on you survives independently.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *If you are the platform others build on, what happens to them when you have a bad year?*

KEY TAKEAWAYS

1. Reefs exist because of a partnership between coral polyps and photosynthetic algae.
2. The physical platform one organism builds enables an entire dependent ecosystem.
3. Platform position brings dependency risk in both directions, not only power.

SUMMARY

Coral reefs support enormous biodiversity in nutrient-poor water because coral and zooxanthellae form a partnership that builds physical structure for thousands of other species. The ecosystem is entirely dependent on that platform, and collapses when bleaching occurs. Business ecosystems offer great value but concentrate systemic risk on whoever provides the platform.

SCIENCE CHECK

ESTABLISHED

Coral-algae symbiosis supplies most of the coral's energy; bleaching is an active stress response to heat, not passive death.

INFERRED

That platform dependencies concentrate risk on the party everything else depends on.

UNCERTAIN

Heat tolerance varies significantly between coral populations and species, so no single reef's fragility generalizes to all reefs.

VOCABULARY & PHRASAL VERBS

polyp (n.) the small individual animal that builds coral. The coral polyp hosts algae inside its own tissue.

zooxanthellae (n.) algae living inside coral tissue. Zooxanthellae supply most of the coral's energy.

symbiotic (adj.) involving a close mutually beneficial relationship. The relationship is symbiotic in both directions.

to bleach (v.) (of coral) to turn white after expelling its algae. The coral bleaches when it expels its algae under heat stress.

nursery (n.) here: a place where young animals grow safely. The reef acts as a nursery for many other species.

platform (n.) a base that others build upon. The coral is the platform the whole ecosystem depends on.

build on (phrasal verb) to develop something using an existing base. Thousands of species build on the structure coral provides.

die off (phrasal verb) to disappear gradually through deaths. Large sections can die off after prolonged heat.

depend on (phrasal verb) to rely on for survival or success. Both partners depend on each other with no good alternative.

PRACTICE

Where the Analogy Breaks

Coral cannot renegotiate its partnership. Platform businesses set the terms of dependence, which makes the relationship political as well as ecological.

Use the Language

Make sentences using these three words: platform, build on and depend on.

Discuss

If you are the platform others depend on, what happens to them in your worst year?

Write

Describe a dependency in your organization that runs in both directions, not just one.

SOURCES

A Muscatine, L. & Porter, J. W., "Reef corals: mutualistic symbioses adapted to nutrient-poor environments", BioScience 27, 454–460 (1977).

A Hughes, T. P. et al., "Global warming and recurrent mass bleaching of corals", Nature 543, 373–377 (2017).

THIS CONNECTS TO

Story 24: Fungi: The Network Under the Forest SAME MECHANISM
Both describe a partnership between two different organisms that trades complementary capabilities, producing a result — nutrient access, reef abundance — that neither partner could achieve alone.

Story 25: Forests: Spare Parts That Differ LIMITATION
Functional redundancy protects a forest against an unpredictable future stress. Coral reef abundance depends on a single, non-substitutable partnership, which means it has far less of this kind of protection when conditions change.

Story 41: Forests: Competition and Shelter SAME MECHANISM
Both describe an ecosystem whose overall abundance depends on a mix of

cooperative resource-sharing and ordinary competitive pressure operating simultaneously, rather than one replacing the other.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Coral reefs support extraordinarily dense and diverse marine life despite growing in tropical waters that are typically nutrient-poor — a paradox long known in marine ecology as Darwin's paradox, since abundant ecosystems normally require nutrient-rich water.

What is the mechanism?

Coral polyps host photosynthetic algae called zooxanthellae directly within their tissue; the algae receive shelter and access to the polyp's waste compounds as raw material for photosynthesis, and in return supply the coral with the large majority of its energy needs directly, creating an unusually efficient, tightly closed nutrient cycle that lets the reef sustain abundant life on a nutrient budget that would support very little in open water.

What organizational problem does it explain?

Does your organization's apparent resource scarcity actually reflect a lack of resources, or a lack of the tight, efficient internal partnership needed to cycle the resources you already have far more effectively?

Level 2 — Make It Your Story

> *Tell me about a time an unusually tight, efficient internal partnership let your organization achieve more with limited resources than the resource level alone would predict.*

Use STAR to answer. Think about: two teams whose close collaboration multiplied a limited budget's impact, an efficient internal exchange that outperformed simply having more resources.

TAKE THE IDEA FURTHER

> *Is your organization's sense of resource scarcity actually about the total resources available, or about how inefficiently they currently cycle between the people and teams who could use them?*

43

Mangroves: Making Their Own Ground

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A mangrove forest stands in salt water on stilts. The prop roots of a red mangrove arch out of the trunk and enter the mud in dozens of places, producing a dense, flexible thicket at exactly the height where waves arrive.

Field measurements repeatedly show that wave energy falls as it passes through this structure, with substantial reductions in wave height over relatively short distances of forest — though the effect depends strongly on species, root density, water depth and the width of the belt. Nothing here is rigid. A sea wall opposes a wave and eventually fails catastrophically; a mangrove belt lets the water through while taking energy out of it, and if some trees are lost, the rest continue working.

The second half of the story is better still. Those same roots slow the water enough that suspended sediment drops out and accumulates, so the forest gradually builds the ground it stands on and can, under the right conditions, keep pace with a rising sea. The mangrove is not merely tolerating its environment. It is manufacturing it.

There is a design principle worth extracting from the failure rather than only the success. A single mangrove tree provides negligible protection; the effect scales with the width and density of the belt, which is why restoration projects that plant a thin fringe for visual reassurance rather than a substantial band typically deliver little of the wave attenuation the ecosystem is capable of. Buffering capacity is a property of the system's extent, not of its presence.

What happened in south Florida after Hurricane Irma in 2017 shows exactly how a buffer fails. Storm surge pushed water inland faster than the site's drainage could remove it, and the resulting prolonged ponding — not the wave energy itself — killed large areas of forest that had survived the storm's initial impact intact. The buffer absorbed the blow it was built for and was undone by a second, slower mechanism nobody had designed it to handle.

There is one more comparison worth making, because it sharpens what 'buffering' actually means as a design category. A sea wall is a single point of failure: engineered to a specific load, and catastrophic once that load is exceeded. A mangrove belt degrades gradually — losing trees at the front while the rest continues absorbing energy — which is a fundamentally different failure curve even when the total protection delivered is comparable. Organizations building resilience should ask which curve they are actually constructing, not only how much protection it provides on an average day.

For organizations, this is the most precise definition of buffering capacity available anywhere in nature. Resilience is not a wall. It is a wide, permeable zone that absorbs disturbance gradually, degrades in parts rather than all at once, and — in the best case — accumulates something useful from the very forces that were supposed to destroy it.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where is your organization defended by a wall that would fail all at once?*

KEY TAKEAWAYS

1. Mangroves reduce wave energy by absorbing it through a permeable structure rather than resisting it.
2. A buffer degrades in parts and keeps working, unlike a wall, which fails all at once.
3. The protective effect is conditional on structure, drainage and species, not automatic.

SUMMARY

Mangrove forests absorb wave energy through dense flexible prop roots and trap sediment that builds the ground beneath them, though their protective capacity depends heavily on local conditions and can fail, as post-hurricane dieback in Florida showed. Buffering works by absorbing disturbance gradually rather than opposing it. Organizations should design permeable buffers rather than rigid defences.

SCIENCE CHECK

ESTABLISHED

Mangrove root structure measurably reduces wave height, though the effect depends on belt width, density and drainage.

INFERRED

That resilient systems should be built as wide, permeable buffers rather than rigid walls.

UNCERTAIN

Buffering scales with the extent of the forest; thin restoration plantings deliver little of the protection a substantial belt provides.

VOCABULARY & PHRASAL VERBS

prop root (n.) a root that grows from the trunk and supports the tree above ground.

buffer / buffering capacity (n.) something that absorbs shock; the ability to do so.

permeable (adj.) allowing liquid or force to pass through.

sediment (n.) fine material carried by water that settles out.

accretion (n.) gradual growth by accumulation of material.

dieback (n.) the progressive death of trees in an area.

drop out (phrasal verb) to fall out of suspension; to leave.

keep pace with (phrase) to change at the same rate as something else.

hold back (phrasal verb) to restrain or absorb a force.

PRACTICE

Where the Analogy Breaks

A mangrove has no fixed cost per tree and no payroll. Buffers in business are visible expenses in good years and invisible savings in bad ones.

Use the Language

Make sentences using these three words: permeable, buffering capacity and hold back.

Discuss

Where is your organization defended by a wall that would fail all at once rather than gradually?

Write

Compare two ways of protecting something you rely on: one that fails suddenly, one that degrades slowly.

SOURCES

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THE CONTRAST

Story 43: Mangroves: Making Their Own Ground vs. Story 26: Rivers: Around, Not Through

Story 43: Mangroves: Making Their Own Ground: Survival in a hostile environment means actively absorbing and dissipating force through a permeable structure, rather than avoiding it.

Story 26: Rivers: Around, Not Through: Survival means total flexibility about route, finding the path of least resistance around any obstacle rather than confronting it.

Should an organization facing a hostile force build structure to absorb it directly, or stay flexible enough to route around it entirely?

THIS CONNECTS TO

Story 16: Birds: The Nest Fits the Conditions LIMITATION
Bird nests adapt their design to whatever local materials and conditions already exist. Mangroves go a step further — actively changing the physical conditions around them, building new ground rather than just adapting to existing terrain.

Story 37: Termite Mounds: Walls That Breathe SAME MECHANISM
Both show a structure that manages a harsh environmental force — extreme temperature swings, or wave energy — passively, through its own physical architecture, without any active regulation.

Story 23: Penguins: Warmth by Taking Turns LIMITATION
Mangroves survive a harsh environment through a structural, not behavioural, solution — a fixed physical design that absorbs stress rather than a rotating group behaviour like penguin huddling.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Mangrove forests grow directly in salt water along unstable, wave-exposed coastlines — a combination most plants cannot tolerate at all — and over time the mangrove forest itself builds new, stable land out of what was previously open water or unstable mudflat.

What is the mechanism?

The dense, arching prop roots of a mangrove tree slow incoming wave energy and water flow across a wide area rather than blocking it outright, which

causes waterborne sediment to settle and accumulate around the roots instead of being carried away; over years, this slow, distributed accumulation builds genuinely new, stable ground, meaning the mangrove doesn't just tolerate a hostile coastal environment, it actively converts it into a more stable one.

What organizational problem does it explain?

Is your organization treating a hostile market or operating condition purely as something to survive, when a structure absorbing and redirecting that force over time might actually convert the condition itself into something more stable?

Level 2 — Make It Your Story

> *Tell me about a time your organization didn't just survive a difficult market or operating condition, but built something durable directly out of dealing with it.*

Use STAR to answer. Think about: a capability built specifically to handle a hostile condition that then became a lasting asset, infrastructure built during a crisis that outlasted it.

TAKE THE IDEA FURTHER

> *Is there a hostile condition your organization is currently just enduring, that a deliberately absorbing structure — built to slow and redirect the force rather than block it — might actually convert into a lasting advantage?*

PART VI

Life, Strategy & Personal Success

What can nature teach us about living well?

44

Butterflies: One Species, Two Lives

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Inside a chrysalis, most of the caterpillar dissolves. Enzymes break its tissues down into a nutrient-rich fluid, and from that fluid small clusters of cells called imaginal discs — present and idle since early larval life — build the adult insect. Remarkably, something survives the dissolution: caterpillars trained to avoid an odour produced adult moths that still avoided it, so certain neural structures persist through the reorganization.

That is how it happens. The better question is what the transformation is actually for.

A caterpillar and a butterfly are the same animal, and they have almost nothing in common. The caterpillar has chewing mouthparts and eats leaves. The butterfly has a proboscis — a coiled tube — and drinks nectar. The caterpillar cannot travel; the butterfly can cross a continent. The caterpillar cannot reproduce; for many species the adult barely feeds and exists largely to reproduce and disperse. Biologists point out that this division lets one species occupy two entirely separate ecological niches without the two life stages ever competing with each other for food.

There is a reason the metaphor of 'wearing many hats' fails to describe this kind of change, and naming the failure sharpens the lesson. Wearing hats implies the head underneath stays the same between them. What a caterpillar becomes is not a butterfly wearing a caterpillar's old competence as one option among several — the old competence is gone, structurally, and cannot

be reassembled. Few career changes are truly like this, though many are described that way in job interviews.

The niche difference is not vague. The two forms disagree about almost every dimension of survival at once: what to eat, how to move, how to avoid being eaten, and how and when to reproduce. Trying to change only one of those dimensions while keeping the others fixed does not produce a butterfly. It produces something that eats like a caterpillar and tries to fly, which is roughly what an organization gets when it promotes its best specialist and asks them to keep doing the specialist's job at the same time.

That is the design purpose. The transformation is not self-improvement. It is a change of niche — and it is total, because you cannot drink nectar with jaws built for leaves.

The uncomfortable implication for careers is that some changes cannot be made additively. People trying to move from expert to leader, or from operator to owner, often attempt to keep every existing skill and simply add new ones. But the excellent engineer who becomes a manager and still writes most of the code has not changed niche; they have become a slower engineer with meetings.

There is a reason this feels like loss rather than achievement, and it is worth naming rather than managing around. The caterpillar was genuinely good at being a caterpillar — it was not failing, it was complete — and the butterfly does not experience its old competence as a stepping stone it has now surpassed. The old form simply stops being the right instrument. People asked to make an equivalent change rarely describe it as growth while it is happening. They describe it as loss, and the description is accurate.

Growth of this kind is not learning more. It is accepting that the mouthparts that served you extremely well are the wrong instrument for the thing you are becoming — and letting them go.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What skill made you successful that is now the wrong instrument for where you are going?*

KEY TAKEAWAYS

1. Caterpillar and butterfly occupy completely different niches with different anatomy.
2. The transformation exists to change role entirely, not to improve the existing one.
3. Some growth requires giving up capabilities, not accumulating them.

SUMMARY

The butterfly's metamorphosis allows one species to occupy two separate ecological niches, with the adult eating differently, moving differently and reproducing — the larval form's anatomy simply cannot do those things. Change of this kind is substitution rather than addition. Real growth sometimes means abandoning the skills that previously defined you.

SCIENCE CHECK

ESTABLISHED

Complete metamorphosis lets one species occupy two ecological niches with radically different anatomy for each.

INFERRED

That some career or organizational transitions require substitution of capability rather than addition to it.

UNCERTAIN

Whether niche-partitioning explains the diversification of these insect lineages is a documented correlation, not proof of evolutionary intention.

VOCABULARY & PHRASAL VERBS

proboscis (n.) a long tube-like mouthpart used for drinking. The adult drinks nectar through a coiled proboscis.

chrysalis (n.) the hard case in which a caterpillar transforms. Inside the chrysalis, most of the old body dissolves.

imaginal disc (n.) a group of cells that becomes an adult body part. Imaginal discs build the wings and legs of the adult.

niche (n.) the role and position of an organism in its environment. The two forms occupy completely different niches.

to disperse (v.) to spread over a wide area. Adults disperse to find mates and lay eggs.

additively (adv.) by adding to what already exists. The change cannot be made additively.

substitution (n.) replacing one thing with another. Growth here means substitution, not addition.

mouthparts (n.) the structures an insect uses to feed. The old mouthparts are the wrong instrument now.

let go of (phrasal verb) to release or abandon. She had to let go of the skill that made her valuable.

hold on to (phrasal verb) to keep, often unnecessarily. He held on to the old role and slowed everyone down.

grow out of (phrasal verb) to stop needing something as you develop. Some capabilities you have to grow out of, not build on.

PRACTICE

Where the Analogy Breaks

A butterfly cannot go back. Humans can, which makes half-transformations possible — and common.

Use the Language

Make sentences using these three words: niche, let go of and hold on to.

Discuss

What skill made you successful that may now be the wrong instrument for where you are going?

Write

Describe a change you experienced as loss before you experienced it as growth, and say why.

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THE CONTRAST

Story 44: Butterflies: One Species, Two Lives vs. Story 52: Cicadas: Built for One Loud Season

Story 44: Butterflies: One Species, Two Lives: The same organism occupies two completely different niches in sequence, with a total anatomical rebuild between them.

Story 52: Cicadas: Built for One Loud Season: The same organism spends most of its life in one hidden form, then briefly emerges into a totally different, unmistakable one before dying.

When an organization needs to become something structurally different, is it better to rebuild completely, as a caterpillar does, or to prepare quietly for one brief, unmistakable expression, as a cicada does?

THIS CONNECTS TO

Story 52: Cicadas: Built for One Loud Season SAME MECHANISM

Both describe an organism whose life is divided into two structurally distinct stages, and whose apparent transformation is not a malfunction or a waste but the intended design.

Story 19: Axolotls: Regrowth Instead of Scar LIMITATION

A caterpillar's transformation into a butterfly is a complete, scheduled anatomical rebuild that happens regardless of injury. Axolotl regeneration is a targeted rebuild triggered specifically by damage — both rebuild structure completely, but for very different reasons.

Story 45: Scrub-Jays: Planning for a Future Need LIMITATION

A caterpillar's transformation is genetically scheduled regardless of current conditions. Scrub-jays actively plan and cache resources based on anticipating a specific future need — a more flexible, situation-dependent form of preparing for what comes next.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A caterpillar and the butterfly it becomes occupy almost entirely different ecological niches — different food sources, different predators, different modes of movement, different roles in the ecosystem — despite being, genetically, the exact same individual organism.

What is the mechanism?

During metamorphosis inside the chrysalis, most of the caterpillar's larval tissue is broken down and reorganized into entirely new adult structures, guided by the same genome expressing a completely different developmental program; this is not gradual growth but a wholesale structural rebuild, allowing one organism to fully specialize for two unrelated functions — eating and growing as a larva, dispersing and reproducing as an adult — in sequence rather than compromising on a single body plan that had to do both.

What organizational problem does it explain?

When your organization needs to serve two genuinely different functions across its lifecycle — like a startup phase and a scale phase — is it trying to

compromise with one structure for both, when a complete rebuild between phases might actually serve each one properly?

Level 2 — Make It Your Story

> *Tell me about a time your organization or team had to completely rebuild its structure to serve a new phase, rather than gradually adjusting the old one.*

Use STAR to answer. Think about: a startup that restructured entirely to scale, a role that required becoming someone functionally different rather than a gradual evolution.

TAKE THE IDEA FURTHER

> *Is your organization trying to gradually adjust one structure to serve two genuinely different functions, when a complete rebuild between phases — like a caterpillar's metamorphosis — might serve each function better?*

45

Scrub-Jays: Planning for a Future Need

Evidence Level 1
Biological claim: Strong scientific evidence
Business application: Analogy

Almost everything an animal does can be explained as a response to how it feels now. Hunger produces foraging; cold produces shelter. The interesting question is whether any animal acts for a state it is not currently in — and western scrub-jays appear to.

In an experiment at Cambridge, jays spent nights in one of two compartments. In one they received breakfast; in the other they received nothing. After they had learned the pattern, they were given food in the evening and allowed to store it wherever they liked. They cached significantly more in the compartment where they would wake up hungry — although at the moment of caching they were not hungry at all. In a second version, jays given only one food type in each room cached each food where it would not otherwise be available — arranging a variety they could not have been craving at the time.

The species itself is worth a note, because it changes how surprising the finding is. Western scrub-jays are corvids, a family that includes crows and ravens and is generally associated with unusually flexible problem-solving among birds — so the caching result sits alongside a broader pattern of corvid cognition rather than standing alone as an isolated curiosity. That context matters for how much weight the finding can bear: a surprising result from a family already known for cognitive flexibility is more credible than the same result from a species with no other such evidence.

The design of the second experiment is worth walking through, because it rules out the obvious alternative explanation. If jays simply preferred variety, they should have distributed both foods evenly regardless of the rooms. Instead they concentrated each food specifically in the room where it would otherwise be unavailable the next morning — a pattern that tracks future access rather than a general appetite for change, which is precisely the distinction the experiment was built to isolate.

The interpretation remains debated, as it should be: distinguishing genuine future planning from sophisticated learned association is hard, and researchers continue to argue about it. But the behaviour itself is not in doubt.

The business version is not the platitude "plan ahead". Almost every organization optimizes for its current condition — demand, margins, staffing, technology — and does so with real discipline, because those things are measurable today. The jay's discipline is different: it spends effort now, while nothing is wrong, on a state it can only anticipate.

It is worth being honest about what the jay is not doing, since the popular framing overreaches easily. There is no evidence the bird is running scenarios or comparing named futures the way a strategist might; the mechanism could be a highly specific, evolved response to a particular recurring problem rather than general foresight applied flexibly to new situations. What survives careful reading of the data is narrower and still useful: behaviour organized around an anticipated future state that the animal is not currently experiencing.

The practical form of this is optionality. A cache is not a prediction. It is a small, cheap position that will matter enormously in one possible tomorrow and cost very little in all the others.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What are you doing today only because of conditions that already exist?*

KEY TAKEAWAYS

1. Scrub-jays cache food according to a future need they are not currently experiencing.
2. Whether this is true planning or learned association is still scientifically debated.
3. Optionality means paying a small cost now for a position that only matters in some futures.

SUMMARY

In controlled experiments, western scrub-jays stored food where and in the form they would need it the next morning, despite not being hungry when they cached it, though researchers still debate whether this constitutes genuine future planning. The behaviour spends present effort on an anticipated state. Organizations that optimize only for measurable present conditions forgo exactly this kind of cheap optionality.

SCIENCE CHECK

ESTABLISHED

Scrub-jays cache food according to a specific future need in controlled experiments, ruling out a simple preference for variety.

INFERRED

That cheap present investment in a foreseeable future is systematically under-supplied by organizations optimizing for the present.

UNCERTAIN

Whether this constitutes genuine future planning or a highly specific evolved response to a recurring problem remains actively debated.

VOCABULARY & PHRASAL VERBS

to cache (v.) to store something in a hidden place for later use. The jay caches food it will need tomorrow, not today.

compartment (n.) a separate enclosed section of a space. Each compartment offered a different breakfast.

association (n.) a learned link between two things. Researchers ruled out simple learned association.

optionality (n.) the value of having a choice available later. Optionality is a small cost now for a possible future gain.

to anticipate (v.) to expect and prepare for.

The behaviour anticipates a hunger the bird does not yet feel.

put aside / set aside (phrasal verb) to save something for later use. We should set aside a little capacity for the unlikely case.

stock up on (phrasal verb) to accumulate a supply in advance. It makes sense to stock up before the need is certain.

plan ahead (phrasal verb) to make preparations for the future. Planning ahead here means paying a real cost today.

fall back on (phrasal verb) to use a reserve when the main option fails. Keep something to fall back on if conditions change.

PRACTICE

Where the Analogy Breaks

A jay's cache costs almost nothing and expires slowly. Human options usually require ongoing investment and can expire quietly.

Use the Language

Make sentences using these three words: optionality, set aside and plan ahead.

Discuss

What are you doing today only because of a future you can already foresee?

Write

Identify one cheap option worth taking now for a future that may or may not arrive.

SOURCES

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THIS CONNECTS TO

Story 21: Plants: Roots Before Visible Growth SAME MECHANISM

Both describe investing effort now — storing energy underground, caching food — against a future need that isn't currently being experienced, based on anticipation rather than immediate reward.

Story 12: Elephants: What the Oldest Female Knows LIMITATION

Elephant matriarchs draw on accumulated past experience to guide present decisions. Scrub-jays draw on a different cognitive capacity entirely —

anticipating a future need not currently being experienced — showing memory and foresight are separate mechanisms, not the same one.

Story 47: Tardigrades: Stopping Instead of Dying LIMITATION
Scrub-jays actively prepare in advance for a specific anticipated future condition. Tardigrades handle an unpredictable future condition through a completely different strategy — suspending activity reactively once the condition arrives, rather than anticipating it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Western scrub-jays cache food in specific locations based not only on current hunger but on anticipating a specific future need they are not currently experiencing — for instance, caching food specifically in a location where they will wake up hungry the next morning, before that hunger exists.

What is the mechanism?

Experiments have shown scrub-jays adjust their caching behaviour based on anticipated future conditions — including which locations they'll have access to later and what they'll want when they get there — rather than caching purely in response to their current motivational state, indicating a genuine capacity to plan for a future need distinct from what they need right now.

What organizational problem does it explain?

Does your organization only invest resources in response to a need that already exists, or does it have a mechanism for genuinely anticipating and preparing for a need it hasn't experienced yet?

Level 2 — Make It Your Story

> *Tell me about a time you prepared for a future need before you were actually experiencing any pressure related to it.*

Use STAR to answer. Think about: a skill developed before it was needed, a resource stockpiled ahead of an anticipated shortage, a hire made before the workload justified it.

TAKE THE IDEA FURTHER

> *What future need can you already reasonably anticipate that your organization is currently not preparing for, because the pressure of that need hasn't arrived yet?*

46

Ants: Millions of Small Fragments

> **Evidence Level 1**
>
> **Biological claim:** Strong scientific evidence
>
> **Business application:** Analogy

A leafcutter ant carries a fragment of leaf many times its own body weight, along a cleared trail that can run for over a hundred metres. She is not carrying it home to eat. Leafcutters cannot digest leaves at all. The material is taken underground to feed a cultivated fungus, and the fungus feeds the colony — an agricultural system estimated to have been running for tens of millions of years. The fungus is not a passive food source either. The ants cultivate a single specialized fungal lineage, weed out contaminating species, and in return the fungus produces swollen structures that seem to exist mainly to feed the colony; neither partner can now survive without the other.

The scale that emerges from this is remarkable. Mature colonies of Atta species can contain millions of workers, excavate nests containing thousands of chambers, and move a very large volume of soil in the process. Studies of tropical forests describe leafcutters as among the dominant herbivores of the ecosystem, processing an appreciable share of the vegetation in some habitats. Each contribution to that total is a single fragment of leaf.

The scale of the operation makes the individual fragment easy to lose sight of. A mature Atta colony can consume, in leaf material processed through the fungus, a quantity comparable to the forage of a large grazing mammal — a fact usually reported to provoke astonishment at the insects, when the more useful reaction is astonishment at the arithmetic: millions of workers, each carrying a few milligrams, adding up to that.

Some of that effort goes into the road rather than the load. Leafcutters clear and maintain their trunk trails, removing leaf litter and obstacles, and the cleared surface measurably increases how fast loaded ants can travel. A colony therefore spends part of its labour on infrastructure that carries no leaves at all — and gets it back on every subsequent journey.

No ant sees the fungus garden as a whole. No ant knows the colony's annual biomass. Each simply cuts, carries and returns, over and over.

It is worth adding a limit to the analogy, since ant persistence is unchosen in a way human persistence is not. A worker ant cannot become discouraged, cannot compare her contribution with a colleague's, and has no alternative use for her attention. Human consistency has to survive exactly those pressures, which is why it is a harder achievement than the insect version and deserves more credit than the tidy phrase 'small actions add up' usually gives it.

The lesson for individuals is easy to state and genuinely hard to believe while you are doing it: the size of the result and the size of the action are unrelated. A daily action feels identical on the day when it will amount to something and on the day when it will not. You cannot tell from inside the process. The only variable you actually control is whether the trail keeps running.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What have you done consistently for long enough that you can no longer see its effect?*

KEY TAKEAWAYS

1. Leafcutter colonies achieve ecosystem-scale effects entirely through individual leaf fragments.
2. No participant perceives the aggregate result of their own contribution.
3. Small repeated actions feel the same whether or not they will accumulate into something.

SUMMARY

Leafcutter ants sustain colonies of millions and rank among the dominant herbivores in some tropical forests, using nothing but individual fragments of leaf carried underground to feed a cultivated fungus. The aggregate is invisible to every participant. Consistency produces outcomes that no single action ever reveals.

SCIENCE CHECK

ESTABLISHED

Leafcutter ants sustain colonies of millions through individually tiny leaf fragments, cultivating a fungus neither party can survive without.

INFERRED

That consistent small actions can produce results no single action, or participant, can perceive.

UNCERTAIN

Ant persistence is unchosen and cannot be discouraged; human consistency has to survive comparison, doubt and fatigue that ants do not face.

VOCABULARY & PHRASAL VERBS

to cultivate (v.) to grow and care for crops or fungus. The colony cultivates a single fungal lineage.

fragment (n.) a small piece broken off something. Each ant carries one fragment of leaf.

herbivore (n.) an animal that eats plants. Leafcutters rank among the dominant herbivores in some forests.

biomass (n.) the total mass of living material. The colony processes an enormous biomass of leaves.

aggregate (n.) the total made up of many parts. No ant perceives the aggregate result of her own work.

appreciable (adj.) large enough to be noticed or significant.

The colony's effect on the forest is appreciable.

add up (phrasal verb) to accumulate into a significant total. Small contributions add up in ways no participant can see.

keep at it (phrase) to continue doing something persistently. Consistency is harder for us because we can get discouraged.

carry on (phrasal verb) to continue. She carried on long after any visible progress appeared.

PRACTICE

Where the Analogy Breaks

An ant cannot become discouraged. Humans need visible progress to continue, which is why consistency is psychologically hard rather than technically hard.

Use the Language

Make sentences using these three words: aggregate, add up and carry on.

Discuss

What have you done consistently for long enough that you can no longer see its effect?

Write

Write 120 words about a small daily action and what made it hard to continue without visible progress.

SOURCES

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THE CONTRAST

Story 46: Ants: Millions of Small Fragments vs. Story 24: Fungi: The Network Under the Forest

Story 46: Ants: Millions of Small Fragments: Ecosystem-scale impact is achieved by millions of small, individually modest actions accumulating through a coordinated colony.

Story 24: Fungi: The Network Under the Forest: Ecosystem-scale exchange is achieved by one continuous physical network connecting many organisms directly.

Is large-scale impact better achieved by aggregating countless small individual actions, or by building one continuous connecting structure?

THIS CONNECTS TO

Story 24: Fungi: The Network Under the Forest LIMITATION
Leafcutter ants achieve ecosystem-scale effects by aggregating millions of small individual actions into a cultivated resource. Mycorrhizal fungi achieve a comparable scale of resource exchange through one continuous physical network — two very different structures producing outsized impact.

Story 1: Ants: Order Without a Manager SAME MECHANISM
Both show a leafcutter or foraging ant colony achieving sophisticated collective results — a cultivated food source, an efficient route — through the accumulated effect of many individually simple actions, not central direction.

Story 42: Coral Reefs: Abundance From Poor Water SAME MECHANISM
Both describe a tightly coupled partnership between two organisms — ants and a cultivated fungus, coral and algae — where neither partner can survive without the other, and the relationship itself produces the outsized result.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A leafcutter ant colony can strip a significant portion of a tree's leaves in a single day, with the material carried underground not to be eaten directly —

leafcutters cannot digest raw leaves — but to feed a cultivated fungus that the ants have farmed for tens of millions of years, longer than humans have practiced agriculture.

What is the mechanism?

Each individual leafcutter ant carries a single fragment of leaf, often many times its own body weight, along a cleared trail; the ecosystem-scale effect — measurable changes in forest nutrient cycling — comes entirely from the aggregation of millions of these individually modest actions, combined with a genuine domestication relationship in which the ants actively cultivate a single fungal lineage and the fungus has become dependent on the ants for survival.

What organizational problem does it explain?

Is your organization looking for one large lever to produce ecosystem-scale impact, when millions of small, individually modest, well-coordinated actions might actually be the more realistic and durable path to the same scale of result?

Level 2 — Make It Your Story

> *Tell me about a time a large-scale result came from the accumulation of many small, individually modest actions, rather than from one big decisive move.*

Use STAR to answer. Think about: a marketing effort built from many small touches, a culture built from countless minor consistent actions rather than one grand initiative.

TAKE THE IDEA FURTHER

> *Is your organization currently searching for one large lever to produce a big result, when the more realistic path might be coordinating millions of small actions the way a leafcutter colony does?*

47

Tardigrades: Stopping Instead of Dying

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A tardigrade — a water bear — is half a millimetre long, lives in moss and lake sediment, and is not especially strong. What it can do is stop.

When its environment dries out, a tardigrade pulls in its legs, contracts into a barrel-shaped form called a tun, and enters cryptobiosis — a state in which measurable metabolism falls to almost nothing. In this condition tardigrades have survived temperatures near absolute zero, enormous pressures, doses of radiation that would kill a human many times over, and, in a 2007 orbital experiment, direct exposure to the vacuum of space. Add water and, hours later, they walk away.

It is worth being clear about what dormancy costs even when it works perfectly, since the survival stories tend to skip this part. Revival from cryptobiosis is not instantaneous — it can take hours, during which the organism is doing nothing else — and studies tracking tardigrades through repeated desiccation-rehydration cycles have found evidence of accumulated cellular damage and reduced future performance. The tun state is a genuine solution to an otherwise fatal problem, not a costless one.

The molecular detail is what turned an oddity into a research field. As the animal dries, it produces proteins that remain disordered rather than folding into a fixed shape, and these form a glass-like matrix inside the cell as the last water leaves — a structure that immobilizes and protects delicate cell machinery in something close to physical suspension, roughly analogous to

how sugar syrups can vitrify around biological material during freeze-drying. Rehydration dissolves the glass and, if the transition was managed correctly, everything underneath resumes as though no time had passed.

It is worth being exact about the caveats, because the popular version of this story tends to flatten them into simple toughness. Speed of drying matters enormously: tardigrades that experience gradual moisture loss survive substantially better than the same species desiccated rapidly, since rapid drying resembles ordinary heat or chemical death rather than controlled dormancy — though tolerance for both rate and depth of drying varies considerably between species, and some are markedly more resistant to fast desiccation than others. The organism is not simply hard to kill. It is capable of a specific, gradual transition that has to be allowed to complete.

Two caveats matter, and they are the whole business lesson. The transition must be made before the crisis is complete — dried too fast, a tardigrade dies. And dormancy is not free: revival takes time, and repeated cycles take a toll.

Most organizations under stress do the opposite. Revenue falls, and activity increases: more launches, more meetings, more initiatives, all funded from reserves that were the actual survival asset. The tardigrade's strategy is the unglamorous one — reduce activity deliberately and early, protect the core, accept that nothing will grow for a while, and wait for conditions that no amount of effort can create.

Compare this with the sunflower later in this part. The sunflower knows when to keep adjusting. The tardigrade knows when to stop entirely. Strategy is largely the ability to tell which situation you are in.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *If revenue fell by half, would your instinct be to do more — or to reduce and protect the core?*

KEY TAKEAWAYS

1. Tardigrades survive extremes by suspending metabolism, not by resisting the conditions.
2. The transition must be made early; dormancy entered too late does not work.
3. Reducing activity to protect the core can be the correct response to a downturn.

SUMMARY

Tardigrades endure desiccation, radiation and even vacuum by contracting into a tun and entering cryptobiosis, with disordered proteins forming a glass-like matrix that preserves cell structures until water returns. Survival depends on entering dormancy early and on accepting the cost of revival. Sometimes the correct strategic response is to reduce activity and wait rather than to work harder.

SCIENCE CHECK

ESTABLISHED

Tardigrades survive extreme desiccation, radiation and vacuum by entering cryptobiosis, provided the transition happens gradually.

INFERRED

That reducing activity deliberately and early can be the correct response to severe conditions.

UNCERTAIN

Revival is slow and repeated cycles cause measurable cumulative damage — dormancy is a genuine trade-off, not a costless survival trick.

VOCABULARY & PHRASAL VERBS

cryptobiosis (n.) a state in which metabolism is suspended almost completely. Cryptobiosis suspends metabolism almost entirely.

dormancy (n.) a period of inactivity. Dormancy has to begin before the crisis is complete.

desiccation (n.) the process of drying out completely. Rapid desiccation kills; gradual desiccation does not.

tun (n.) the barrel-shaped form a tardigrade contracts into. The animal contracts into a barrel-shaped tun.

to take a toll (phrase) to cause gradual damage or cost. Repeated cycles of dormancy take a toll on the animal.

shut down (phrasal verb) to stop operating. Sometimes the right response is to shut down deliberately.

scale back (phrasal verb) to reduce the size or level of an activity. We scaled back everything except the core.

ride out (phrasal verb) to survive a difficult period without serious damage. The company rode out the downturn by doing less, not more.

wait out (phrasal verb) to wait until something unpleasant has finished. Sometimes the strategy is simply to wait out the conditions.

PRACTICE

Where the Analogy Breaks

A tardigrade loses nothing by stopping. A dormant business loses customers, staff and market position, so dormancy has costs no tun state has.

Use the Language

Make sentences using these three words: dormancy, scale back and ride out.

Discuss

If conditions became severe, would your instinct be to do more or to protect the core and wait?

Write

Describe what your organization's core is, and what you would deliberately stop doing to protect it.

SOURCES

A Jönsson, K. I., Rabbow, E., Schill, R. O., Harms-Ringdahl, M. & Rettberg, P., "Tardigrades survive exposure to space in low Earth orbit", Current Biology 18, R729–R731 (2008).

A Boothby, T. C. et al., "Tardigrades use intrinsically disordered proteins to survive desiccation", Molecular Cell 65, 975–984 (2017).

THE CONTRAST

Story 47: Tardigrades: Stopping Instead of Dying vs. Story 22: Camels: Tolerance, Not Reserves

Story 47: Tardigrades: Stopping Instead of Dying: Survival of an extreme condition comes from suspending activity almost entirely until the condition passes.

Story 22: Camels: Tolerance, Not Reserves: Survival of a harsh environment comes from continuing to function while tolerating a wider range of internal conditions.

When conditions become extreme, is it better to keep functioning by tolerating a wider range, or to stop entirely and wait the extreme condition out?

THIS CONNECTS TO

Story 22: Camels: Tolerance, Not Reserves LIMITATION
Camels tolerate extreme conditions by widening their internal operating range while staying active. Tardigrades tolerate extreme conditions through a different strategy entirely — suspending activity almost completely rather than continuing to function through it.

Story 19: Axolotls: Regrowth Instead of Scar LIMITATION
Axolotls respond to damage by actively rebuilding the lost structure. Tardigrades respond to extreme conditions by doing close to the opposite — suspending activity almost entirely rather than actively resisting or repairing.

Story 45: Scrub-Jays: Planning for a Future Need LIMITATION
Scrub-jays actively prepare in advance for a specific anticipated future condition. Tardigrades handle an unpredictable future condition through a completely different strategy — suspending activity reactively once the condition arrives, rather than anticipating it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Tardigrades can survive conditions that would kill nearly any other animal — complete desiccation, extreme temperatures, high radiation doses, and even, in tested cases, the vacuum of space — not by resisting these conditions while remaining active, but by entering a state called cryptobiosis in which nearly all measurable metabolic activity stops.

What is the mechanism?

Facing extreme desiccation, a tardigrade expels almost all its body water and produces protective proteins that stabilize its cellular structures in a glass-like state, effectively suspending its metabolism to a rate too low to measure; in this state, the extreme condition that would normally cause fatal damage during active metabolism has little to actively damage, and the tardigrade can remain in suspension for years, resuming normal activity once conditions become survivable again.

What organizational problem does it explain?

When your organization faces a genuinely extreme, temporary condition, is continuing to operate at full activity actually the safest option, or would deliberately suspending non-essential activity until conditions improve reduce more risk than it costs?

Level 2 — Make It Your Story

> *Tell me about a time deliberately pausing or suspending an activity, rather than pushing through an extreme condition, turned out to be the safer and smarter choice.*

Use STAR to answer. Think about: a project paused during a crisis rather than pushed forward, a deliberate slowdown that prevented worse damage.

TAKE THE IDEA FURTHER

Is your organization currently pushing to stay fully active through a genuinely extreme, temporary condition, when a deliberate, temporary suspension of non-essential activity might actually preserve more long-term capability?

48

Owls: Hearing Where Something Is

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A barn owl can locate a mouse under snow, in complete darkness, using sound alone. Roger Payne's classic experiments showed owls striking prey accurately in a light-tight room, and the mechanism has since been mapped in detail.

The design is worth appreciating. A barn owl's ear openings are placed asymmetrically — one slightly higher than the other — so a sound arrives at the two ears with a tiny difference in both timing and intensity. Horizontal position is computed from the time difference, vertical position from the intensity difference, and the auditory system contains a map of space itself. The facial disc of stiff feathers functions as a parabolic collector, funnelling sound towards the ears.

The owl is not simply listening harder than other animals. Its entire head is an instrument for locating things.

It is worth adding the trade-off, because silence is not free for the owl either. The same feather modifications that break up turbulent noise also reduce aerodynamic efficiency, so silent-flying owls pay a measurable energetic cost compared with birds whose wings are optimized purely for efficient flight. The adaptation is a genuine trade, not a free upgrade — quiet was worth more to a hunter that locates prey by sound than speed or fuel economy was, and the owl's whole wing reflects that priority being chosen over the alternative.

The comb-like fringes on the leading edge of an owl's flight feathers are the detail that makes the silence possible, and they solve a specific acoustic problem rather than a general one. A conventional wing generates a burst of turbulent noise at the leading edge as air passes over it; the fringes break that flow into smaller, quieter eddies before it can generate the frequencies a mouse's ear would detect. The owl is not flying quietly in general. It has specifically engineered away the part of the noise spectrum its prey can hear.

Everything else about the bird is subordinated to this. Fringed flight feathers make its approach nearly silent — which is partly about not alerting the prey and partly about not masking the sound it is still listening to during the attack. The owl keeps gathering information right up to the moment of contact.

It is worth being precise about the limits of the acoustic system too, since a fair account names them. The horizontal-timing, vertical-intensity map works well for locating a source directly ahead but degrades towards the edges of the bird's hearing field, and very low-frequency sounds are harder to localize precisely by this method than the mid and high frequencies a rustling mouse actually produces. The owl is not omniscient. It is exceptionally good at exactly the problem it evolved to solve, which is a narrower and more useful claim.

The lesson is not the usual advice to slow down and be present, pleasant though that is. It is more specific. The owl invests structurally in observation, and it does not stop observing once it has committed to action. Most poor decisions in work and life are not made by people who thought too little. They are made by people who stopped looking at the moment they started moving.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What have you already decided about, without having properly looked at it?*

KEY TAKEAWAYS

1. Barn owls locate prey in darkness through asymmetric ears and a sound-collecting facial disc.
2. The animal is structurally built around observation rather than merely attentive.
3. Silent flight lets the owl keep gathering information while it is already acting.

SUMMARY

Barn owls hunt in total darkness by computing a prey's position from tiny differences in when and how loudly a sound reaches each of their asymmetrically placed ears, aided by a parabolic facial disc and silent flight. Observation is built into their anatomy and continues during the strike. Good decisions require sustained attention, including after action has begun.

SCIENCE CHECK

ESTABLISHED

Barn owls locate prey using measurable time and intensity differences between their asymmetrically placed ears.

INFERRED

That sustained observation, continued through action rather than stopping at the decision, improves outcomes.

UNCERTAIN

The auditory system localizes some directions and frequencies better than others; owl hearing is exceptional at a specific task, not omniscient.

VOCABULARY & PHRASAL VERBS

prey (n.) an animal hunted for food. The owl locates its prey using sound alone.

asymmetric (adj.) not the same on both sides. Its ears are placed asymmetrically on its head.

intensity (n.) here: the loudness of a sound. Vertical position is read from a difference in intensity.

facial disc (n.) the round arrangement of feathers on an owl's face. The facial disc funnels sound towards the ears.

to funnel (v.) to direct into a narrow channel. Stiff feathers funnel sound like a dish.

to mask (v.) to cover or hide something, e.g. a sound.

Silent flight avoids masking the sound it still needs to hear.

home in on (phrasal verb) to move directly towards a target. The owl homes in on the faintest rustle.

pick up (phrasal verb) to detect a signal or sound. It picks up a signal most predators would miss.

slow down (phrasal verb) to reduce speed. Slowing down is what makes the detection possible.

PRACTICE

Where the Analogy Breaks

An owl's attention costs it nothing socially. Human observation competes with the pressure to appear decisive.

Use the Language

Make sentences using these three words: asymmetric, home in on and pick up.

Discuss

What have you already decided about without properly looking at it first?

Write

Describe a decision that improved once someone was willing to look longer before acting.

SOURCES

A Payne, R. S., "Acoustic location of prey by barn owls (Tyto alba)", Journal of Experimental Biology 54, 535–573 (1971).

A Knudsen, E. I. & Konishi, M., "A neural map of auditory space in the owl", Science 200, 795–797 (1978).

THIS CONNECTS TO

Story 49: Predators: The Cost Is in the Search SAME MECHANISM

Both describe a predator's effectiveness coming from a specialized capability that reduces wasted effort — precise location-finding, or early abandonment of a bad pursuit — rather than from raw physical power.

Story 33: Beetles: Drinking From Fog LIMITATION

Barn owls solve a location problem using a specialized anatomical structure

(asymmetric ears) tuned to a specific physical signal (sound). Namib beetles solve a resource-collection problem using a specialized surface structure tuned to a different physical signal (fog condensation) — different senses, same principle of purpose-built precision.

Story 50: Sunflowers: Turning Before the Sun LIMITATION

Barn owls locate prey through passive but highly specialized sensory anatomy. Sunflowers track the sun through an active, energy-costly growth mechanism — passive precision versus active tracking, as two different ways of orienting accurately toward something.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A barn owl can locate and capture prey in complete darkness, using hearing alone, with an accuracy precise enough to strike a target it has never seen.

What is the mechanism?

A barn owl's ears are asymmetrically positioned on its skull — one slightly higher and angled differently than the other — so that a sound reaches each ear at a very slightly different time and intensity depending on its exact source location; the owl's brain processes this minute difference, combined with a facial disc of feathers that funnels sound like a satellite dish, to triangulate a sound's precise vertical and horizontal location well enough to strike accurately without any visual information at all.

What organizational problem does it explain?

Does your organization have a genuinely precise mechanism for locating

where a problem or opportunity actually is, or is it relying on a general sense of direction that leaves it acting on approximate rather than precise information?

Level 2 — Make It Your Story

> *Tell me about a time a precise diagnostic or listening mechanism let you locate a problem's exact source, rather than just its general direction.*

Use STAR to answer. Think about: root-cause analysis that pinpointed an exact issue, a feedback mechanism precise enough to locate a specific failure point.

TAKE THE IDEA FURTHER

> *Where is your organization currently acting on a general sense of where a problem is, when a more precise diagnostic mechanism — like the owl's asymmetric hearing — could locate it exactly?*

49

Predators: The Cost Is in the Search

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A cheetah is the fastest land animal and hunts remarkably rarely. Studies fitting cheetahs with accelerometer collars produced a surprising finding: the largest energy cost in a cheetah's day is not the chase but the searching, and successful predation depends heavily on avoiding wasted pursuits. Cheetahs abandon hunts frequently and early, often before reaching full speed, because a failed chase costs energy that must be recovered from a kill that did not happen.

The mathematics of this is brutal and clarifying. A predator that catches one animal in three and pursues everything will starve, while one that pursues only the situations it can win survives on far fewer attempts. What looks like laziness — a lion sleeping through most of the day, a cheetah watching a herd without moving — is energy management. The decision not to hunt is a decision.

There is a further wrinkle in the cheetah data worth including, because it complicates a purely energetic account. Cheetahs are themselves subject to kleptoparasitism — larger predators such as lions and hyenas frequently steal their kills — and some of the observed caution in initiating chases appears related to minimizing time near a carcass as much as to conserving energy for the chase itself. The decision to hunt selectively is not only about affording the pursuit. It is also about not being caught vulnerable immediately afterwards.

Comparable accelerometer studies of lions in the Serengeti reinforce the picture from a different angle: they typically hunt at night, in cooperation, and even successful hunts are a small minority of attempts — with researchers estimating success rates in the range of one in five or fewer for many prey types, figures that vary by prey species and hunting conditions. A predator surviving on those odds cannot afford to treat every opportunity as worth the chase; the arithmetic only works if most opportunities are declined before real energy is spent on them.

The transfer to human working life is direct, and it is not really about time management. Most people are reasonably good at deciding how to spend hours. Very few are good at deciding what not to pursue, because refusing an opportunity feels like a loss while starting one feels like progress.

The predator's discipline is to evaluate before committing: is this catchable, from this position, at this cost? Every pursuit you begin and abandon has already spent the energy you needed for the one you could have won.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What are you currently pursuing that you would not start today?*

KEY TAKEAWAYS

1. Cheetahs abandon most hunts early because failed pursuits carry serious energy costs.
2. Apparent inactivity in predators is deliberate energy management, not idleness.

3. Choosing what not to pursue is a harder and more valuable skill than working faster.

SUMMARY

Research with accelerometer collars shows that cheetahs spend most of their energy searching rather than sprinting, and that abandoning unpromising hunts early is essential to their survival. Selectivity, not maximum effort, sustains predators. Choosing your battles is an energy calculation rather than a matter of ambition.

SCIENCE CHECK

ESTABLISHED

Accelerometer studies show cheetahs spend most daily energy on searching for prey rather than on the chase, and abandon most pursuits early.

INFERRED

That selectivity, not maximum effort, is what sustains a predator — and by analogy, a strategy.

UNCERTAIN

Some caution in cheetah hunting may reflect avoiding kleptoparasitism by larger predators as much as conserving energy for the chase.

VOCABULARY & PHRASAL VERBS

pursuit (n.) the act of chasing something. Most pursuits are abandoned before they begin in earnest.

accelerometer (n.) a device that measures movement and acceleration. Accelerometer collars recorded where the energy actually went.

predation (n.) the act of hunting and killing prey. Successful predation depends on avoiding wasted chases.

to abandon (v.) to give up and stop. Cheetahs abandon most hunts early, before full speed.

idleness (n.) the state of doing nothing. What looks like idleness is energy management.

selectivity (n.) the practice of choosing carefully. Selectivity, not effort, is what sustains the predator.

give up on (phrasal verb) to stop trying with something. Give up on the pursuits you cannot win.

hold back (phrasal verb) to refrain from acting. Holding back is sometimes the correct decision.

go after (phrasal verb) to pursue something you want. Going after everything available is a losing strategy.

pass up (phrasal verb) to decline an opportunity. Passing up a bad opportunity protects the good one.

PRACTICE

Where the Analogy Breaks

A cheetah's calculation is metabolic and immediate. Humans continue losing pursuits because of sunk cost and reputation, which cheetahs do not have.

Use the Language

Make sentences using these three words: pursuit, give up on and pass up.

Discuss

What are you currently pursuing that you would not choose to start today?

Write

Write 120 words identifying a pursuit worth abandoning, and what makes stopping difficult.

SOURCES

A Scantlebury, D. M. et al., "Flexible energetics of cheetah hunting strategies provide resistance against kleptoparasitism", Science 346, 79–81 (2014).

A Wilson, A. M. et al., "Locomotion dynamics of hunting in wild cheetahs", Nature 498, 185–189 (2013).

THE CONTRAST

Story 49: Predators: The Cost Is in the Search vs. Story 10: Wolves: The Alpha That Never Existed

Story 49: Predators: The Cost Is in the Search: Corrects a myth about effort and reward by showing the real cost sits in the search and pursuit decision, not the dramatic chase itself.

Story 10: Wolves: The Alpha That Never Existed: Corrects a myth about dominance and reward by showing real leadership doesn't come from fighting for status.

How much of what organizations believe about effort and reward in competitive situations comes from focusing on the dramatic, visible moment rather than the real cost, which sits somewhere quieter?

THIS CONNECTS TO

Story 10: Wolves: The Alpha That Never Existed LIMITATION
Wolves corrects a myth about dominance and reward. Predators corrects a myth about effort and reward from a different angle — showing that the visible, dramatic moment is usually not where the real cost sits.

Story 26: Rivers: Around, Not Through SAME MECHANISM
Both describe a system optimizing for the goal itself rather than for any particular fixed method of reaching it, abandoning a specific approach the moment it stops being the cheapest path forward.

Story 48: Owls: Hearing Where Something Is SAME MECHANISM
Both describe a predator's effectiveness coming from a specialized capability that reduces wasted effort — precise location-finding, or early abandonment of a bad pursuit — rather than from raw physical power.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Cheetahs, despite being the fastest land animal, hunt successfully only a modest fraction of the time and frequently abandon a chase within seconds of starting it, well before exhaustion would explain the decision.

What is the mechanism?

Studies using accelerometer collars on wild cheetahs found that the largest single energy cost in a hunt is not the high-speed chase itself but the travelling needed to find prey in the first place; because that cost is paid immediately and heavily, cheetahs that sense early — through the prey's positioning or

reaction — that a chase is unlikely to succeed abandon it fast, before paying the full cost of a low-probability pursuit.

What organizational problem does it explain?

Is your organization measuring effort by the length or visibility of a pursuit — a sales cycle, a project, a chase — when the real cost is actually concentrated in the early, high-intensity decision to commit, which deserves far more scrutiny than the pursuit itself gets?

Level 2 — Make It Your Story

> *Tell me about a time you or your team abandoned a pursuit early, once it became clear the cost of continuing wasn't justified by the odds of success.*

Use STAR to answer. Think about: a deal you walked away from early, a project cut before the sunk cost grew larger, a pursuit you correctly judged wasn't worth finishing.

TAKE THE IDEA FURTHER

> *Where is your organization currently committing fully to a pursuit — a deal, a project, a market — before properly pricing in the early, concentrated cost of the initial commitment itself?*

50

Sunflowers: Turning Before the Sun

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

Young sunflowers track the sun. The head faces east at dawn, follows the sun west through the day, and turns back overnight to face east again before sunrise. The behaviour is called heliotropism, and it works through differential growth: cells on one side of the stem elongate faster during the day and the other side catches up at night, so the plant bends by growing rather than by moving.

It is worth adding why the plant bothers reorienting at all overnight, since the return journey costs energy the forward tracking does not obviously repay. The leading explanation is that facing east at dawn lets the flower head warm rapidly as soon as the sun rises, which in turn affects both the plant's own metabolic activity and its attractiveness to pollinators arriving in the cool morning — an advantage that would be lost entirely if the plant simply stayed facing west at the end of each day, however much energy that omission would save.

The experiment that pinned down the mechanism is elegant in its simplicity. Researchers rotated potted sunflowers, physically preventing them from tracking the sun, and the plants continued to grow in a rhythmic east-west pattern anyway — evidence that the movement is generated internally on roughly a daily cycle rather than being a direct mechanical response to sunlight at all. Light does not drive the motion. It resets a clock that would keep running in the dark.

The most interesting part is the ending. As a sunflower matures, the tracking stops and the head settles permanently facing east. The same study found that east-facing mature heads warm faster in the morning and attracted significantly more pollinators than heads artificially turned west. The plant stops responding to conditions and commits to one position — because at that stage, the payoff comes from being reliably in the right place rather than from continuing to adjust.

It is worth being careful about intention in this account, since nothing here involves a plant weighing options. What happens is that the differential growth rates driving the tracking motion decline with maturity as the stem stiffens and its capacity for further elongation falls, and the fixed east-facing orientation is simply what remains once active tracking is no longer mechanically possible. The plant is not choosing to stop. It is running out of the flexibility that tracking required, and the outcome happens to be advantageous.

Two things follow. Strategy is a continuous response to conditions, not a document you write once. And there comes a point where continuing to adjust costs more than it earns, and the correct move is to commit — knowing which phase you are in is itself the strategic question.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Are you in the phase that rewards adjusting, or the phase that rewards committing?*

KEY TAKEAWAYS

1. Young sunflowers track the sun through circadian-driven differential growth, not passive response.
2. The plant anticipates tomorrow's sunrise rather than merely reacting to today's light.
3. Mature plants stop tracking and commit to east, gaining warmth and more pollinators.

SUMMARY

Sunflowers follow the sun using an internal circadian rhythm that drives differential stem growth and even turns them back east before dawn, then stop tracking at maturity and settle facing east, which warms them faster and attracts more pollinators. Continuous adjustment gives way to deliberate commitment. Strategy means responding to conditions — and knowing when responding should stop.

SCIENCE CHECK

ESTABLISHED

Sunflower heliotropism is driven by an internal circadian rhythm, and mature plants stop tracking and settle facing east.

INFERRED

That strategy is a continuous response to conditions, and that some phases call for commitment rather than continued adjustment.

UNCERTAIN

The mechanism halting tracking may be mechanical — reduced stem flexibility with maturity — rather than an adaptive 'decision' to stop.

VOCABULARY & PHRASAL VERBS

heliotropism (n.) growth or movement in response to the sun. Heliotropism is growth or movement that tracks the sun.

circadian rhythm (n.) an internal biological cycle of about 24 hours. An internal circadian rhythm drives the tracking, not light alone.

to elongate (v.) to become longer. Cells on one side elongate faster than the other.

anticipatory (adj.) acting in advance of an expected event. The turn back to the east is anticipatory, not reactive.

pollinator (n.) an animal that carries pollen between flowers. East-facing heads attract more pollinators.

payoff (n.) the reward or benefit gained. The payoff comes from being reliably in the right place.

catch up (phrasal verb) to reach the same level as something ahead. Mature stems can no longer catch up with the sun's movement.

settle on (phrasal verb) to decide on and stay with something. The plant settles on one orientation permanently.

keep up with (phrasal verb) to stay level with a changing situation. Adjustment has a cost; not every phase should keep up with change.

PRACTICE

Where the Analogy Breaks

A sunflower's transition is genetically timed. Nobody tells a business when to stop adjusting, which is the entire difficulty.

Use the Language

Make sentences using these three words: anticipatory, settle on and keep up with.

Discuss

Are you in a phase that rewards continued adjustment, or one that rewards committing to a position?

Write

Describe something your team keeps adjusting that may be ready for a permanent decision.

SOURCES

A Atamian, H. S. et al., "Circadian regulation of sunflower heliotropism, floral orientation, and pollinator visits", Science 353, 587–590 (2016).

THE CONTRAST

Story 50: Sunflowers: Turning Before the Sun vs. Story 20: Trees: Wind Is What Makes Them Strong

Story 50: Sunflowers: Turning Before the Sun: Orientation toward a predictable daily cycle is achieved through active, internally-driven, energy-costly tracking.

Story 20: Trees: Wind Is What Makes Them Strong: Structural strength against a variable force is achieved through passive, responsive growth adaptation to whatever stress actually occurs.

Should an organization actively track and anticipate a predictable cycle, or passively strengthen itself in response to whatever variable stress actually arrives?

THIS CONNECTS TO

Story 20: Trees: Wind Is What Makes Them Strong LIMITATION

Trees respond to mechanical stress by building targeted structural strength. Sunflowers respond to a predictable environmental cycle with active, energy-costly tracking — two different strategies for dealing with a recurring environmental force.

Story 48: Owls: Hearing Where Something Is LIMITATION

Sunflowers track the sun through an active, energy-costly growth mechanism. Barn owls locate prey through passive but highly specialized sensory anatomy — active tracking versus passive precision, as two different ways of orienting accurately toward something.

Story 52: Cicadas: Built for One Loud Season LIMITATION

Sunflowers actively track a fast, predictable daily cycle throughout their growth. Cicadas orient their entire life cycle around one slow, multi-year cycle culminating in a single brief window — very different timescales of the same underlying idea, timing effort to a cycle.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Young sunflowers track the sun's movement across the sky each day, facing east in the morning and gradually turning to face west by evening, then reorienting back to face east again overnight in anticipation of the next sunrise — a behaviour long assumed to be a passive response to light.

What is the mechanism?

Research has shown this tracking is driven by an internal circadian rhythm producing differential growth on opposite sides of the stem — the side facing away from the sun grows faster during the day, bending the plant toward the light — and critically, the plant continues this rhythmic movement even under constant artificial light with no directional cue at all, confirming the behaviour is actively driven by an internal clock, not a passive reaction, and stops once the plant matures and flowers fully face east permanently.

What organizational problem does it explain?

Is your organization treating a predictable daily or cyclical pattern as something to react to as it happens, when actively anticipating and orienting ahead of the cycle — the way a sunflower turns before dawn, not after — would perform better?

Level 2 — Make It Your Story

> *Tell me about a time your team actively anticipated and prepared for a predictable cycle, rather than simply reacting to it once it arrived.*

Use STAR to answer. Think about: a seasonal business cycle you prepared for in advance, a recurring deadline your team anticipated rather than scrambled to meet.

TAKE THE IDEA FURTHER

> *What predictable cycle does your organization currently just react to as it happens, when actively orienting ahead of it — anticipating rather than responding — could improve the outcome?*

51

Nature: What Survival Actually Takes

> **All Three Levels**
>
> **Biological claim:** mixed — the chapter says which is which
>
> **Business application:** synthesis; the reader judges

For fifty weeks this course has looked at individual organisms. This closing article asks a larger question: what strategies keep appearing, in unrelated lineages, across roughly four billion years?

Twelve recur throughout this series. Adaptation — form follows the environment, as bird nests and camel physiology show. Diversity — mixed groups of cockroaches decide differently from uniform ones. Cooperation — dolphins, penguins, coral. Competition — real, constant, and coexisting with cooperation in every forest. Feedback — the bee's wait at the handover point. Decentralization — starlings, octopus arms, ant colonies. Redundancy — several species doing one job. Resource efficiency — nacre's toughness from 5% organic material. Specialization — bees moving through roles. Interdependence — mycorrhizal networks. Experimentation — the slime mould overbuilding and pruning. Regeneration — the axolotl's limb, the serotinous pine.

It is worth pausing on why a list this size is the right length rather than an arbitrary one. Fewer principles would flatten distinctions this book has spent fifty chapters drawing carefully — decentralization and redundancy are related but not the same claim, and collapsing them would lose exactly the precision the evidence framework was built to protect. A longer list would start including things that are true of some systems and not others, which is a different kind of claim from a genuine recurring pattern. Twelve is not a

magic number. It is roughly where the evidence in this book stopped supporting more.

Notice what is missing. Many living systems do not depend on central planning, on permanent hierarchy, on optimization for a single outcome, or on an assumption of stable conditions — and those recurring across the widest range of environments tend to depend on them least. They have endured enormous uncertainty through distributed intelligence, diversity, cooperation and continual reorganization.

The course should end honestly rather than triumphantly. Nature is not wise. It has no intentions and offers no guarantees: over 99% of all species that have ever existed are extinct, and evolution optimizes for reproduction, not for fairness or human flourishing. Everything in these 52 articles is a strategy that worked in a particular context — not a moral instruction.

It is worth naming the risk in a chapter like this one directly, since drawing twelve principles from fifty different organisms invites exactly the error this book has tried to guard against throughout: selecting the examples that fit a conclusion decided in advance. The defence is not that every case here is equally certain — several chapters have said plainly that they are not — but that the same handful of patterns kept recurring across genuinely independent lineages, studied by different researchers, for different reasons, over more than a century. That recurrence is evidence of something. It is not evidence that the something applies to your organization.

So the closing question is not what should we copy? It is: What has life learned over billions of years that human organizations still struggle to understand — and which of those lessons actually apply to us?

That second half is your work. It cannot be borrowed from an ant.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Of the twelve principles, which three genuinely apply to your organization — and which do not?*

KEY TAKEAWAYS

1. The same strategies recur independently across unrelated lineages and enormous timescales.
2. Many living systems rely more on distribution, diversity and reorganization than on central control.
3. Nature is a source of strategies, not of moral guidance; over 99% of species have gone extinct.

SUMMARY

Twelve principles recur across living systems — adaptation, diversity, cooperation, competition, feedback, decentralization, redundancy, resource efficiency, specialization, interdependence, experimentation and regeneration — and none of them requires central planning or stable conditions. These are strategies that persisted in particular contexts, not moral lessons, and the vast majority of species have still gone extinct. The reader's task is to judge which of them genuinely apply to human organizations.

SCIENCE CHECK

ESTABLISHED

Twelve strategic patterns recur across independent biological lineages studied over more than a century.

INFERRED

That certain organizational strategies — distribution, redundancy, feedback — are worth considering seriously because of that recurrence.

UNCERTAIN

Recurrence across biology is evidence the pattern is a viable strategy somewhere. It is not evidence the pattern applies to any particular human organization.

VOCABULARY & PHRASAL VERBS

lineage (n.) a line of descent. The pattern recurs across unrelated lineages.

to recur (v.) to happen repeatedly. The same handful of strategies keep recurring.

decentralization (n.) the distribution of control away from a centre. Decentralization appears in ants, starlings and octopus arms alike.

to endure (v.) to survive over a long period. Living systems have endured enormous uncertainty.

extinct (adj.) no longer existing as a species. Over 99% of all species that have ever existed are extinct.

flourishing (n.) the state of thriving and living well. Evolution optimizes for reproduction, not human flourishing.

sum up (phrasal verb) to summarize. Twelve principles sum up what recurs across the book.

come down to (phrasal verb) to be reducible to one essential point. The final question comes down to which lessons apply to you.

hold true (phrase) to remain valid. Not every lesson here will hold true for your organization.

PRACTICE

Where the Analogy Breaks

This is the question the whole course has been preparing you to answer. Nature supplies mechanisms; only you can judge which transfer.

Use the Language

Make sentences using these three words: lineage, endure and hold true.

Discuss

Of the twelve principles in this book, which three genuinely apply to your organization, and which do not?

Write

Choose one principle from the course and argue for 150 words why it does not transfer to your own context.

SOURCES

C World Economic Forum, "Nature can teach us about leadership and building resilience" (2026). https://www.weforum.org/stories/2026/04/nature-leadership-resilience-fractured-world/

B Raup, D. M., Extinction: Bad Genes or Bad Luck?, W. W. Norton (1991) — source of the widely cited figure that over 99% of species that have ever lived are extinct.

A Novacek, M. J. & Cleland, E. E., "The current biodiversity extinction event: scenarios for mitigation and recovery", PNAS 98, 5466–5470 (2001).

THIS CONNECTS TO

Story 7: Octopuses: Thinking at the Edges SAME MECHANISM
One of the twelve recurring principles the capstone names — decentralization — is illustrated directly by this chapter, among others.

Story 24: Fungi: The Network Under the Forest SAME MECHANISM
One of the twelve recurring principles the capstone names — interdependence — is illustrated directly by this chapter's mycorrhizal networks.

Story 19: Axolotls: Regrowth Instead of Scar SAME MECHANISM
One of the twelve recurring principles the capstone names — regeneration — is illustrated directly by this chapter, alongside wildfire-adapted pine.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

Twelve strategies recur across genuinely unrelated organisms, studied independently, over roughly four billion years of evolution.

What is the mechanism?

The recurrence itself, across lineages with no shared ancestry near enough to explain the similarity, is treated as evidence that these strategies solve real, general problems rather than being arbitrary or coincidental.

What organizational problem does it explain?

Which of these twelve recurring patterns genuinely applies to your

organization's actual circumstances, and which ones merely sound appealing without being tested against your specific situation?

Level 2 — Make It Your Story

> *Tell me about a time you had to decide which of several appealing general principles actually applied to your specific situation, and which ones didn't.*

Use STAR to answer. Think about: a framework or best practice you tested against your own context and rejected, a general principle you correctly adapted rather than copied.

TAKE THE IDEA FURTHER

> *Of the twelve principles this course has named, which one have you been assuming applies to your organization without ever actually testing whether it does?*

52

Cicadas: Built for One Loud Season

> **Evidence Level 1**
> **Biological claim:** Strong scientific evidence
> **Business application:** Analogy

A cicada spends the overwhelming majority of its life as a nymph, underground, feeding slowly on fluid drawn from tree roots, in near-total darkness. Depending on the species, this stage lasts anywhere from one or two years to, in the case of North America's periodical cicadas, a precisely documented thirteen or seventeen. Popular accounts of Japan's common species round the figure to seven years, and the number is worth treating carefully: it is repeated far more confidently than the underlying entomology actually supports, since duration varies with soil temperature, species and individual, and has been rigorously tracked for far fewer Japanese species than for their North American relatives. What is well established, regardless of the exact figure, is the ratio. Years of unseen development are followed by a final moult into a winged adult that will, in most species, live for somewhere between two and four weeks.

That adult phase is not quiet. Male cicadas produce, proportionally, some of the loudest sounds in the insect world, using a pair of ribbed membranes called tymbals that buckle in and out to generate a call that can exceed 100 decibels at close range — loud enough to cause temporary hearing effects in humans standing directly beside the source. The entire purpose of the adult stage is to make that sound, find a mate, and reproduce, all within a window that closes in days. There is no second season. A cicada that emerges, calls and dies within three weeks has not failed to live a full life. It has completed the only life its species has ever had.

In Japan, the cicada's short, overwhelming summer chorus is one of the most recognizable seasonal markers in the language, tied closely to the aesthetic idea of mono no aware — roughly, the poignancy of things precisely because they do not last. The cicada's call marks the peak of summer and its coming end in the same breath, which is exactly why it appears constantly in classical poetry: not as a symbol of failure or waste, but as proof that something can matter enormously without needing to last. One of the most famous haiku in the language, written by Matsuo Bash at a mountain temple, does nothing more than notice a cicada's cry sinking into the stillness of the rock around it. The moment is the whole point. Nothing about it asks to be extended.

There is a second image from the same insect that Japanese literature has used for a different purpose. When a cicada nymph makes its final climb up a tree trunk, it splits its old exoskeleton down the back and leaves it behind entirely intact, still gripping the bark, perfectly shaped, completely empty. This shed shell — utsusemi — appears throughout classical Japanese writing, including as the title of an early chapter of The Tale of Genji, as an image of a self that has been outgrown and left in plain view while the actual creature has already moved on, changed, and become something that can no longer fit inside its old form. The shell is not evidence that something went wrong. It is the clearest possible evidence that a transformation finished.

Organizations tend to apply a single measure to almost everything they do: did it last? Judged that way, a research program that never shipped a product, a campaign that ran for one quarter, or a team that was deliberately dissolved once its purpose was met all look like failures, or at best like unfinished business. The cicada suggests a different question. Some efforts are built for duration, and their success is measured in years of compounding. Others are built for a single, unrepeatable, high-intensity window, and judging them by whether they persisted afterward is applying the wrong instrument to the wrong kind of work.

The years underground are the part most easily undervalued, precisely because nothing about them is visible or immediately convertible into a result. A nymph feeding slowly on root fluid for several years produces no report, no milestone, no interim output that a quarterly review would recognize as progress. But the loud, brief, unmistakable adult phase is not separate from that invisible period — it is what the invisible period was for. A launch, a public campaign, a once-a-year sales window, a live performance: these are frequently the visible tip of years of preparation that no dashboard tracked, and treating the noisy part as the whole story, or the quiet part as wasted time when it isn't immediately followed by permanence, misreads both halves of the same process.

The shed shell adds a second, smaller lesson, usually most relevant at the level of a team or a system rather than a whole strategy. When an organization outgrows a structure — an old team shape, a retired product, a process nobody uses anymore — the artifact left behind can look, to someone arriving late, like evidence of failure: an empty shell, no longer occupied. It is worth asking, before dismantling it or apologizing for it, whether it is actually a failure or simply an utsusemi: the visible proof that whatever used to live inside it has already grown past the point where it fit.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What in your organization was built for one brief, unrepeatable season — and are you judging it by a standard meant for things built to last?*

KEY TAKEAWAYS

1. A long invisible development period and a short, intense visible outcome can be the same process rather than two separate ones; the second does not need to outlast the first to justify it.
2. Not every valuable effort is meant to persist. Some are built for a single high-intensity window, and their success should be measured against that window, not against duration.
3. A structure an organization has outgrown and left behind is not automatically evidence of failure. It can be the visible proof that a transformation actually completed.

SUMMARY

Cicadas spend years underground as nymphs, then live briefly above ground as adults, calling loudly for a matter of weeks before dying, having completed the only life their species has. Japanese literature has long used this insect for two connected images: the brief, loud adult season as proof that something can matter without lasting (mono no aware), and the shell nymphs leave behind (utsusemi) as evidence of a finished transformation rather than a loss. Organizations that judge every initiative by whether it persisted risk misjudging both the quiet preparation behind a brief intense effort and the structures they have deliberately, successfully outgrown.

SCIENCE CHECK

ESTABLISHED

Cicada nymphs spend the great majority of the species' lifespan underground, with adult life typically lasting only a few weeks; male cicadas produce mating calls among the loudest of any insect, using tymbal organs.

INFERRED

That brief, high-intensity organizational efforts should be evaluated by different criteria than sustained ones.

UNCERTAIN

Exact nymphal duration by species and region, which has been rigorously documented for North American periodical cicadas far more precisely than for most Japanese species.

VOCABULARY & PHRASAL VERBS

nymph (n.) an immature insect stage, before the final adult moult. A cicada nymph spends years underground before its final moult.

moult (v./n.) to shed an old exoskeleton in order to grow or transform. The nymph moults one final time into a winged adult.

tymbal (n.) a ribbed membrane cicadas use to produce sound. The tymbal buckles rapidly to generate the mating call.

decibel (n.) a unit measuring the loudness of sound. Cicada calls can exceed 100 decibels at close range.

poignancy (n.) a quality of being emotionally moving, often because something is brief or fragile. The poignancy of the season lies in knowing it will end.

exoskeleton (n.) a hard external skeleton, such as an insect's outer shell. The nymph splits its exoskeleton down the back to emerge.

high-intensity (adj.) involving a large amount of effort or energy over a short time. A product launch is a high-intensity, short-duration effort.

outgrow (v.) to grow beyond the size or fit of something. A fast-scaling team can outgrow its original structure within a year.

leave behind (phrasal verb) to not take something with you as you move on or progress. The nymph leaves its old shell behind, still gripping the bark.

PRACTICE

Where the Analogy Breaks

A cicada has no say in the length of either phase; the timing is fixed by its biology. Organizations choose, or at least influence, whether an effort is built for duration or for a single intense window, which makes the judgment about which kind of effort you are looking at a live decision, not a fact to be discovered afterward.

Use the Language

Make sentences using these three words: high-intensity, outgrow and leave behind.

Discuss

Think of an effort in your organization that was brief and intense by design. Was it judged as a success on its own terms, or against a standard meant for things built to last?

Write

Describe one structure, team or product your organization has outgrown. Argue for whether it should be read as a failure or as an utsusemi — evidence that a transformation already completed.

SOURCES

A Williams, K. S. & Simon, C., "The ecology, behaviour, and evolution of periodical cicadas", Annual Review of Entomology 40, 269-295 (1995).

A Sanborn, A. F. & Phillips, P. K., "Scaling of sound pressure level and body size in cicadas", Annals of the Entomological Society of America (2013).

C Keene, D., Seeds in the Heart: Japanese Literature from Earliest Times to the Late Sixteenth Century, Henry Holt (1993) - on utsusemi and mono no aware as recurring classical images.

THIS CONNECTS TO

Story 44: Butterflies: One Species, Two Lives CONTRAST

See The Contrast in that chapter. In short: a butterfly rebuilds its entire structure to become something functionally different; a cicada spends most of its life preparing, then briefly, loudly becomes fully itself, once, before dying.

Story 21: Plants: Roots Before Visible Growth SAME MECHANISM

Both describe a long, invisible period of preparation whose value only becomes visible in a short, intense period afterward — and both warn against judging the invisible phase as wasted time.

Story 27: Wildfires: Renewal That Needs Burning LIMITATION

Wildfire ecosystems depend on periodic, externally-triggered disruption for renewal. Cicadas depend on an internally-timed cycle of preparation and release — both involve a long buildup followed by a triggering event, but one is imposed from outside and the other is scheduled from within.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Explain the Principle

What happens in nature?

A cicada spends the vast majority of its life underground as a nymph, then lives briefly above ground as a winged adult — weeks, not years — calling loudly, mating, and dying, having completed the only life its species has.

What is the mechanism?

Years of slow, invisible underground development culminate in a single, unrepeatable, high-intensity adult phase; the loud calling season is not separate from the invisible preparation before it; it is what the preparation was for, and the shed exoskeleton left behind on the bark is evidence the transformation completed, not evidence of loss.

What organizational problem does it explain?

What in your organization was built for one brief, unrepeatable season — and are you judging it by a standard meant for things built to last?

Level 2 — Make It Your Story

> *Tell me about a time you judged a brief, intense effort by the wrong standard — expecting it to last, when lasting was never the point.*

Use STAR to answer. Think about: a campaign, a launch, a live event, a project deliberately built to be short and intense rather than sustained.

TAKE THE IDEA FURTHER

> *What structure has your organization outgrown and left behind — and is it actually a failure, or proof, like a cicada's shed shell, that a transformation already completed?*

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Every word and phrase taught in the book, with the chapters where it appears.

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